From Agriculture to Services: Development without Industrialization

Tancredi Rapone†

UAB, BSE

July 23, 2026

Abstract

This paper studies the long-run consequences of developing without industrialization. Countries with higher historical peaks in manufacturing employment grow faster, accumulate more tertiary education, experience faster fertility declines, and develop more skill-intensive service sectors. The same pattern appears within the United States: counties with larger manufacturing booms later moved into service occupations with higher educational content. I interpret these facts through a quantitative growth model with structural change, endogenous fertility, education, and heterogeneous producer services. In the model, manufacturing creates demand for high-skill producer-service tasks that build the skill base of the service economy that follows. Calibrated to the United States over 1850–2019, I vary a manufacturing-demand shifter to match the cross-country distribution of manufacturing peaks and compare the resulting model gradients to the data. This exercise accounts for 37 percent of cross-country log GDP per capita gaps, 46 percent of GDP per capita growth differences, 55 percent of tertiary education differences, and 22 percent of cumulative population growth differences since 1850. Countries can move into services without industrializing, but the service economy they build is less skill-intensive and grows more slowly.

Keywords: structural change, service-led growth, industrialization, unified growth, fertility, human capital.

JEL Classification: O14, O15, O41, J13, J23, O33, E24.

† First and foremost, I am deeply grateful to my advisors, Raül Santaeulàlia-Llopis and André Gröger, for their continued support, patience, and guidance throughout this project. A special thanks also goes to my PhD evaluation committee members for their valuable insight and continued guidance: Jordi Caballe, Marti Mestieri, Pau Roldan, and Luis Rojas. I also thank Tasso Adamopoulos, Gabriela Araujo, Sophie Brochet, Francisco Buera, Marcela Eslava, Alejandro Gobierno, Jacob Hess, Luca Looser, Enoch Ntsiful, Michael Peters, Luis Felipe Sáenz, Etienne Wasmer, and seminar participants at the BSE PhD Jamboree and the Bellaterra Macro Club for their valuable comments and suggestions. All errors are my own. For comments, suggestions, or critiques, you can reach me at `tancredi.rapone@icloud.com`.

I Introduction

Modern economic growth began with industrialization. The British Industrial Revolution transformed a largely agrarian economy into one organized around manufacturing, mechanized production, and increasingly complex networks of suppliers, managers, clerks, engineers, accountants, and other producer-service occupations. In the vast majority of today's rich countries, the long-run movement of labor out of agriculture was accompanied by a substantial manufacturing phase before services came to dominate employment. Yet this path has become increasingly uncommon. Many developing countries now move directly from agriculture into services, reaching lower manufacturing employment peaks and deindustrializing at lower income levels than earlier industrializers (Rodrik, 2016).

Can countries develop without manufacturing? A narrow answer is yes: labor can leave agriculture and move into services. But this paper argues that the path out of agriculture matters for the kind of service economy that emerges. Manufacturing is not only a sector that produces goods. It also creates demand for high-skill producer services: accounting, management, technical supervision, engineering, logistics, purchasing, quality control, and other coordination tasks. These tasks organize production and generate transferable routines, standards, management systems, and supplier-coordination capabilities. I call this stock of transferable routines and know-how *codified organizational knowledge*. Once created, it can become the foundation for a high-skill service economy.[1] By contrast, economies that bypass manufacturing may enter services earlier, but with employment concentrated in lower-skill consumer-facing activities such as retail, restaurants, hospitality, and personal services. Manufacturing therefore does not just delay the rise of services; it changes the composition of service work, the demand for skills, and in so doing also the education and fertility choices made during the transition out of agriculture.

I provide evidence for this mechanism in two steps. First, I document cross-country facts linking the size of the manufacturing phase to later development outcomes. Countries with higher peak manufacturing employment shares grow faster even after they have largely transitioned into services, accumulate more tertiary education, experience faster fertility declines, and develop more skill-intensive service sectors. These patterns continue to hold among economies with high service employment shares, suggesting that manufacturing matters not only because it absorbs labor during the transition, but because it shapes the service economy that follows. Second, I use full-count U.S. Census data from 1850 to 1940 to study the mechanism within a single economy. Counties with larger manufacturing booms later developed service occupations with higher educational content, measured using occupation-level education scores from the 1950 Census. The effects appear after the manufacturing boom, are strongest for counties with the largest manufacturing peaks, and persist for over a century until the last available data in the early 2000s. IV estimates based on geography-driven manufacturing exposure point to the same conclusion.

1 A historical example is the diffusion of factory-born management methods into services. Scientific management was developed to make repeated industrial work measurable and reorganizable; Lillian Gilbreth later applied these methods to department-store work at Macy's and to other service settings. Similarly, management accounting and budgeting tools developed for complex industrial and procurement problems later became portable tools for retail and service firms through the work of John McKinsey.

Motivated by this evidence, I build a growth model with structural change, fertility, education, and heterogeneous producer services. In the model, manufacturing raises demand for organizational and technical producer-service tasks that are intensive in skilled labor. This raises the return to education and builds codified organizational knowledge that persists after manufacturing declines. Economies with larger manufacturing phases therefore enter the service economy with more human capital, lower fertility, and a more skill-intensive producer-service sector. Calibrated to the U.S. transition from 1850 to 2019, the model matches the decline of agriculture, the manufacturing hump, the rise of services, the fertility transition, and the high-skill content of producer-service inputs. Counterfactual economies with smaller manufacturing peaks, obtained by varying a preference shifter on manufacturing consumption, end up with larger but less skilled populations and grow more slowly even in their asymptotic growth path. Quantitatively, the mechanism accounts for a sizable fraction of the dispersion in development outcomes in 2019: 37 percent of the cross-country relationship between peak manufacturing and relative income, 46 percent of the relationship with GDP per capita growth, 55 percent of the relationship with tertiary education, and 22 percent of the relationship with cumulative population growth. It also nearly matches the cross-country relationship between peak manufacturing employment shares and the high-skill producer-service intensity of modern services. Although the model reproduces the positive association between manufacturing phases and later development outcomes, a policy extension shows that manufacturing subsidies are not welfare improving. They are fiscally costly and, once income and price effects are pulling the economy toward services, ineffective at raising manufacturing employment. By contrast, subsidies to high-skill producer services raise welfare because they operate directly on the margin through which manufacturing has persistent effects.

This paper contributes to three literatures. First, it contributes to work on structural transformation and premature deindustrialization. Classic and modern theories of structural change study why labor moves from agriculture to manufacturing and services, emphasizing non-homothetic preferences, sectoral productivity differences, and relative prices (Kuznets, 1966, 1973; Lewis, 1954; Rostow, 1959; Matsuyama, 1992; Kongsamut et al., 2001; Ngai and Pissarides, 2007; Herrendorf et al., 2014; Boppart, 2014; Comin et al., 2021). Recent work studies why manufacturing peaks differ across countries, emphasizing premature deindustrialization, trade, global specialization, technology gaps, and heterogeneous sectoral productivity growth (Rodrik, 2016; Matsuyama, 2009; Uy et al., 2013; Sposi et al., 2021; Huneeus and Rogerson, 2024; Fujiwara and Matsuyama, 2024). I focus instead on the consequences of these differences. I ask whether missing the manufacturing phase changes the development path itself by altering skill demand, fertility choices, knowledge accumulation, and the long-run composition of services.

Second, the paper contributes to work on service-led growth, industrial policy, and the organization of production. A growing literature asks whether services can substitute for manufacturing as an engine of development and emphasizes that services are highly heterogeneous (Fan et al., 2023; Chen et al., 2023; Peters et al., 2026; Herrendorf et al., 2022; Buera and Kaboski, 2012a,b; Buera et al., 2022; Duernecker and Herrendorf, 2022; Duernecker et al., 2024). Recent work argues that, as manufacturing becomes less labor-absorbing, developing countries must find ways to raise productivity in labor-absorbing services (Rodrik and Sandhu, 2025; Rodrik and Stiglitz, 2025). This paper asks a complementary question: why do some economies enter the service era with the skills and organizational capabilities needed for such upgrading, while others do not? Recent work further shows that modern services can generate productivity growth when firms are able to scale operations across markets and when the economy has sufficient human capital and managerial capacity (Hsieh and Rossi-Hansberg, 2023). Other work shows that development involves the reorganization of production toward larger, more skill-intensive firms and HSPS tasks (Engbom et al., 2025). My contribution is to connect this organizational margin to the sectoral path of development. Manufacturing matters not simply because it has high productivity growth, but because it helps determine whether the service economy expands into high-skill producer services or into low-skill consumer-facing services. This mechanism also connects the paper to work on urbanization without industrialization and consumption cities, where labor leaves agriculture without necessarily entering modern manufacturing (Gollin et al., 2016; Jedwab et al., 2022).

Third, the paper contributes to unified growth theory. Canonical unified growth models explain how technological progress, human capital accumulation, and fertility decline interact in the transition from stagnation to modern growth (Galor and Weil, 2000; Hansen and Prescott, 2002; Galor, 2005, 2011; Becker et al., 1990; Doepke, 2004; Galor and Moav, 2004). I embed this transition in a model of structural change across agriculture, manufacturing, and services. This matters because sectors differ in their demand for skills and in their contribution to codified organizational knowledge. The path out of agriculture therefore affects not only sectoral employment shares, but also fertility, education, and the organizational frontier inherited by the service economy. In this sense, the paper links two literatures that have often been studied separately: structural change across sectors and the joint evolution of fertility, human capital, and income.

The rest of the paper is structured as follows. Section II documents the cross-country facts relating peak manufacturing employment to GDP per capita growth, education, service sector composition, and fertility. Section III uses full-count U.S. Census data to study the effect of historical manufacturing booms on the skill content of services across counties. Section IV develops the model. Section V describes the calibration to the United States over 1850–2019. Section VI presents the counterfactual results. Section VII studies forward-looking manufacturing and high-skill producer-service subsidies in country-calibrated economies. Section VIII concludes. Appendix A describes the data construction, Appendix B reports additional cross-country evidence, Appendix C reports additional U.S. county evidence, Appendix D contains model details, Appendix E reports calibration details and robustness, and Appendix F describes additional policy exercises.

II Cross-Country Evidence

This section documents three cross-country facts linking the size of the manufacturing phase to later development outcomes. Countries with larger manufacturing employment peaks grow faster after moving into services, accumulate more tertiary education by developing more skill-intensive service sectors, and experience faster fertility declines. The analysis combines data on GDP per capita, fertility, population, and broad sectoral employment shares from Gapminder with longer sectoral series from the GGDC 10-Sector Database, tertiary education measures from Barro and Lee (2013), and service-sector occupation data from ILOSTAT. The baseline sample includes 174 countries with employment shares over 1990–2018 and long-run GDP per capita, population, and fertility data. Appendix A describes the data sources, sample construction, and variable definitions in detail.

FACT 1*: Countries with higher peak manufacturing employment shares grow faster as they transition into services.*

The rise of services is often viewed as a drag on aggregate growth. In the standard Baumol cost-disease logic, services experience slower productivity growth than agriculture or manufacturing and, as labor reallocates toward them, aggregate productivity growth declines (Baumol, 1967; Duernecker and Sanchez-Martinez, 2023; Duernecker et al., 2024). This logic, however, does not distinguish between different paths into services. In particular, it does not ask whether service economies that emerge after a large manufacturing phase differ from service economies that bypass manufacturing.

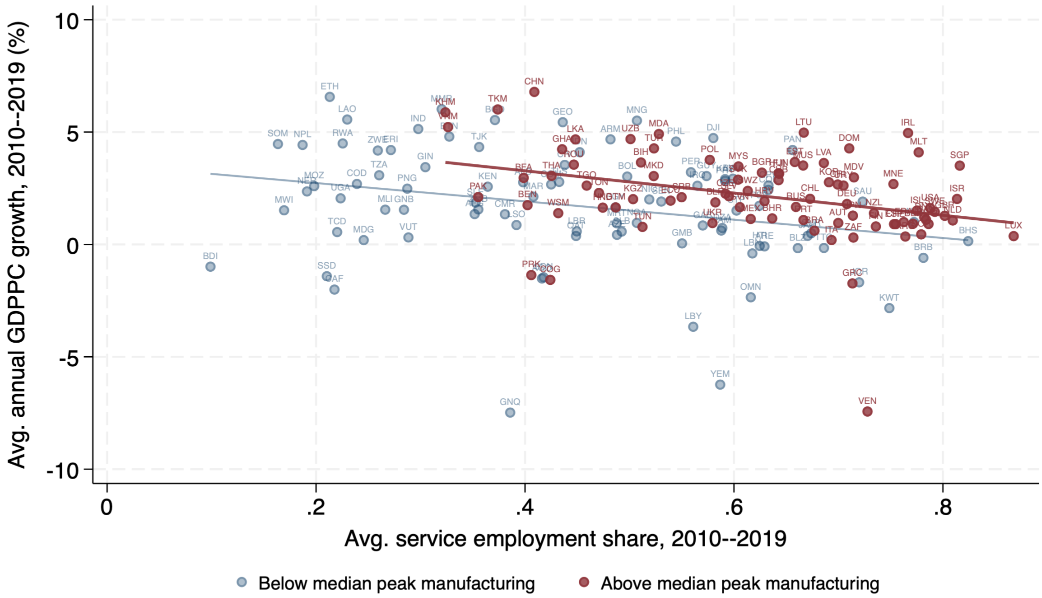


Figure 1: Growth in service economies by peak manufacturing exposure. Countries with above-median peak manufacturing employment shares continue to have higher growth rates as they move into services. Data source: ILO and World Bank via Gapminder.

Figure 1 plots average annual growth in PPP real GDP per capita over 2010–2019 against the average service employment share in the same period. Countries are split by whether their peak manufacturing employment share is above or below the cross-country median. The peak manufacturing employment share is the maximum manufacturing employment share observed over the sample period. Service employment is negatively related to growth, but countries with above-median manufacturing peaks grow faster at nearly all levels of service employment.

To summarize this relationship, I estimate

$$y_c = \alpha_r + \alpha_i + \Gamma' X_c + \beta^s \text{Serv. Emp. Share}_c + \beta^m \text{Manu. Peak}_c + \epsilon_c,$$

where y_c is average GDP per capita growth over 2010–2019, α_i and α_r are income-quintile and region fixed effects, and X_c includes institutional quality, average GDP per capita over 2010–2019, and years since the manufacturing peak.

Table 1: Dep. Var. is Avg GDPPC growth rate 2010-2019

	(1)	(2)	(3)	(4)
Avg. Serv. Emp. Share, 2010–2019	-2.742***	-5.171***	-7.566***	-9.962***
	(-2.84)	(-4.49)	(-3.59)	(-4.17)
Peak Emp. % Man.		9.658***	5.455**	5.511**
		(4.59)	(2.16)	(2.09)
N	174	174	174	174
R2	0.0469	0.130	0.215	0.284
Income Quintile FE	no	no	yes	yes
Region FE	no	no	yes	yes
Controls	no	no	no	yes

t statistics in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 1 reports the results. The coefficient on peak manufacturing employment is positive and statistically significant across specifications. In the most saturated specification, a one percentage point higher manufacturing peak offsets more than half of the growth slowdown associated with a one percentage point higher service employment share. The next fact shows that this is accompanied by entry into a higher-skilled service economy.

FACT 2*: Countries with higher peak manufacturing employment shares accumulate more tertiary education and develop more skill-intensive services.*

If manufacturing-intensive development raises demand for organizational, technical, financial, and professional tasks, then countries with larger manufacturing phases should accumulate more skill during the transition. I summarize this relationship by estimating

$$\Delta e_c = \alpha + \gamma \Delta \log y_c + \sum_{q=2}^{5} \beta_q \, \mathbb{1}\left\{\text{Manu. Peak}_c \in Q_q\right\} + \varepsilon_c.$$

Here $\Delta e_c = e_{c,T_c} - e_{c,t_c}$ is the change in tertiary education for country c between the first and last years in which both education and GDP per capita are observed over 1960–2020, and $\Delta \log y_c = \log y_{c,T_c} - \log y_{c,t_c}$ is the corresponding change in log GDP per capita. The omitted category is the lowest peak-manufacturing quintile, so the coefficients compare education gains in higher peak-manufacturing quintiles to those in the lowest quintile, conditional on income growth. Figure 2 shows that education gains rise with the size of the manufacturing phase. Countries in the highest peak-manufacturing quintile experience substantially larger increases in tertiary education than countries in the lowest quintile, even after conditioning on income growth. The corresponding regression coefficients are reported in Appendix B. This evidence points to the skill-accumulation side of the mechanism: economies with larger manufacturing phases enter the service era with larger skilled workforces.

I next ask whether this education pattern is reflected in the composition of services. For the subset of countries with harmonized service-sector occupation data from ILOSTAT, I measure

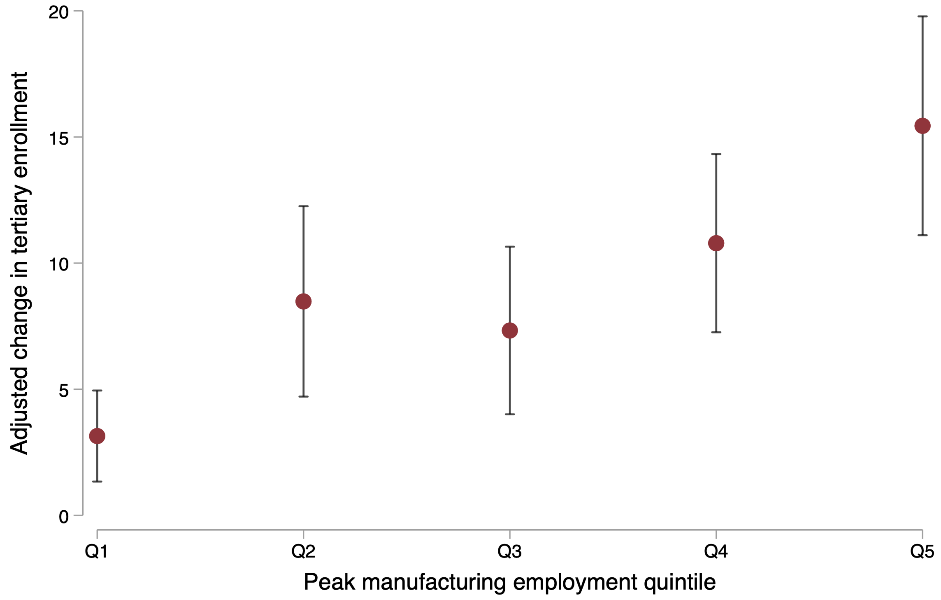


Figure 2: Adjusted education gains by peak manufacturing quintile. The figure reports predicted changes in tertiary enrollment by peak-manufacturing quintile, holding the change in log GDP per capita at its sample mean. Vertical bars denote 95 percent confidence intervals. Data sources: Barro and Lee (2013), GGDC, and Gapminder.

the share of service employment in high-skill producer-service occupations. These occupations, consisting of ISCO codes 1-4, include managerial, professional, technical, clerical, financial, and administrative work performed within service industries.[2] Figure 3 shows a positive relationship between peak manufacturing and the high-skill producer-service share of services. Taken together, the education and service-composition evidence point to two sides of the same mechanism. Countries with larger manufacturing phases accumulate more tertiary education, even conditional on income growth, and their modern service sectors are more tilted toward high-skill producer-service work. I next examine the demographic counterpart of this pattern. In standard quantity-quality models, higher returns to education induce households to invest more in child quality and reduce fertility (Becker et al., 1990; Galor and Weil, 2000; Galor, 2005, 2011). If larger manufacturing phases are associated with higher returns to education, they should therefore also be associated with faster fertility declines.

FACT 3*: Countries with higher peak manufacturing employment shares experience faster demographic transitions, exhibiting lower fertility at comparable levels of development.*

Figure 4 plots fertility over log GDP per capita for countries above and below the median peak manufacturing employment share, using data on fertility and PPP GDP per capita from a balanced panel of 189 countries over 1800-2020. Countries with higher manufacturing peaks have lower fertility rates even at relatively low levels of GDP per capita. For comparison, a log GDP per capita of 6.5 corresponds roughly to that of a median low-income country today. To establish whether this comparison merely reflects differences across regions or comparisons of

2 Also known in the literature as "white collar" occupations (e.g. Engbom et al., 2025) plus clerical.

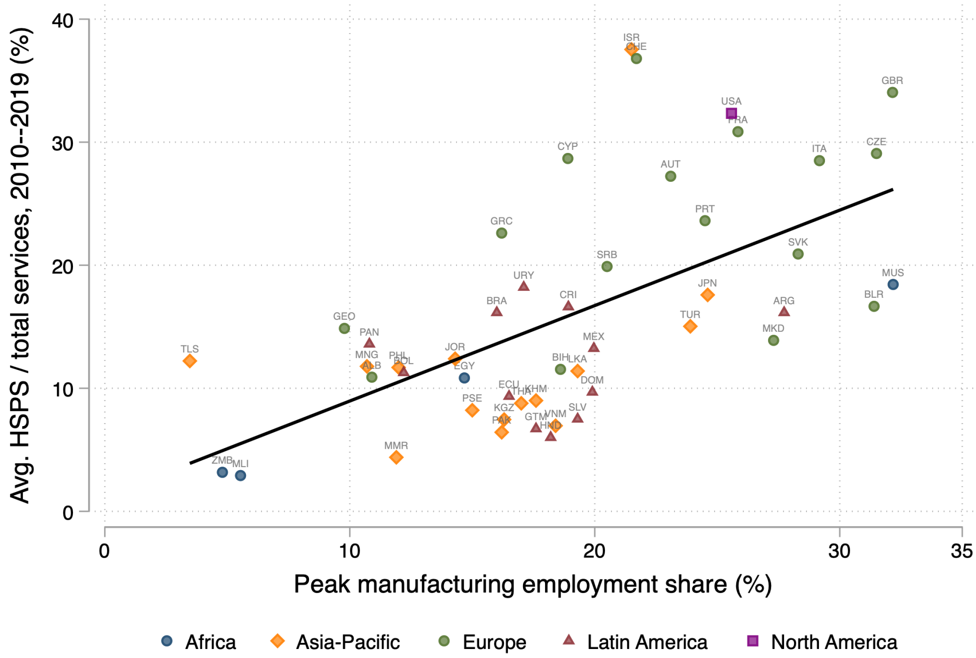


Figure 3: Peak manufacturing and high-skill producer services. The figure plots the share of service employment in high-skill producer-service occupations against each country's peak manufacturing employment share. The fitted line is an OLS regression. Data sources: ILOSTAT, GGDC, and Gapminder.

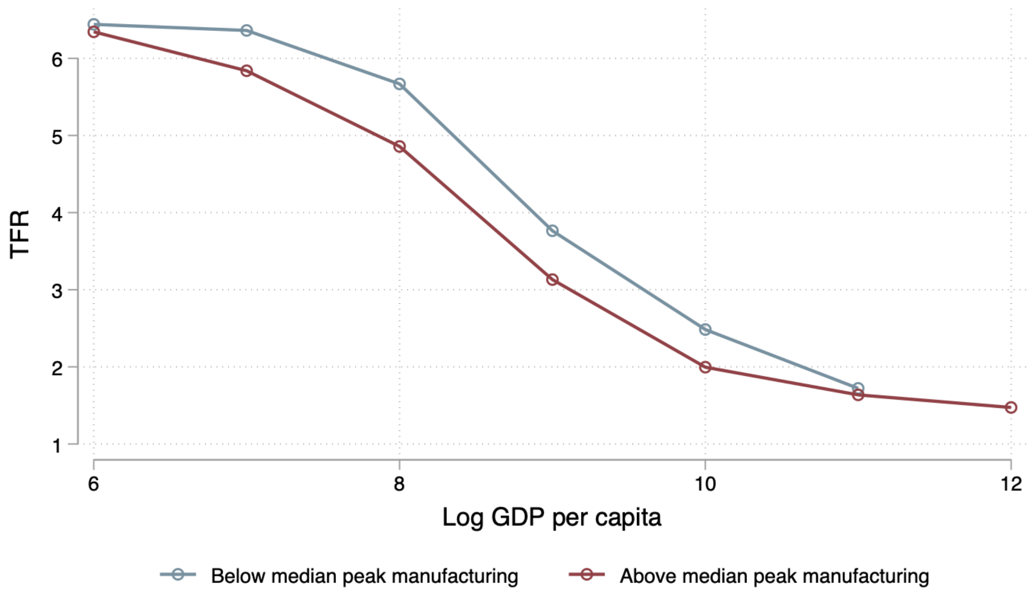


Figure 4: Fertility and GDP per capita. The figure plots the total fertility rate against log GDP per capita for countries above and below the median peak manufacturing employment share. Data source: Gapminder.

late vs early developers, I estimate:

$$\overline{TFR}_c = \alpha + \beta \,\text{Manu. Peak}_c + \gamma_1 TFR_{c,1800} + \gamma_2 \log GDPpc_{c,1800} + \gamma_3 \text{Peak Year}_c + \delta_r + \varepsilon_c.$$

The dependent variable, $\overline{TFR}_c$, is the average fertility rate observed for country c over 1800–2020, and Manu. Peak$_c$ is the country's peak manufacturing employment share. I control for initial fertility, initial log GDP per capita, the year the country reached its manufacturing peak, and region fixed effects δ_r.

Table 2: Dep. Var. is Average Fertility Rate 1800-2020

	(1)	(2)	(3)	(4)
Peak Emp. % Man.	-1.788*	-1.036***	-1.184***	-0.700***
	(-1.80)	(-2.75)	(-3.81)	(-2.82)
Fertility rate 1800		0.598***	0.387***	0.316***
		(13.97)	(8.70)	(5.80)
log GDP per capita 1800			-0.810***	-0.574***
			(-5.54)	(-4.28)
Year of manufacturing peak			0.00701*	0.00451
			(1.75)	(1.28)
N	189	189	176	176
R2	0.0465	0.513	0.642	0.760
Region FE	no	no	no	yes

t statistics in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 2 reports the results. Across specifications, the coefficient on peak manufacturing is negative and statistically significant. This relationship is not simply a comparison between rich and poor countries: it remains after controlling for initial income and initial fertility, and after adding region fixed effects. These cross-country relationships are descriptive, but they establish a benchmark for what a theory of how manufacturing shapes the development process should be able to explain. The next section turns to U.S. county evidence to provide evidence for the mechanism, namely that manufacturing increases demand for high-skill producer services activities.

III U.S. County Evidence

The cross-country evidence in Section II suggests that economies with larger manufacturing phases later develop more skill-intensive service sectors, accumulate more human capital, and grow faster. I now study the same mechanism within the United States during the main period of American industrialization. The county setting holds fixed national institutions, aggregate shocks, and measurement systems, while allowing me to observe the timing and size of local manufacturing booms and follow counties into the modern service economy.

The evidence has four parts. First, counties with larger manufacturing peaks experience an increase in the education content of service employment after the peak. Second, this increase is concentrated in organizational and business service occupations, rather than in all service work. Third, historical manufacturing peaks predict persistent differences in modern outcomes: patenting, college attainment, income, and the high-skill producer-service share of service employment in 2000. Fourth, the modern relationship between manufacturing peaks and high-skill services is largely absorbed by the county's own historical exposure to these service occupations, while controlling for other confounders capturing market size or migration spillovers leaves the coefficient unchanged. Hence, the relationship between manufacturing and long-run outcomes appears to be absorbed by the exposure to high skill producer service occupations during the manufacturing boom.

I measure county-level manufacturing shares of population using county tabulations from the Census of Manufactures over 1880–1940 compiled by Haines (2001). Since consistent county employment denominators are not available before 1950, I define a county's manufacturing peak as the maximum manufacturing employment-to-population ratio observed over this period. Earlier manufacturing censuses are omitted because of well-known reliability issues (Vickers and Ziebarth, 2024). I combine these manufacturing data with the full-count U.S. Censuses from 1850 to 1940. The main historical outcome is the skill content of service employment. Because individual education is not observed before 1950, I use an occupation-based education score. For each three-digit occupation, IPUMS reports the share of workers in that occupation in 1950 who had completed some tertiary education. I average this score across workers employed in service industries in each county-year. The resulting variable, $EDSCOR50_{ct}$, measures whether local services shift toward occupations that were education-intensive in 1950.[3]

Let T_c^{peak} denote the decade in which county c reaches its manufacturing peak. For each peak-manufacturing group $\mathcal{P}$, I estimate

$$EDSCOR50_{ct} = \delta_c + \gamma_t + \sum_k \beta_k \mathbf{1}\{t - T_c^{peak} = k\} \times \mathbf{1}\{\text{ManuPeak}_c \in \mathcal{P}\} + \varepsilon_{ct}, \qquad (1)$$

where δ_c are county fixed effects and γ_t are census-decade fixed effects. I implement the event study using the imputation estimator of Borusyak et al. (2024). The coefficients trace whether counties in different parts of the manufacturing-peak distribution experience differential changes in service-sector occupations before and after the peak.

Figure 5 shows that counties in the top decile of the manufacturing-peak distribution experience a clear increase in the education score of service occupations after their peak. The pre-peak estimates are close to zero. The post-peak estimates rise and remain positive several decades after the peak. Thus, larger manufacturing booms are followed by a persistent shift of the local services industry toward more education-intensive occupations. Appendix C.C shows that the pattern is robust to using finer bins of the manufacturing-peak distribution and Appendix C.D shows that the result is essentially unchanged after controlling for contemporaneous urban share and log county population, as well as when using a geography-predicted manufacturing peak to group counties. Next, I decompose the post-peak effect across mutually exclusive service-occupation groups, excluding health, education, religion, and welfare from the denominator which are non-market facing and have consistently high educational scores. For each group g, I define its contribution to the service education score as

$$Contribution_{ct}^{g} = \frac{\sum_{i \in g} w_i \, EDSCOR50_i}{N_{ct}^{\neg HERW}}, \qquad (2)$$

3 I use a common sample to compute county-by-year enrollment rates, education scores, and fertility rates. Appendix A describes the data construction in detail. For the $EDSCOR50_{ct}$ variable this is computed using all fathers of children aged 6-20.

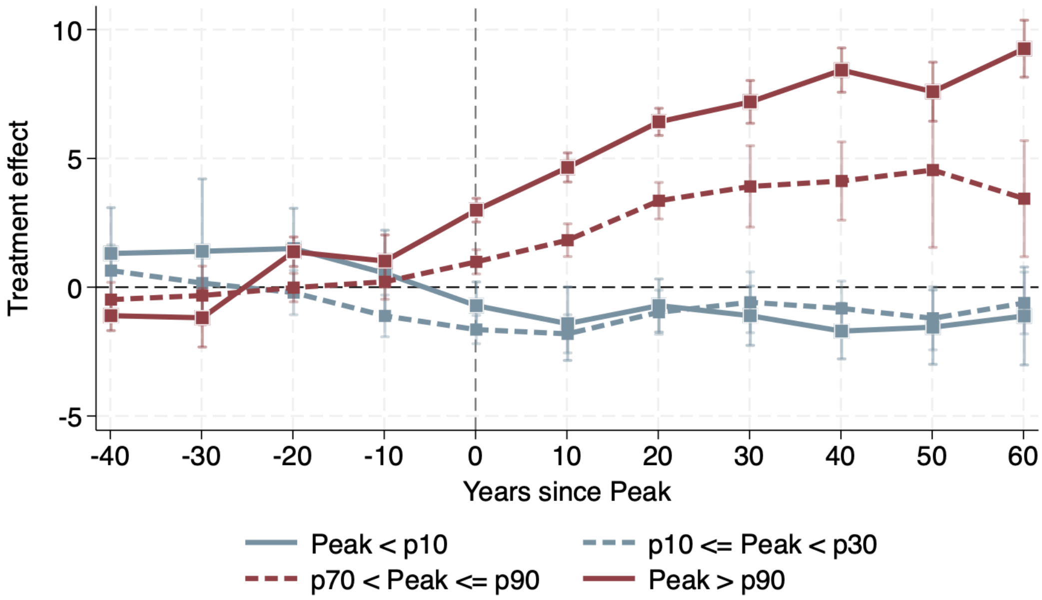


Figure 5: Event-study estimates by mutually exclusive peak-manufacturing bins. The outcome is the mean 1950 occupation education score among service-sector workers in county c and decade t. Counties are grouped by their position in the cross-county distribution of ManuPeak_c. Event time is measured in years relative to the county manufacturing peak. Estimates are obtained using the Borusyak, Jaravel, and Spiess imputation estimator with county and census-decade fixed effects.

where $N_{ct}^{\neg HERW}$ is service employment excluding health, education, religion, and welfare. The groups are organizational/business services, technical services, consumer/personal services, and other services. Organizational/business services include managers, accountants, bookkeepers, clerks, secretaries, purchasing agents, legal and financial intermediaries, and related coordination workers. Figure 6 shows that the increase is concentrated in organizational and business occupations. Technical occupations, such as engineers and chemists, contribute positively but not nearly as much as occupations related to firm organization. Consumer and residual service occupations do not contribute positively. Thus, the post-peak increase is not a broad upgrading of all service work. It is concentrated in occupations connected to firm organization, administration, coordination, finance, law, accounting, and business intermediation. Appendix C.E reports the corresponding decomposition for individual occupations.

I next ask whether these differences persist into the modern service economy. To do this, I estimate cross-sectional specifications of the form

$$y_{c,s,2000} = \delta_s + \beta\,\text{ManuPeak}_{c,s} + \Gamma' X_{c,s} + \varepsilon_{c,s}, \tag{3}$$

where $y_{c,s,2000}$ is a county outcome measured in 2000, δ_s are state fixed effects, and $\text{ManuPeak}_{c,s}$ is the historical manufacturing peak. The controls $X_{c,s}$ include log population, urbanization, school enrollment, colleges, enslaved population in 1850, and the manufacturing peak decade. The main modern mechanism outcome is the high-skill producer-service share of service employment. Throughout the paper, I define high-skill producer services as employment in information, professional/scientific/technical services, and management. The denominator is total

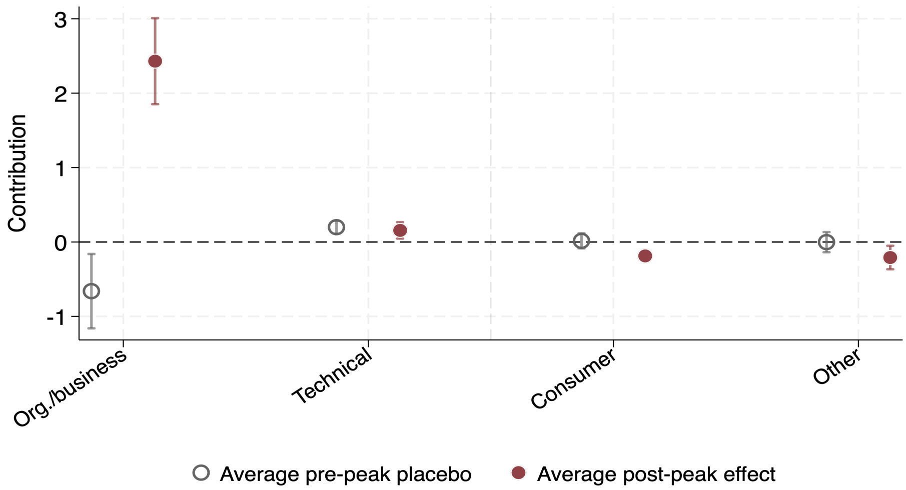


Figure 6: Occupational decomposition of the post-peak effect among service occupations excluding health, education, religion, and welfare. Each point reports the average event-study effect for a mutually exclusive occupation group. The outcome is the group's contribution to the service education score, defined as education-score-weighted employment in the group divided by total service employment excluding health, education, religion, and welfare. Red markers report the average post-peak effect; gray markers report the average pre-peak effect. Data sources: U.S. Census of Population 1850–1940 and Census of Manufactures 1880–1940.

service employment. To address the concern that realized manufacturing peaks are endogenous to other county characteristics, I also construct a geography-based predicted manufacturing peak. The prediction uses the fact that the geographic determinants of manufacturing changed over time: proximity to the fall line mattered more during the water-power phase, while proximity to coal became more important later. I estimate

$$\widehat{\text{ManuPeak}}_c = \sum_d \mathbf{1}\{\text{peak decade}_c = d\} \Big[\widehat{\beta}_d^C \ln(\text{coal}_c) + \widehat{\beta}_d^F \ln(\text{fall}_c) + \widehat{\beta}_d^O \ln(\text{ohio}_c) + \widehat{\beta}_d^{GL} \text{GreatLakes}_c \Big]. \tag{4}$$

I use this geography-based prediction as an instrument for the realized manufacturing peak. The identifying variation comes from whether a county's fixed geographic advantages were useful for manufacturing in the decade when manufacturing peaked, not from whether the county is located in a generally favorable place.[4] This identification strategy exploits the fact that the characteristics of a county which were conducive to manufacturing changed substantially over the course of the American industrial revolution. In the early decades, water power was by far the most important source of power in the manufacturing industry, which is counties which peaked in 1880 and were close to the fall-line had substantially higher peaks than those that were in the mid-west and peaked at the same time. Only a few decades later, when access to coal and trade networks became more important the situation was reversed: counties close to the fall-line which peaked in 1920 had significantly lower peaks than those close to the great lakes, coal mines or the Ohio river. This variation was driven by the aggregate

4 Data sources are described in Appendix A.

changed in the technology and infrastructure used by the manufacturing sector over the sample period, and is hence unlikely to affect economic outcomes through a channel other than increased manufacturing exposure.[5]

Table 3: Historical Manufacturing Peaks and Persistent County Outcomes in 2000

	Log patents pc	Frac. college+	Log income pc	HSPS / services
Panel A: OLS				
Peak Emp. % Man.	0.031***	0.127***	0.488***	0.151***
	(0.006)	(0.036)	(0.105)	(0.023)
State FE	Yes	Yes	Yes	Yes
Historical controls	Yes	Yes	Yes	Yes
Observations	1600	1600	1600	1599
R2	0.192	0.317	0.401	0.209
Panel B: IV				
Peak Emp. % Man.	0.042**	0.421***	0.774	0.453***
	(0.018)	(0.156)	(0.471)	(0.127)
State FE	Yes	Yes	Yes	Yes
Historical controls	Yes	Yes	Yes	Yes
Kleibergen–Paap	83.91	83.91	83.91	83.91
Observations	1600	1600	1600	1599
R2	0.189	0.285	0.396	0.128

Notes: Conley standard errors in parentheses with a cutoff of 100km. All specifications include state fixed effects, peak year, and 1850 controls for log population, urbanization, school enrollment, colleges, and enslaved population. The IV panel instruments the realized manufacturing peak using geography-predicted manufacturing exposure. HSPS/services is the share of service employment in high-skill producer services. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3 shows that historical manufacturing peaks predict persistent differences in county outcomes. Counties with larger manufacturing peaks have higher patenting, higher college attainment, higher income, and more high-skill producer-service-intensive service sectors in 2000. The OLS estimates are positive and statistically significant for all four outcomes. The IV estimates point in the same direction and are especially strong for college attainment and the high-skill producer-service share. The last column is central for the mechanism: historical manufacturing predicts not only more development, but a different composition of modern services. I next ask whether this relationship between historical manufacturing peaks and modern outcomes is driven by the channel emphasized in this paper, rather than a broad market size or income growth effect. The model predicts that manufacturing matters because it creates demand for producer-service tasks that organize complex production. These tasks can be performed inside manufacturing firms during industrialization, but they are not manufacturing-specific: accounting, clerical systems, supervision, technical control, procurement, engineering support, and management are organizational inputs that can later be used by service firms. I therefore examine whether the long-run relationship between manufacturing peaks and modern outcomes is absorbed by variables that proxy for the creation and diffusion of these organizational capabilities. I construct three channel variables. The first is the county's own HSPS share of manufacturing employment in the decade when it peaked. The second is the maximum peak among neighboring counties within a 300km radius from the county's centroid. This captures regional spillovers and local market-size effects. The third channel is migrant-origin exposure,

5 See Appendix C to see the first stage results from 4.

Table 4: Mechanism: Local Organizational Capabilities and Persistent Outcomes

	Added channel controls				
	Baseline	Manu HSPS	Neighbor	Migration	All
Panel A. Dependent variable: HSPS / services, 2000					
Peak Emp. % Man.	0.151***	0.043*	0.143***	0.132***	0.040
	(0.023)	(0.023)	(0.026)	(0.024)	(0.026)
Manufacturing HSPS		1.972***			1.926***
		(0.223)			(0.226)
Neighbor peak, 100km			0.021		0.000
			(0.029)		(0.023)
Migrant-origin exposure				1.944***	0.494
				(0.697)	(0.628)
Observations	1599	1599	1576	1599	1576
R^2	0.209	0.270	0.206	0.216	0.265
Panel B. Dependent variable: log income per capita, 2000					
Peak Emp. % Man.	0.488***	0.032	0.264**	0.321***	-0.172
	(0.105)	(0.102)	(0.112)	(0.112)	(0.105)
Manufacturing HSPS		8.386***			7.656***
		(0.718)			(0.641)
Neighbor peak, 100km			0.536***		0.349***
			(0.126)		(0.112)
Migrant-origin exposure				17.096***	8.945***
				(2.896)	(2.828)
Observations	1600	1600	1577	1600	1577
R^2	0.401	0.472	0.423	0.438	0.500
State FE	Yes	Yes	Yes	Yes	Yes
Historical controls	Yes	Yes	Yes	Yes	Yes

Notes: Standard errors in parentheses. All regressions include state fixed effects, peak year, and 1850 historical controls. Neighbor columns exclude counties with no 1850 county centroid within 300km. Channel variables are post-treatment, so the table is an attenuation exercise rather than a causal mediation design. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

measured using linked Census records and the manufacturing structure of migrants' origin counties. In particular, the migration control for county c measures the average HSPS share of manufacturing employment in the origin counties among all migrants received by county c over 1880-1940. This captures the diffusion of manufacturing capabilities through migration spillovers. I then estimate:

$$\begin{aligned} y_{c,s,2000} = \delta_s + \beta\,\text{ManuPeak}_{c,s} + \theta\,\text{ManuHSPS}_{c,s} + \rho\,\text{NeighborPeak}_{c,s} \\ + \lambda\,\text{MigrantOrigin}_{c,s} + \Gamma' X_{c,s} + \varepsilon_{c,s}. \end{aligned} \tag{5}$$

These specifications are descriptive. The channel variables are themselves outcomes of the historical manufacturing process, so the coefficients should not be interpreted as causal mediation effects. The exercise asks a narrower question: which observable variables absorb the persistent relationship between manufacturing peaks and modern outcomes?

Table 4 reports the attenuation exercise. Panel A uses the high-skill producer-service share of service employment as the dependent variable. The baseline relationship with historical manufacturing peaks is positive and statistically significant. Adding the county's own HSPS manufacturing structure reduces the manufacturing-peak coefficient by more than two thirds, from

0.151 to 0.043. Adding neighboring manufacturing exposure leaves the coefficient almost unchanged. Adding migrant-origin exposure reduces it only modestly. When all channels are included, the manufacturing-peak coefficient becomes small and statistically insignificant, while the own HSPS manufacturing share channel remains large and precisely estimated. Panel B shows a similar pattern for income per capita. The baseline relationship between manufacturing peaks and modern income is strong, but it is almost entirely absorbed by the county's own HSPS manufacturing structure. Neighboring manufacturing exposure and migrant-origin exposure are themselves predictive of income, but they do not absorb the manufacturing-peak coefficient in the same way. This pattern reveals that the persistent relationship between historical manufacturing and modern outcomes is not explained mainly by broad regional spillovers or by migration from industrial places, but is rather most closely tied to the organizational and HSPS tasks created locally during the manufacturing phase. Taken together, the U.S. evidence supports the mechanism developed in the model. Larger manufacturing booms are followed by increases in the skill content of service employment. The increase appears after the manufacturing peak, is concentrated in organizational and business occupations, and persists into the modern service economy. By 2000, counties with larger manufacturing peaks are more educated, more innovative, richer, and have more high-skill producer-service-intensive service sectors. The modern service-composition relationship is largely absorbed by the county's own historical HSPS manufacturing structure.

IV Model

This section develops a parsimonious growth model in which the sectoral path of development shapes the accumulation of organizational knowledge and, through it, the skill intensity and productivity of the service economy. Firms in every sector face similar organizational problems. They must manage inventories, record information, budget costs, assign workers, coordinate suppliers, and monitor quality. The circumstances of a particular firm determine the exact solution it requires, but the underlying problems are common across firms and sectors. A hospital, a retailer, and a manufacturing plant produce different outputs, yet all require these managerial and organizational inputs which I will refer to as "producer services". This means that developing new methods to perform these tasks more efficiently occurring in one sector can potentially spillover to the rest of the economy. An improvement in the technology to produce a pencil cannot be applied in a hospital, but a new inventory management system can because all firms, regardless of their sector benefit from more productive management.

The primitive difference across sectors is that they differ in how easily their organizational problems can be addressed using general methods. Formal methods are particularly useful when production is repeated, standardized, and observable; they are less useful when production is highly idiosyncratic, relevant outcomes are difficult to measure, or the appropriate response depends heavily on local circumstances. The model represents these differences through a sector-specific learning process. In sectors where methods require less learning to be developed a larger share of producer-service work can be performed formally by skilled workers. Thus the same learning process governs both the assignment of workers to tasks and the creation of reusable methods, which I call *organizational knowledge*.

The rest of the model's machinery, including non-homothetic demand, input-output linkages, fertility and education embed this mechanism in a quantitative account of the US economy's transition out of agriculture.

IV.A Environment

Demographics. Time is discrete. The economy is populated by overlapping generations who live for two periods, childhood and adulthood. During childhood, members of the next cohort receive education and may acquire skill. During adulthood, they work, consume, choose fertility, and invest in the education of their children.

Final sectors and producer-service tasks. There are three final-good sectors, indexed by $j \in \{a, m, c\}$: agriculture, manufacturing, and consumer services. Final-good sectors produce output using value added, agricultural and manufacturing intermediates, and producer-service inputs. Each sector uses a continuum of producer-service tasks. Every task can be performed informally using low-skill producer-service labor or formally using skilled producer-service labor. Informal execution relies on task-specific experience and rules of thumb. Formal execution requires a skilled worker to search for and apply an organizational method suited to the task.

Common learning space and task complexity. Organizational methods are located on a common ordered learning space

$$\mathcal{P} = \mathbb{R}_+.$$

A location $h \in \mathcal{P}$ measures the effective learning required to discover, understand, and codify the corresponding method. Larger values of h therefore represent methods that lie farther along the learning space.

For each producer-service task s used by sector j, methods applicable to that task occur along $\mathcal{P}$ according to a Poisson counting process. Let $M^T_{j,s}(x)$ denote the number of methods applicable to task s encountered within the first x units of effective learning. Sector j is characterized by a locally integrable arrival intensity

$$\iota_j : \mathbb{R}_+ \to \mathbb{R}_+,$$

with cumulative intensity

$$\Lambda_j(x) \equiv \int_0^x \iota_j(u)\, du. \tag{6}$$

I assume

$$\lim_{x\to\infty} \Lambda_j(x) = \infty,$$

so that an applicable method is eventually encountered with probability one. The task-specific counting process satisfies

$$M^T_{j,s}(x) \sim \text{Poisson}\big(\Lambda_j(x)\big)\,, \qquad x \geq 0. \tag{7}$$

The complexity of task s is the learning distance to its first applicable organizational method:

$$z_s \equiv \inf\left\{x \geq 0 : M^T_{j,s}(x) \geq 1\right\}. \tag{8}$$

Hence,

$$\begin{aligned}\Pr(z_s > x \mid j) &= \Pr\Big(M^T_{j,s}(x) = 0\Big) \\ &= \exp\big[-\Lambda_j(x)\big]\,, \end{aligned} \tag{9}$$

and the sector-specific distribution of task complexity is

$$F_j(z) = 1 - \exp\big[-\Lambda_j(z)\big]\,. \tag{10}$$

For the quantitative model, I assume a homogeneous Poisson process:

$$\iota_j(h) = \frac{1}{\kappa_j}, \qquad\qquad \Lambda_j(x) = \frac{x}{\kappa_j}. \tag{11}$$

Task complexity is therefore exponentially distributed:

$$F_j(z) = 1 - \exp\left(-\frac{z}{\kappa_j}\right), \qquad z \geq 0, \tag{12}$$

$$f_j(z) = \frac{1}{\kappa_j}\exp\left(-\frac{z}{\kappa_j}\right). \tag{13}$$

Because

$$\mathbb{E}[z_s \mid j] = \kappa_j,$$

the parameter κ_j is the expected amount of effective learning required to encounter an organizational method applicable to a sector-j producer-service task. A lower κ_j means that applicable methods occur more densely along the learning space. Such a sector therefore has a larger concentration of tasks for which an applicable method can be reached with relatively little learning. Because the learning space is common across sectors, methods are not sector specific.

Organizational knowledge. The stock of reusable methods available at the beginning of period t is

$$\mathcal{K}_t = [0, H_t]. \tag{14}$$

The frontier H_t is the location of the most advanced method known to the economy. Because the hierarchy is nested, knowledge of the method at H_t includes all prerequisite methods indexed below it.

Education and knowledge creation. Let $\bar{e}_t$ denote average effective education among adults. Let $\ell_{j,t}$ denote the share of total employment attributed to final-sector activity j, including producer-service workers allocated according to the final sector they serve, with

$$\sum_{j\in\{a,m,c\}} \ell_{j,t} = 1.$$

Production in sector j exposes the economy to that sector's organizational problems, while education determines the amount of effective learning that can be applied to this experience. Let

$$X_{j,t}^K = \bar{e}_t \ell_{j,t} \tag{15}$$

denote sector-j effective learning. Applying the same sector-specific learning process that governs task complexity, the number of codifiable method positions supported by sector-j learning satisfies

$$N_{j,t}^K \sim \text{Poisson}\Big(\Lambda_j\left(X_{j,t}^K\right)\Big). \tag{16}$$

Sectoral processes refer to the production environment in which a method is learned, not to the sectors in which it may subsequently be used. Once codified, all methods enter the common economy-wide knowledge stock. Assuming independent sector-of-origin processes, Poisson superposition gives

$$\begin{aligned} N_t^K &= \sum_{j\in\{a,m,c\}} N_{j,t}^K \\ &\sim \text{Poisson}\left(\sum_{j\in\{a,m,c\}} \Lambda_j\left(X_{j,t}^K\right)\right). \end{aligned} \tag{17}$$

Organizational methods occupy fixed positions on a common ordered ladder. By assumption, if current learning supports n method positions, the economy knows the entire initial segment up to position n. Sectoral arrivals are pooled before positions are assigned on this common ladder, so sectoral counts represent distinct contributions to the aggregate supported segment. For tractability, the quantitative model replaces the realized supported count by its conditional expectation and measures the deterministic frontier in expected method-position units:

$$\widetilde{H}_{t+1} \equiv \mathbb{E}_t[N_t^K].$$

Hence:

$$\begin{aligned} \widetilde{H}_{t+1} &= \mathbb{E}_t\left[N_t^K\right] \\ &= \sum_{j\in\{a,m,c\}} \Lambda_j\left(\bar{e}_t \ell_{j,t}\right). \end{aligned} \tag{18}$$

Current learning therefore supports the prerequisite-complete candidate knowledge set $[0, \widetilde{H}_{t+1}]$. Since inherited knowledge is $[0, H_t]$ and previously codified methods are never forgotten,

$$
\begin{aligned}
\mathcal{K}_{t+1} &= [0, H_t] \cup [0, \widetilde{H}_{t+1}] \\
&= \left[0, \max\left\{H_t, \widetilde{H}_{t+1}\right\}\right].
\end{aligned} \tag{19}
$$

The law of motion is therefore

$$
H_{t+1} = \max\left\{H_t,\, \bar{e}_t \sum_{j\in\{a,m,c\}} \frac{\ell_{j,t}}{\kappa_j}\right\}. \tag{20}
$$

Skilled workers and task-specific search. Skilled producer-service workers take $\mathcal{K}_t$ as given when solving current production problems. Conditional on task complexity and inherited organizational knowledge, task-specific solutions arrive through a separate search process. This process makes two fundamental assumptions: (1) more complex tasks take longer to solve and (2) the stock of reusable methods H_t reduces the time needed to solve all tasks.

Conditional on task complexity z, a skilled worker who devotes q effective units of time to the task generates solutions according to:

$$
N^S(q, z, H_t) \sim \text{Poisson}(q\, a(z, H_t)), \tag{21}
$$

where

$$
a(z, H_t) = \exp\left(-\frac{z}{\phi(H_t)}\right) \tag{22}
$$

is the arrival rate of applicable task-specific solutions per unit of search. The function $\phi : \mathbb{R}_+ \to \mathbb{R}_{++}$ is continuous and strictly increasing. The probability of finding at least one applicable solution is hence

$$
\Pr\Big(N^S(q, z, H_t) \geq 1\Big) = 1 - \exp\left[-q\, a(z, H_t)\right]. \tag{23}
$$

IV.B Households

Adults are indexed by their sector of employment, $i \in \{a, m, s\}$, where s denotes services and includes consumer-service and producer-service employment. This household-origin index is distinct from the final-good sector index $j \in \{a, m, c\}$, where c denotes consumer services. Let $\lambda_{i,t}$ denote the share of adults of origin i, with $\sum_{i\in\{a,m,s\}} \lambda_{i,t} = 1$. The origin shares are the corresponding shares of total employment: $\lambda_{a,t}$ and $\lambda_{m,t}$ are the agricultural and manufacturing employment shares, and $\lambda_{s,t} = 1 - \lambda_{a,t} - \lambda_{m,t}$ includes all consumer- and producer-service workers. Sectoral employment affects the fixed time cost of raising children, τ_i. Apart from this cost, households face a common wage and education environment.

A household of type i chooses consumption, fertility $n_{i,t}$, and education investment per child $e_{i,t}$. Preferences are

$$U_{i,t} = (1-\gamma)\log C_{i,t} + \gamma \log\left(n_{i,t}h_{i,t}\right). \tag{24}$$

Manufacturing is the numeraire. Final consumption is allocated across agriculture, manufacturing, and consumer services using the non-homothetic CES demand system of Comin et al. (2021):

$$\sum_{j\in\{a,m,c\}} \omega_j^{\frac{1}{\sigma}} \left(\frac{C_{ij,t}}{C_{i,t}^{\varepsilon_j}}\right)^{\frac{\sigma-1}{\sigma}} = 1. \tag{25}$$

Here $C_{ij,t}$ is type-i consumption of final good j, $C_{i,t}$ is the corresponding consumption aggregate, ω_j is the demand weight, σ is the elasticity of substitution, and ε_j governs the income elasticity. Aggregate final consumption is $C_{j,t} = L_t \sum_{i\in\{a,m,s\}} \lambda_{i,t} C_{ij,t}$.

Let $w_{u,t}$ and $w_{s,t}$ denote unskilled and skilled wages, and let S_t denote the skilled share of employment. The wage scale entering the household problem is

$$w_t(S_t) = (1-S_t)w_{u,t} + S_t w_{s,t}. \tag{26}$$

I assume complete pooling of skilled and unskilled labor income within the representative household. Consequently, the fertility and education problem depends on the average wage $w_t(S_t)$, rather than on a skill-specific wage.

Each adult has one unit of time. Raising a child requires time τ_i, while education requires an additional time cost $\tau_e(S_t)e_{i,t}$, where

$$\tau_e(S_t) = (1-S_t)\tau_e^u + S_t\tau_e^s, \qquad \tau_e^s < \tau_e^u. \tag{27}$$

The household budget constraint is

$$p_{a,t}C_{ia,t} + C_{im,t} + p_{c,t}C_{ic,t} \le \left[1 - n_{i,t}\left(\tau_i + \tau_e(S_t)e_{i,t}\right)\right] w_t(S_t). \tag{28}$$

Child human capital follows Lagerlöf (2006):

$$h_{i,t} = \frac{e_{i,t}+\tau_h}{e_{i,t}+\tau_h+g_t}, \tag{29}$$

where g_t is aggregate productivity growth. Letting $\chi_t \equiv \tau_e(S_t)$, the household solution is

$$n_{i,t} = \frac{\gamma}{\tau_i + \chi_t e^*_{i,t}}, \tag{30}$$

$$e^*_{i,t} = \max\left\{\sqrt{\frac{g_t(\tau_i - \chi_t \tau_h)}{\chi_t}} - \tau_h, 0\right\}. \tag{31}$$

Appendix D derives these expressions.

Skill acquisition. The share of the next cohort that acquires skill depends on the expected skill premium

$$\pi_{t+1} \equiv \frac{w_{s,t+1}}{w_{u,t+1}}. \tag{32}$$

With logistic idiosyncratic acquisition costs, the skilled share of the next cohort is

$$p_{s,t} = \frac{1}{1 + \exp\left(-\frac{\beta_\pi \mathbb{E}_t[\log \pi_{t+1}] - c_0}{\sigma_{\text{skill}}}\right)}. \tag{33}$$

Under perfect foresight, $\mathbb{E}_t[\log \pi_{t+1}] = \log \pi_{t+1}$.

IV.C Production and producer services

Final-good production. Value added in sector j is produced using skilled and unskilled labor:

$$VA_{j,t} = A_{j,t}\left[\alpha_j (h_s L_{s,j,t})^{\frac{\sigma_L - 1}{\sigma_L}} + (1-\alpha_j)(h_u L_{u,j,t})^{\frac{\sigma_L - 1}{\sigma_L}}\right]^{\frac{\sigma_L}{\sigma_L - 1}}. \tag{34}$$

Gross output combines value added, agricultural and manufacturing intermediates, and producer-service inputs:

$$Y_{j,t} = A_{go,j,t} VA_{j,t}^{\theta_{j,t}} M_{a,j,t}^{\gamma_{a,j,t}} M_{m,j,t}^{\gamma_{m,j,t}} PS_{j,t}^{\gamma_{ps,j,t}}, \tag{35}$$

$$\theta_{j,t} + \gamma_{a,j,t} + \gamma_{m,j,t} + \gamma_{ps,j,t} = 1. \tag{36}$$

Producer-service supplies. Formal and informal producer-service firms operate economy-wide and sell efficiency units of producer-service labor to all final-good sectors. Aggregate supplies are

$$LS_t = A_{ls,t} h_u L_{u,ls,t}, \qquad HS_t = A_{hs,t} h_s L_{s,hs,t}. \tag{37}$$

Their competitive efficiency-unit prices are

$$p_{hs,t} = \frac{w_{s,t}}{A_{hs,t}h_s}, \qquad\qquad p_{ls,t} = \frac{w_{u,t}}{A_{ls,t}h_u}. \tag{38}$$

IV.D Search and static task assignment

Task aggregation. Final-good sector j uses a continuum of producer-service tasks indexed by complexity z, with task measure $dF_j(z)$. Let $y_{j,t}(z)$ denote the quantity of task z. Producer services are

$$PS_{j,t} = \left[\int_0^\infty y_{j,t}(z)^{\frac{\sigma_{ps}-1}{\sigma_{ps}}}\, dF_j(z)\right]^{\frac{\sigma_{ps}}{\sigma_{ps}-1}}, \qquad \sigma_{ps} > 1. \tag{39}$$

Informal execution requires one efficiency unit of low-skill producer-service labor. Formal execution requires sufficient skilled-worker search to find and implement an applicable task-specific solution.

Required search effort. Formal execution requires the skilled worker to find at least one applicable solution with a common success probability $\alpha \in (0, 1)$. Using (21)–(23), the minimum required search effort satisfies

$$1 - \exp\left[-q_\alpha(z, H_t)a(z, H_t)\right] = \alpha, \tag{40}$$

and is therefore

$$\begin{aligned} q_\alpha(z, H_t) &= -\log(1-\alpha)a(z, H_t)^{-1} \\ &= -\log(1-\alpha)\exp\left[\frac{z}{\phi(H_t)}\right]. \end{aligned} \tag{41}$$

Normalize effective-search units so that $-\log(1-\alpha) = \mathrm{e}^{-1}$, where e is Euler's number. Then

$$q_\alpha(z, H_t) = \exp\left[\frac{z}{\phi(H_t)} - 1\right]. \tag{42}$$

Task assignment. Informal and formal unit costs are

$$c_t^{LS}(z) = p_{ls,t}, \qquad\qquad c_t^{HS}(z, H_t) = p_{hs,t}\exp\left[\frac{z}{\phi(H_t)} - 1\right]. \tag{43}$$

A task is performed formally when $c_t^{HS}(z, H_t) \le c_t^{LS}(z)$. The resulting cutoff is

$$z_t^* = \phi(H_t) \max \left\{ 1 + \log\left(\frac{p_{ls,t}}{p_{hs,t}}\right), 0 \right\}. \tag{44}$$

The cutoff is common across sectors because all sectors face the same producer-service prices and organizational knowledge. Sectoral formalization differs because sectors draw tasks from different complexity distributions.

Under the general sector-specific learning process, the share of sector j's producer-service tasks performed formally is

$$\begin{aligned} x_{j,t} &= F_j(z_t^*) \\ &= 1 - \exp\left[-\Lambda_j(z_t^*)\right]. \end{aligned} \tag{45}$$

Under the homogeneous specification $\Lambda_j(z) = z/\kappa_j$, this becomes

$$x_{j,t} = 1 - \exp\left[-\frac{\phi(H_t) \max\{1 + \log(p_{ls,t}/p_{hs,t}), 0\}}{\kappa_j}\right]. \tag{46}$$

Formalization rises with organizational knowledge, falls with the relative price of skilled producer services, and is greater in sectors with lower κ_j. Appendix D derives the producer-service price index, sectoral formal and informal labor demands, market clearing, and the mapping from $x_{j,t}$ to producer-service employment shares.

Proposition 1 (Common learning order)**.** *Consider two sectors j and k. Suppose that*

$$\Lambda_j(x) \geq \Lambda_k(x) \qquad \textit{for every } x \geq 0.$$

Then:

1. *task complexity in sector j is first-order stochastically smaller than task complexity in sector k:*

$$F_j(z) \geq F_k(z) \qquad \textit{for every } z \geq 0;$$

2. *for any common amount $X \geq 0$ of effective learning,*

$$\mathbb{E}\left[N_{j,t}^K \mid X_{j,t}^K = X\right] = \Lambda_j(X) \geq \Lambda_k(X) = \mathbb{E}\left[N_{k,t}^K \mid X_{k,t}^K = X\right];$$

3. *holding organizational knowledge and producer-service factor prices fixed,*

$$x_{j,t} \geq x_{k,t}.$$

Under the homogeneous specification, these conditions are equivalent to

$$\kappa_j \leq \kappa_k.$$

Proof. Because

$$F_r(z) = 1 - \exp[-\Lambda_r(z)], \qquad r \in \{j, k\},$$

the inequality $\Lambda_j(z) \geq \Lambda_k(z)$ implies $F_j(z) \geq F_k(z)$ for every z. The second result follows directly from the conditional mean of a Poisson random variable. The task cutoff z_t^* is common across sectors, so the first result implies $x_{j,t} = F_j(z_t^*) \geq F_k(z_t^*) = x_{k,t}$. Under homogeneity, $\Lambda_r(x) = x/\kappa_r$, and the stated ordering is equivalent to $\kappa_j \leq \kappa_k$. □

Proposition 1 formalizes the central restriction linking the static and dynamic margins of the model. A sector in which effective learning encounters applicable methods more readily has both a larger share of tasks that can be performed formally and a larger expected contribution to organizational knowledge for a given amount of effective learning.

IV.E Dynamics

The aggregate state is

$$\Omega_t = \Big(L_t,\, S_t,\, \bar{e}_t,\, \{A_{j,t}\}_{j \in \{a,m,c\}},\, H_t,\, g_t \Big). \tag{47}$$

Population evolves according to

$$L_{t+1} = n_t L_t, \qquad\qquad n_t = \sum_{i \in \{a,m,s\}} \lambda_{i,t} n_{i,t}. \tag{48}$$

Average education of the next adult cohort is

$$\bar{e}_{t+1} = \frac{\sum_{i \in \{a,m,s\}} \lambda_{i,t} n_{i,t} e_{i,t}^*}{n_t}. \tag{49}$$

The skilled share evolves according to

$$S_{t+1} = p_{s,t}. \tag{50}$$

Sectoral value-added productivity evolves according to

$$\frac{A_{j,t+1}}{A_{j,t}} = 1 + H_t (S_t L_t)^{\varphi_j}, \qquad j \in \{a, m, c\}. \tag{51}$$

The productivity growth inherited by the next period is

$$g_{t+1} = \sum_{j\in\{a,m,c\}} \ell_{j,t} \left(\frac{A_{j,t+1}}{A_{j,t}} - 1 \right). \tag{52}$$

This specification adapts the semi-endogenous idea-production technology of Jones (1995) to a multisectoral environment while integrating the insight from Klette and Kortum (2004) that managerial capacity increases the arrival rate of productivity improvements.

IV.F Limiting behavior and the service-economy frontier

I now characterize the model's limiting behavior. Structural change makes GDP per capita growth non-stationary during the transition, so the relevant object is the asymptotic growth path, or AGP, reached once sectoral employment shares have converged (Buera et al., 2020). I focus on the service-dominated branch,

$$\ell_{c,t} \to 1, \qquad \ell_{a,t} \to 0, \qquad \ell_{m,t} \to 0.$$

I take this service-employment limit as an assumption selecting the asymptotic branch characterized below. It is distinct from the weaker statement that the consumer-service share of final expenditure converges to one. Once consumer services absorb employment, manufacturing can matter only through the states it leaves behind.[6] Let H^* denote the organizational frontier inherited by the service economy.

Proposition 2 (Development path and terminal service AGP)**.** *Fix primitives, including ω_m. Under the regularity conditions above and in Appendix D.C, and conditional on convergence to the service-employment branch, the model admits a unique terminal AGP for every inherited frontier H^*. In particular, there exists a unique tuple*

$$(S^*, e^*, g^*, L^*)$$

such that

$$S^* = S(H^*), \qquad S'(H) > 0,$$

$$n_t \to 1, \qquad \bar{e}_t \to e^* = \frac{\gamma - \tau_s}{\tau_e(S^*)},$$

$$g^* = \frac{(\gamma - \tau_s + \tau_e(S^*)\tau_h)^2}{\tau_e(S^*)\,(\tau_s - \tau_e(S^*)\tau_h)}, \tag{53}$$

and

$$L^* = \frac{1}{S^*} \left(\frac{g^*}{H^*} \right)^{1/\varphi_c}. \tag{54}$$

Proof. See Appendix D.C.

6 In the quantitative model, limiting employment shares depend on preferences, relative productivity, prices, and input-output linkages. The results below characterize the service-dominated asymptotic branch. This is the empirically relevant case for modern high-income economies, where services account for most employment, and is consistent with structural-change models that characterize asymptotic growth after sectoral reallocation has converged; see Comin et al. (2021) and Buera et al. (2020).

The proposition gives the long-run architecture. The transition determines the inherited organizational frontier H^*. The inherited frontier determines S^*. The skilled share S^* determines g^*. Population scale then adjusts to satisfy

$$g^* = H^*(S^*L^*)^{\varphi_c}.$$

Thus long-run growth is independent of population scale. In Romer (1990), larger research scale raises growth, generating the counterfactual implication that larger economies should grow faster. In a recent survey, Jones (2022) shows that semi-endogenous growth models cannot sustain growth in output per capita once population growth stops, something which is increasingly common in developed countries. In contrast, the model in this paper delivers positive growth, replacement fertility, and finite population. The reason is different from other no-scale models: in Peretto (1998), Laincz and Peretto (2006), and related work, scale effects disappear because entry and product variety adjust with market size; here, long-run growth is selected by the skill content of services, and the skill content of services is selected by the organizational frontier inherited from structural change.

I next characterize when manufacturing raises that frontier. Consider the branching experiment in Figure 7. There are three periods. In each period, one sector absorbs all labor. Both economies start in agriculture and end in consumer services. Economy A passes through manufacturing; economy B moves directly into consumer services.

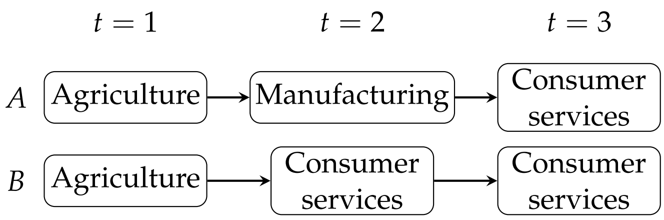


Figure 7: Canonical branching experiment.

Proposition 3 (Manufacturing versus direct services)**.** *Consider the transition in Figure 7. Under the regularity conditions in Appendix D.C, and in the limiting case where $\tau_h \to 0$,*

$$g_A^* > g_B^* \iff H_A^* > H_B^* \iff \underbrace{(S_1 L_1)^{\varphi_m - \varphi_c}}_{growth} \underbrace{\frac{\tau_m}{\tau_s}}_{quantity\text{-}quality} > \underbrace{\left(\frac{\kappa_m}{\kappa_c}\right)^2}_{codification}.$$

Proof. See Appendix D.C.

The condition says that manufacturing raises long-run growth only if it raises the inherited organizational frontier H^*. It can do so in three ways. First, if $\varphi_m > \varphi_c$, manufacturing raises transition growth for a given $S_1 L_1$; higher growth raises education and therefore codified organizational knowledge. Second, if $\tau_m > \tau_s$, manufacturing-origin households face higher fertility costs and choose more child quality, again raising education and H. Third, if $\kappa_m < \kappa_c$, a unit of education generates more codified organizational knowledge in manufacturing than in consumer services.

Evidence that manufacturing households are similar to service households in education and fertility therefore disciplines τ_m/τ_s, but it does not rule out a role for manufacturing in development.[7] Equivalently, for the two paths in Figure 7 to converge to the same AGP, the direct-service path must compensate for any manufacturing advantage in transition growth or codification. If τ_m/τ_s is close to one, as suggested by the household evidence, equalization requires either φ_c to be sufficiently high relative to φ_m, or κ_c to be sufficiently low relative to κ_m. Neither case is supported by the empirical evidence presented in this paper.

Appendix D.C extends Proposition 3 to arbitrary transition paths with interior sectoral employment shares. In general, manufacturing raises long-run service growth if and only if it raises the organizational frontier inherited by services. Services can absorb labor, but an economy that bypasses manufacturing may enter consumer services with a lower organizational frontier, a lower skilled share, and a lower asymptotic growth rate.

V Calibration

I calibrate the model to the United States over 1850–2019. The U.S. transition provides the benchmark economy for the counterfactual exercises in Section VI: over this period, the United States moved out of agriculture, experienced a pronounced manufacturing hump, shifted into services, and completed most of its demographic transition. The calibration is designed to discipline both the aggregate development path and the producer-service mechanism emphasized in the paper. Aggregate moments discipline structural change, fertility, education, and income growth. Producer-service moments discipline the skill content of organizational tasks and, in particular, the fact that the producer-service occupations associated with manufacturing are more education-intensive than those associated with agriculture or consumer services.

A central measurement choice is how to define consumer services. In the baseline, I use a strict industry classification. I classify industries into agriculture, manufacturing, consumer services, and producer services, and define consumer services as the subset of service industries that sell at least half of their output to consumers. This choice has two advantages. First, many service industries, such as finance, information, and professional services, are used both by firms and by households, but their final-consumption share varies across countries and likely changed substantially over the long U.S. transition. A final-expenditure-weighted definition based on modern input-output tables may therefore assign mixed-use services to the consumer bundle in a way that is inappropriate for earlier periods. The strict classification instead isolates service industries that are plausibly consumer-facing throughout the transition. Second, the strict classification allows the mechanism to be measured directly in worker-level Census data. The model's key production-side object is whether the producer-service tasks associated with a sector are high-skill intensive. With a strict industry classification, I can measure this object directly: within each broad final-good sector, I restrict attention to

7 Peters et al. (2026) show that manufacturing households are not systematically different from households in consumer services along several observable dimensions in developing countries. Appendix C reports a similar exercise for U.S. counties: within counties, manufacturing and service households have similar fertility and child enrollment rates, while counties with larger manufacturing exposure differ substantially in both outcomes.

producer-service occupations and compute the share of workers with at least one year of college. This produces sector-specific high-skill producer-service shares for agriculture, manufacturing, and consumer services. If mixed-use industries such as finance or professional services were instead split into consumer-facing and firm-facing components using expenditure weights, the corresponding worker-level skill moments would no longer be directly observed. The baseline strategy therefore disciplines codifiability using the education of workers performing producer-service occupations, rather than proxying for skill intensity with industry expenditure shares.

The targeted moments fall into four groups. First, I target the time-series paths of agricultural, manufacturing, and service employment shares, fertility, and normalized GDP per capita. These moments discipline labor reallocation, the demographic transition, and the aggregate income path. Second, I target three scalar summaries of the transition: the peak manufacturing employment share, average fertility, and the agricultural-to-service fertility ratio. Third, I target final-period skill moments in 2019: the aggregate skilled employment share and the skilled employment shares within value added in agriculture, manufacturing, and consumer services. These moments discipline the supply of skilled labor and sectoral skill intensities in value-added production. Fourth, I target the producer-service moments most directly tied to the mechanism: the high-skill employment share within producer-service inputs used by agriculture, manufacturing, and consumer services in 2019, and the change in the aggregate high-skill producer-service share between 1960 and 2019. These moments discipline the inverse-codifiability parameters κ_j, and therefore the relative ease with which sectoral producer-service tasks are formalized and turned into reusable organizational routines.

Appendix E.G considers an alternative input-output-based calibration of the same mechanism. In that exercise, I replace the baseline Census-based producer-service moments with moments constructed from OECD input-output tables. Specifically, I compute the high-skill service share induced by a \$100 final-demand shock to agriculture, manufacturing, and consumer services, and recalibrate the codification parameters to match those induced service mixes. I consider both the strict consumer-service definition used in the baseline and a final-expenditure-weighted definition in which mixed-use services can partly enter the consumer-service bundle. This exercise provides an alternative discipline on the codification parameters. It is less direct than the baseline calibration because it proxies high-skill services using service subsectors rather than observing the education of workers performing producer-service occupations, but the results are qualitatively stable across definitions and calibration strategies. The calibration proceeds in two steps. I first fix a set of parameters externally, either by normalization or by reference to the literature. I then choose the remaining parameters jointly by simulated method of moments. The model is simulated at an annual frequency and aligned so that model period $t_0 = 60$ corresponds to 1850. Appendix E reports the full target vector, data sources, numerical procedure, and robustness exercises.

V.A Externally set parameters

Table 5 reports the parameters fixed outside the simulated method of moments. These parameters are either normalizations, standard values from the literature, or auxiliary choices that fix units and timing. Manufacturing is the numeraire, unskilled efficiency is normalized to one, and initial productivity levels are normalized to zero in logs. I set a model generation to 20 years, following the quantitative unified-growth literature and the parameterization in Lagerlöf (2006). Final demand follows the non-homothetic CES structure of Comin et al. (2021); I set the elasticity of substitution across final goods to $\sigma = 0.3$, reflecting low substitutability across broad consumption categories.

Table 5: Externally Set Parameters and Normalizations

Parameter	Value	Source / rationale
$p_{m,t}$	1	Manufacturing numeraire
h_u	1	Unskilled efficiency normalization
h_s	1.6	Fixed skill efficiency premium
$\log A_{j,0}$	0	Initial productivity normalization
gen	20	Generation length, Lagerlöf (2006)
ε_m	1	Manufacturing income-effect normalization
w_m	2.5	Manufacturing demand-weight normalization
σ	0.3	Final-goods elasticity, Comin et al. (2021)
β_π	1	Skill-choice index normalization
σ_{skill}	0.1	Benchmark logit dispersion
σ_L	1.2	Skill substitution elasticity
σ_{ps}	1.2	Producer-service mode elasticity
τ_e^U	1	Education-time cost normalization
τ_e^S	0.5	Education-environment complementarity
$\tau_m^{rel}, \tau_s^{rel}$	1, 1	Equal child-rearing costs in manufacturing and services
t_0	60	Model period aligned with 1850
$\theta_{j,1850}$	1	Initial value-added production
t_{GO}	2025	IO coefficients reach modern values

Notes: Appendix E reports the remaining fixed numerical parameters and robustness checks.

The skill-choice parameters fix the scale and smoothness of the transition from a low-skill to a high-skill economy. I normalize $\beta_\pi = 1$. I set the logit dispersion to $\sigma_{\text{skill}} = 0.1$, which implies that historically plausible changes in the log skill premium generate the observed transition from a very low initial skilled share to a terminal skilled share of roughly one third. Appendix E shows the implied calculation and reports robustness. On the production side, I set the elasticities of substitution across skill types and across producer-service modes to 1.2. These near-unit elasticities allow imperfect but nonzero substitution across inputs and are consistent with quantitative work that treats skill groups and tasks as imperfect substitutes in skill-complementary production structures (Krusell et al., 2000; Acemoglu and Autor, 2011). The education-time cost parameters govern the productivity of parental education investment. I normalize $\tau_e^U = 1$, fixing the units of education time in an unskilled environment, and set $\tau_e^S = 0.5$. I interpret

$$\tau_e(S_t) = (1 - S_t)\tau_e^U + S_t\tau_e^S$$

as a reduced-form education-environment complementarity, not as tuition, public education expenditure, or a schooling price index. A lower $\tau_e(S_t)$ means that a unit of parental education time produces more effective child human capital in environments with more educated parents, skilled peers, teachers, and schooling institutions. This is consistent with human-capital technologies featuring self-productivity and dynamic complementarity (Cunha and Heckman, 2007; Cunha et al., 2010). Along the calibrated U.S. path, the terminal skilled share implies $\tau_e(0.34) = 0.83$, so the effective reduction in education time costs is 17 percent relative to an unskilled environment. Consistent with the evidence presented in Appendix E showing that manufacturing and services households exhibit similar fertility education choices over the US transition, I set $\tau_m = \tau_s = 1$, so the baseline calibration shuts down differences in child-rearing costs between manufacturing and service households.[8] Finally, I allow input-output coefficients to evolve over the transition. The economy starts in 1850 with value-added production, $\theta_{j,1850} = 1$, and input-output coefficients gradually converge to their modern values by 2025.[9] This choice is motivated by evidence that changes in U.S. industry output reflect changes in input-output coefficients as well as changes in final demand (Feldman et al., 1987).

V.B Internally calibrated parameters

The remaining parameters are chosen jointly by simulated method of moments. Let

$$\Theta = \left(\gamma, \tau_a, \varphi_a, \varphi_m, \varphi_c, g^{GO}, \kappa_a, \kappa_m, \kappa_c, \eta_H, c_{col}, \alpha_a, \alpha_m, \alpha_c, \varepsilon_a, \varepsilon_c, w_a, w_c\right)$$

denote the internally calibrated vector. These parameters govern fertility preferences, the agricultural child-rearing cost, sectoral productivity growth, gross-output productivity growth, sectoral codifiability, the cost of becoming skilled, value-added skill intensities, and non-homothetic final demand. Table 6 reports the calibrated values. The calibration is joint, so no parameter is identified by a single moment in isolation. The table nevertheless reports the moment most directly associated with each parameter.

The target moments discipline distinct parts of the model. Employment-share paths and final-demand moments discipline the non-homothetic demand parameters and the size of the manufacturing hump. Fertility moments discipline the household block and the agricultural child-rearing cost. Final-period skill moments discipline the cost of becoming skilled and the skill intensities of value-added production. Sectoral productivity growth rates discipline the strength of scale effects in productivity. The producer-service moments discipline the inverse-codifiability parameters κ_j, which govern how easily producer-service tasks associated with each final-good sector can be formalized and turned into reusable organizational routines.

The calibration implies

$$\kappa_m < \kappa_c < \kappa_a, \qquad \text{or equivalently} \qquad \lambda_m > \lambda_c > \lambda_a,$$

where $\lambda_j = 1/\kappa_j$ is codification intensity. Manufacturing-related producer-service tasks are therefore the most codifiable, followed by producer-service tasks associated with consumer services and agriculture. This ordering is not imposed directly. It is disciplined by the 2019 high-skill employment shares within producer-service inputs by final-good sector. In the calibrated economy, manufacturing matters not only because it directly employs workers, but because its producer-service requirements are especially connected to high-skill organizational and technical tasks.

8 I calibrate an economy wide child time cost τ and then impose that the sectoral specific time cost is a fraction of the economy wide cost. Because manufacturing and services households display similar fertility and education choices over the sample period their τ_i^{rel} is equal to one, whereas τ_a^{rel} is calibrated to match the average ratio of fertility in agricultural households relative to services/manufacturing households within counties over the sample period.

9 I construct the modern coefficients from BEA input-output tables by mapping detailed industries into agriculture, manufacturing, consumer services, and producer services, and by computing sector-specific shares of value added, agricultural intermediates, manufacturing intermediates, and producer-service intermediates in gross output. Appendix E describes the mapping and construction in detail.

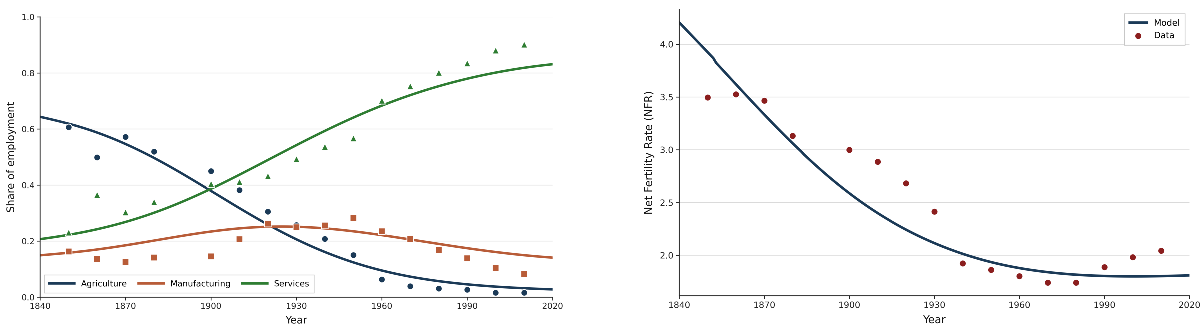
Table 6: Internally Calibrated Parameters

Parameter	Description	Value	Main target	Data	Model
γ	Utility weight on children	0.044	Average fertility	2.474	2.395
τ_a	Agricultural child cost	0.419	Fertility ratio A/S	2.000	1.977
φ_a	Agriculture scale effect	0.931	Agriculture TFP growth	0.037	0.040
φ_m	Manufacturing scale effect	0.828	Manufacturing TFP growth	0.031	0.030
φ_c	Consumer-services scale effect	0.528	Consumer-services TFP growth	0.014	0.012
g^{GO}	Gross-output TFP growth	0.003	GDPpc path deviation	0.000	0.145
κ_a	Inverse codifiability, A	1.007	HS in PS A, 2019	0.39	0.42
κ_m	Inverse codifiability, M	0.415	HS in PS M, 2019	0.65	0.71
κ_c	Inverse codifiability, C	0.731	HS in PS C, 2019	0.53	0.57
η_H	Frontier curvature	4.876	Δ aggregate HS in PS	0.182	0.168
c_0	Skill acquisition cost	0.271	Skill share, 2019	0.345	0.308
α_a	Skill intensity, A	0.180	Within-VA skill A	0.160	0.161
α_m	Skill intensity, M	0.254	Within-VA skill M	0.220	0.229
α_c	Skill intensity, consumer services	0.237	Within-VA skill C	0.200	0.213
ε_a	Agriculture income effect	0.016	Average agriculture share	0.259	0.256
ε_c	Consumer-services income effect	1.087	Average services share	0.560	0.537
w_a	Agriculture demand weight	0.952	Agriculture share, 1850	0.606	0.620
w_c	Consumer-services demand weight	4.035	Services share, 1850	0.231	0.204

Notes: A = agriculture, M = manufacturing, C = consumer services, PS = producer services, HS = high-skill, and VA = value added. The calibration is joint; the target column reports the moment most closely associated with each parameter. Producer-service high-skill targets are constructed from the 2019 IPUMS USA Census sample by restricting to producer-service occupations within each broad industry group and computing the share of workers with at least one year of college.

V.C Model fit

Figure 8 compares the calibrated model to the main U.S. transition moments. The model reproduces the broad path of structural change: agricultural employment declines, manufacturing rises and then falls, and services become the dominant sector. It also captures the long-run demographic transition, with fertility declining from high nineteenth-century levels to modern low fertility. These moments discipline the two aggregate transitions that the model is built to connect: the reallocation of labor across sectors and the movement from high fertility to higher human capital investment.

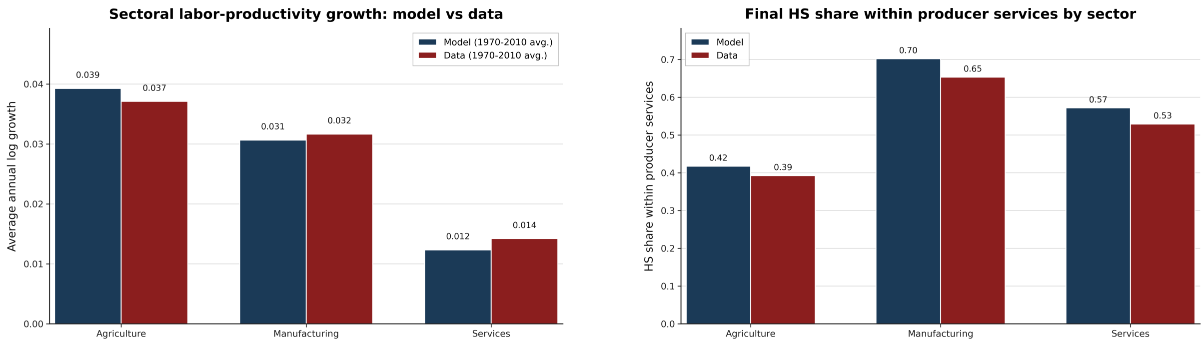


(a) Structural change　　**(b)** Demographic transition

Figure 8: Model fit to targeted U.S. moments, 1850–2019. Dots represent data and lines show model simulations. Panel (a) shows employment shares in agriculture, manufacturing, and services. Panel (b) shows the fertility transition.

Figure 9 reports the fit for the moments most directly tied to the production side of the model. Panel (a) compares sectoral value added productivity growth rates. The model matches the ranking of growth rates across sectors, with faster productivity growth in agriculture and manufacturing than in services. Panel (b) compares the high-skill employment share within producer-service inputs by final sector. The model reproduces the central empirical ranking:

producer services used by manufacturing are the most high-skill intensive, followed by those used by consumer services and agriculture. This ranking is the key empirical discipline on the model's codification mechanism: to match it, the calibration must assign manufacturing the lowest codification resistance. Because the calibrated economy assigns the lowest codification resistance to manufacturing, manufacturing related producer-service tasks are the easiest to reorganize into high-skill form.



(a) VA productivity growth

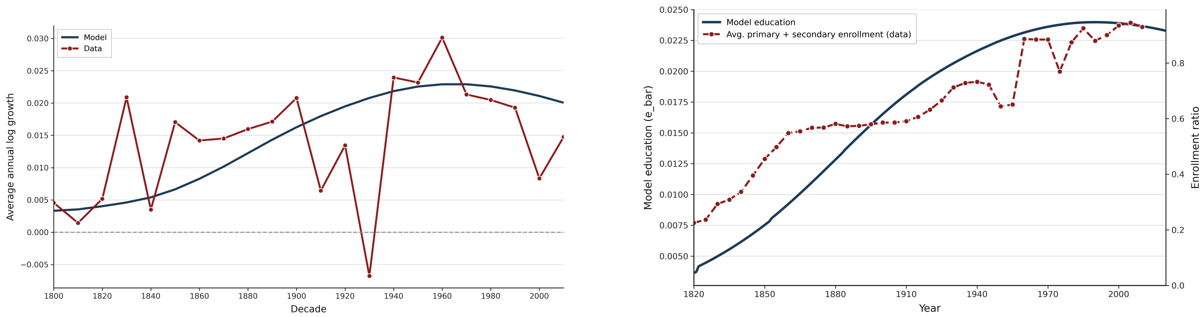
(b) HS shares in producer services

Figure 9: Model fit to mechanism-targeted moments. Panel (a) compares sectoral value added productivity growth rates in the model and the data. Panel (b) compares the final high-skill employment share within producer-service inputs by final sector. Data sources: GGDC 10-sector database for value added productivity growth and the 2019 IPUMS USA Census sample for high-skill employment shares within producer-service inputs.

Table 6 summarizes the fit of the main scalar targets. The model matches average fertility closely, generating an average fertility rate of 2.395 compared with 2.474 in the data, and reproduces the agricultural-to-service fertility ratio, with a model value of 1.977 compared with a target of 2.000. It also matches long-run sectoral productivity growth: agriculture, manufacturing, and services grow at 0.040, 0.030, and 0.012 in the model, compared with 0.037, 0.031, and 0.014 in the data. The model produces a final aggregate skilled share of 0.308, compared with 0.345 in the data. The challenge in matching the aggregate skilled share jointly with the sectoral skilled shares is that, given the separation between consumer and producer services, the aggregate model skilled share will depend on how one constructs the input-output coefficients matrix in the production function from the BEA use tables.[10] For the producer-service targets, the model matches the ranking across sectors and closely tracks the levels: 0.42 versus 0.39 in agriculture, 0.71 versus 0.65 in manufacturing, and 0.57 versus 0.53 in consumer services. The important discipline for the mechanism is that manufacturing related producer services are substantially more high-skill intensive than those used by the other final sectors, and the calibrated model reproduces this pattern.

I next examine non-targeted validation moments. Figure 10 compares model-implied GDP per capita growth and schooling to their empirical counterparts. Although not explicitly targeted, the model captures the broad takeoff and decline in GDP per capita growth over development and generates a rising education path consistent with the long-run increase in schooling. In the model, schooling corresponds to the average education investment chosen by households; in the data, I compare it to average of primary and secondary enrollment rates. This validation is informative because education responds endogenously to the growth-induced obsolescence mechanism rather than being targeted directly as a time series. As a final validation exercise, I

10 Appendix E.C explains this process in detail.



(a) GDP per capita growth

(b) Schooling

Figure 10: Non-targeted validation moments, 1800–2019. Panel (a) compares model-implied GDP per capita growth to the data. Panel (b) compares model education to enrollment rates. Schooling in the model is the average education choice across sectors; in the data it is the average enrollment rate, measured as the average of primary and secondary enrollment. Data sources: Gapminder and U.S. Censuses of Population.

compare the model and the data on the relationship between peak manufacturing employment and observable development outcomes. Figure 11 reports coefficients from regressions of key outcomes on the peak manufacturing employment share, estimated both in the model and in the data. These slopes are not used to estimate the U.S. calibration. The model nevertheless reproduces the qualitative pattern in the data: economies with larger manufacturing peaks have higher college attainment, larger high-skill service shares and faster GDP per capita growth. As GDP per capita growth is not observed at the US county level, for this outcome the figure uses patents per capita in logs, a close empirical counterpart of aggregate productivity growth. In all cases, model slopes are on the same order as those observed in the data, generally sitting between the IV and OLS estimates.

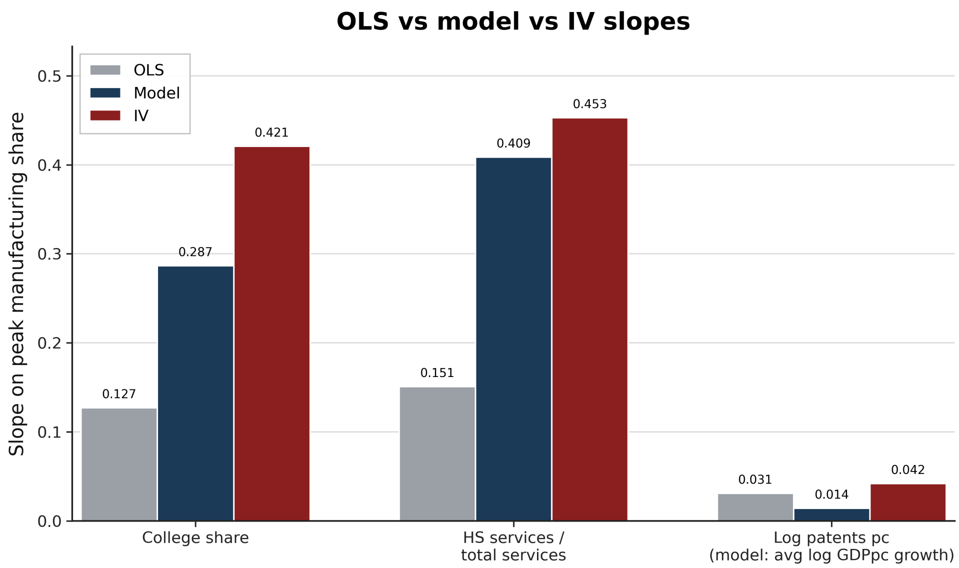


Figure 11: Non-targeted validation: model and data slopes with respect to peak manufacturing employment. The figure compares coefficients from regressions of key observable outcomes on the peak manufacturing employment share in the model and in the data.

The calibrated model therefore reproduces the main U.S. transition and the mechanism-specific moments needed for the counterfactual analysis. It matches the size and timing of the manufacturing phase, the demographic transition, the long-run ranking of sectoral productivity growth, and the high-skill intensity of manufacturing related producer services. It also generates non-targeted movements in schooling and GDP per capita growth, and reproduces the qualitative cross-economy relationship between peak manufacturing and development outcomes. I now use this calibrated economy to study how changing the size of the manufacturing phase affects long-run development.

VI Results

This section uses the calibrated U.S. economy to study how the size of the manufacturing phase affects long-run development. I generate counterfactual economies by varying the manufacturing demand weight w_m in the non-homothetic final-demand system. A higher w_m raises relative demand for manufacturing during the transition and produces a larger manufacturing employment peak. I then ask how this temporary difference in the path out of agriculture affects income, growth, education, population, and the composition of services. The exercise is not meant to evaluate a specific policy instrument. It is a comparative-static experiment that holds fixed the rest of the calibrated economy. Appendix E shows that the same allocations can be implemented through relative-price wedges, while Appendix F studies a more structural source of lower manufacturing exposure through openness to trading partners with a comparative advantage in manufacturing. The goal here is narrower: to isolate whether a smaller manufacturing phase leaves the economy with a different service sector and a different long-run development path.

VI.A The manufacturing phase in the calibrated economy

Figure 12 reports the main comparative statics. Outcomes are normalized to the benchmark U.S. calibration. Economies with larger manufacturing peaks reach higher GDP per capita and higher GDP per capita growth in 2019. They also experience lower cumulative population growth over the transition.

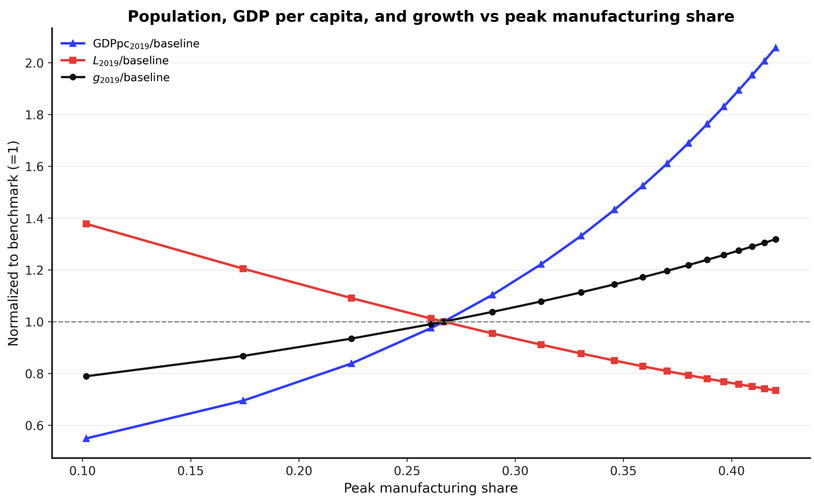


Figure 12: Comparative statics in the calibrated U.S. economy. The figure shows the relationship between the peak manufacturing employment share and GDP per capita, population, and long-run growth in 2019, with outcomes normalized to the U.S. benchmark. Counterfactual economies are generated by varying the manufacturing demand weight w_m in the final-demand system.

The figure shows that a temporary difference in manufacturing exposure is not neutralized once the economy becomes service dominated. Larger manufacturing phases leave the economy with a higher organizational frontier and a larger skilled workforce. These states raise income and growth after manufacturing has declined. The same force accelerates the quantity-quality transition: higher returns to education reduce fertility and lower cumulative population growth. Thus, in the calibrated model, the manufacturing phase has persistent effects because it changes the state variables inherited by the service economy.

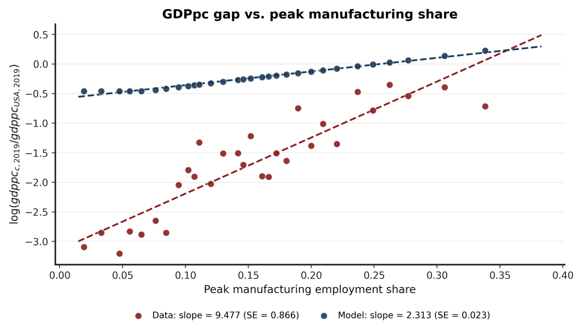
VI.B Accounting for cross-country patterns

I next ask whether the calibrated mechanism can account for the cross-country gradients documented in Section II. For each country, I compute the observed peak manufacturing employment share and assign the country to the model economy with the corresponding peak, interpolating across the simulated grid. The exercise is intentionally one-dimensional. Countries differ only in the size of the manufacturing phase, while all other parameters remain fixed at the U.S. calibration. It is therefore not a variance decomposition. Its purpose is to ask whether the manufacturing margin alone generates gradients of the same sign and economically relevant magnitude as those observed in the data.

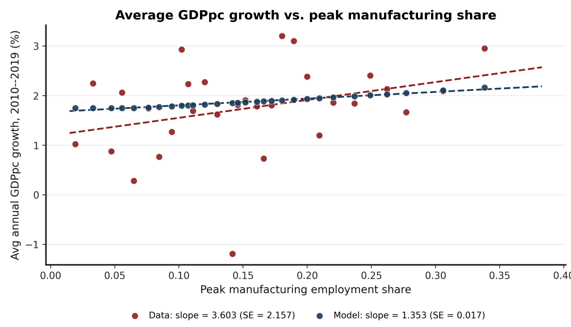
Figure 13 reports the main comparison. The model generates empirical gradients of the correct sign for all four development outcomes. Countries with larger manufacturing peaks are richer relative to the United States, have higher tertiary enrollment, grow faster over 2010–2019, and experience lower cumulative population growth. Table 7 summarizes the corresponding slope comparisons. For relative income, the data slope is 9.477 and the model slope is 2.815; the model therefore accounts for 30 percent of the empirical relationship between peak manufacturing and log GDP per capita relative to the United States. For tertiary education, the model accounts for 50 percent of the data slope. For average GDP per capita growth over 2010–2019, it accounts for 58 percent. For cumulative population growth, it accounts for 39 percent.

These magnitudes are substantial given the discipline of the exercise. The model varies only the manufacturing phase, while the data also reflect institutions, geography, trade exposure, natural resources, education systems, migration, demographic shocks, and measurement error. The model is therefore compressed relative to the data. This is visible in the standard-deviation ratios in Table 7: the counterfactuals generate less unconditional dispersion than the data, but they capture a sizable fraction of the empirical slopes.

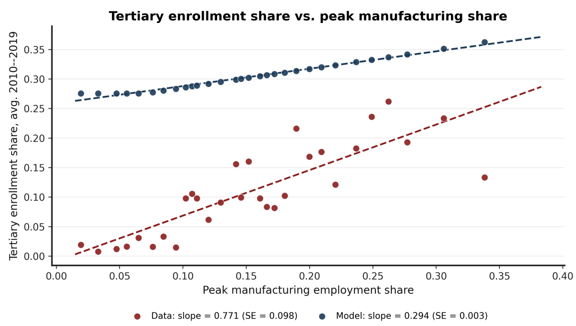
The model is especially successful for the paper's central outcome: the high-skill producer-service content of the service economy. In the data, I measure high-skill producer services using ILOSTAT occupation-by-industry data: ISCO-08 groups 1–4 employed in broad producer-service industries, divided by total market-service employment. The corresponding model object is high-skill producer-service employment divided by total service employment. The data slope is 0.776 and the model slope is 0.753, implying a slope ratio of 0.970. Thus, the calibrated mechanism nearly matches the empirical gradient linking peak manufacturing to the high-skill producer-service content of later services. This is not a targeted cross-country moment. The calibration disciplines producer-service codifiability using U.S. evidence, and the cross-country relationship provides an external check on the mechanism. The model-data correlation is 0.607, implying an R^2 of 0.368.



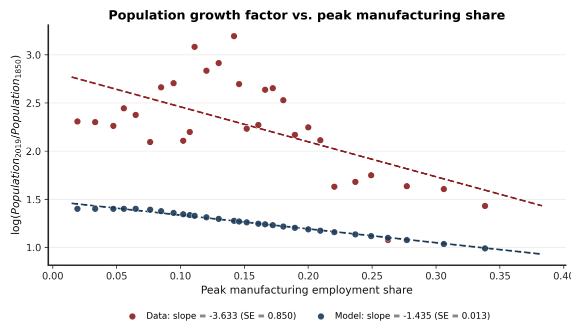
(a) $\log(GDPPC_{c,2019}/GDPPC_{USA,2019})$



(b) Average GDPpc growth



(c) Tertiary education



(d) Cumulative population growth

Figure 13: Peak manufacturing and development outcomes. The figure compares empirical cross-country relationships to model-generated counterfactual economies. Panel (a) plots final log GDP per capita relative to the United States. Panel (b) plots average GDP per capita growth over 2010–2019. Panel (c) plots tertiary enrollment over 2010–2019. Panel (d) plots $\log(Population_{2019}/Population_{1850})$. Red points and lines correspond to the data; blue points and lines correspond to model-generated economies.

The accounting exercise delivers two messages. First, variation in the manufacturing phase alone generates a quantitatively important share of the observed cross-country gradients in income, education, growth, and population. Second, the model performs best for the outcome most closely tied to the mechanism. Countries with smaller manufacturing peaks do not merely become service economies at lower income levels; they enter services with a weaker high-skill producer-service base.

VI.C Which mechanisms matter?

I next isolate which calibrated sectoral heterogeneities generate the model gradients. I run parameter-equalization experiments in which one group of sectoral parameters is set equal to its arithmetic mean, and then repeat the manufacturing-demand counterfactuals. All endogenous choices and equilibrium responses remain active. Table 8 reports each experiment's slope as a percent of the corresponding full-model slope.

The decomposition points to producer-service codification as the central source of the service-composition and education gradients, and as an important source of the income and growth gradients. Equalizing κ_j reduces all five slopes. The income slope falls to 55 percent of its full-model value, the 2010–2019 growth slope to 57 percent, the tertiary-education slope to 33 percent, the population-growth slope to 50 percent, and the high-skill producer-service slope to 41 percent. Thus, the effect of manufacturing does not come only from expanding a sector

Table 7: Country-Level Model Accounting

Outcome	N	N_{slope}	Data β_d	Model β_m	β_m/β_d	σ_m/σ_d	R^2
GDP per capita	172	125	6.835	2.496	36.5%	0.158	0.393
Tertiary enrollment	172	125	0.580	0.320	55.2%	0.199	0.259
GDP per capita growth	172	125	3.278	1.517	46.3%	0.048	0.017
Population growth	172	125	-6.924	-1.496	21.6%	0.124	0.103
High-skill producer services	48	44	0.861	0.558	64.8%	0.411	0.368

Notes: Data and model slopes are from separate regressions of each country-level moment on peak manufacturing employment. Slope comparisons use the common-support sample with peak manufacturing employment shares above 0.100. "Share accounted" is the model slope divided by the data slope. σ_m/σ_d is the ratio of model to data standard deviations. R^2 compares country-level model predictions with the data.

Table 8: Mechanism decomposition: slopes relative to the full model

	Full model	Equal κ_j	Equal φ_j	Equal α_j
GDPpc, 2019	100.0	62.2	73.7	98.1
GDPpc growth, 2010–2019	100.0	62.9	105.8	104.9
Tertiary education	100.0	38.4	61.9	104.8
Population growth	100.0	58.4	27.8	97.5
HS producer services	100.0	46.3	112.2	103.6

Notes: Entries report each experiment's slope as a percent of the full-model slope. Slopes are computed with respect to each experiment's own realized peak manufacturing employment share and are restricted to model economies with peak manufacturing employment shares above 0.100, matching the common-support restriction used in the country-level accounting table. In each equalization experiment, the relevant sectoral parameters are set equal to their arithmetic mean. κ_j governs codification resistance in producer-service tasks, φ_j governs sectoral productivity feedback, and α_j governs value added skill intensity. All endogenous model responses remain active. HS producer services denotes the model high-skill producer-service employment share within total service employment.

with different average productivity. It comes from the type of producer-service tasks that manufacturing generates. When sectors no longer differ in the ease with which these tasks can be codified and reorganized around skilled labor, the link between manufacturing peaks and the later service economy weakens sharply.

Sectoral productivity feedbacks also matter, but mainly through the broader growth and demographic margin. Equalizing φ_j reduces the income slope to 64 percent of its full-model value, the tertiary-education slope to 63 percent, and the population-growth slope to 28 percent. These effects indicate that productivity feedbacks amplify the aggregate development consequences of manufacturing-intensive growth, especially through income, schooling, and fertility. By contrast, the high-skill producer-service slope is not reduced. Productivity dynamics therefore help shape the scale of the transition, but they are not the main force behind the producer-service composition channel.

Equalizing value-added skill intensities, α_j, has little effect. All slopes remain close to their full-model values. The mechanism is therefore not primarily that manufacturing directly employs more skilled labor in value added. It operates through the organizational margin: manufacturing expands codifiable producer-service tasks, raises the incentive to acquire skill, and changes the composition of the service economy that follows.

VII Policy

The results above show that manufacturing mattered historically because it built the skill base and producer-service capabilities inherited by service economies. This does not imply that subsidizing manufacturing is the right forward-looking policy. If the object that matters for long-run development is the organizational capability that manufacturing helped create, then policy may be more effective when it targets that capability directly. Furthermore, if the manufacturing sector has already reached its peak subsidizing it implies fighting against income and relative price effects with a costly policy instrument. I study these possibilities in a country-calibrated policy extension of the model. For 30 countries with the required employment shares, value-added, and input output data from GGDC 10-Sector and OECD-ICIO, I recalibrate the sectoral productivity feedback parameters φ_j, the producer-service codification parameters κ_j, and the manufacturing demand shifter w_m. The targets are sectoral productivity growth, high-skill producer-service input shares, and observed peak manufacturing employment shares. I then introduce permanent labor-cost subsidies in 2026, financed by an income tax, and measure welfare as forward-looking consumption-equivalent variation through 2100. A manufacturing subsidy lowers the firm-side labor cost in manufacturing value added; a high-skill producer-service subsidy lowers the firm-side cost of high-skill producer-service labor. Workers receive the market wage in both cases.

Figure 14 reports mean welfare gains across the 30 recalibrated economies for subsidies introduced in 2026. Shaded areas show the 95% confidence intervals. None of the 30 countries with the available data have a positive optimal subsidy rate for manufacturing industrial policy. In all cases, positive subsidy rates create welfare losses. By contrast, high-skill producer-service subsidies generate positive welfare gains at moderate rates, with a mean-optimal subsidy of about 45 percent. A joint subsidy to manufacturing and high-skill producer services also raises welfare at moderate rates, with an optimum around 40 percent, but it is dominated by the more targeted high-skill producer-service policy. The same ranking holds when the policy is introduced in 1960, the beginning of the main postwar period of infant-industry promotion: manufacturing subsidies remain non-optimal on average, while high-skill producer-service subsidies generate positive welfare gains. The reason is twofold. First, manufacturing subsidies are fiscally expensive. In the country-calibrated experiments, a 15 percent manufacturing subsidy costs about 1.7 percent of GDP per year on average, compared with about 0.5 percent for a high-skill producer-service subsidy of the same rate. Second, for countries that have already passed their manufacturing peak, subsidizing manufacturing works against the income and price effects which push the economy towards services. The policy attempts to preserve or recreate a sectoral structure that the economy is endogenously moving away from, while only indirectly affecting the organizational margin that matters for service-led growth.

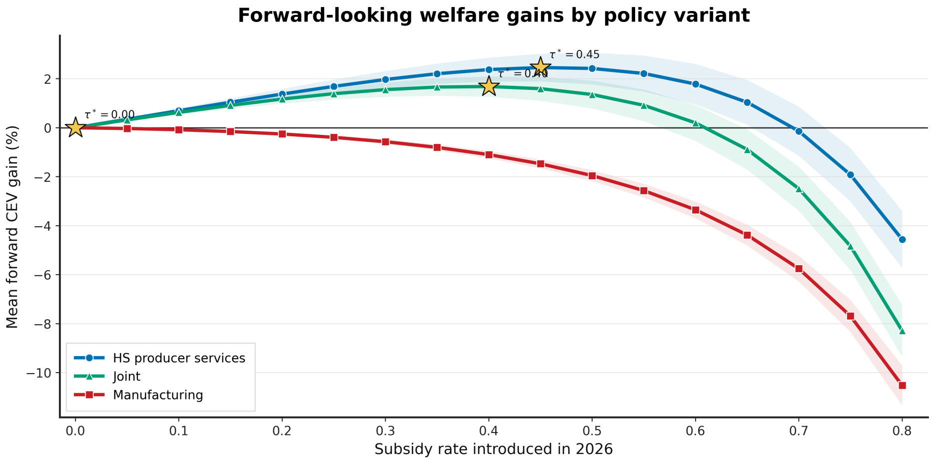


Figure 14: Forward-looking welfare gains by policy variant. The figure reports mean consumption-equivalent welfare gains across 30 country-calibrated economies. Policies are permanent labor-cost subsidies introduced in 2026 and financed with an income tax. Manufacturing subsidies lower the firm-side labor cost in manufacturing value added. High-skill producer-service subsidies lower the firm-side cost of high-skill producer-service labor. Stars mark the mean-optimal subsidy rate for each policy variant.

Hence, the welfare-relevant target is the capability manufacturing helped build, not manufacturing employment per-se. In the model, high-skill producer-service subsidies raise welfare because they lower the cost of formal organizational tasks, increase the use of codified routines, and support the accumulation of skills used in scalable service activities. Manufacturing subsidies are a more indirect instrument: they expand a sector whose historical role was partly to generate these capabilities, but they do so at a higher fiscal cost and, in many economies, after the income effect has already shifted demand toward services. This conclusion is consistent with the argument in Juhász et al. (2024) that industrial policy in a servicized economy should not simply try to reproduce the manufacturing-centered strategies of the past. The relevant policy problem is to identify and support the activities that generate learning, scale, and organizational upgrading in the economy that countries are actually moving into.

VIII Conclusion

This paper studies the long-run consequences of developing without industrialization. Labor can leave agriculture and move into services without a large manufacturing phase, but the service economy that emerges need not be the same. Manufacturing creates demand for organizational, technical, managerial, clerical, and professional producer-service tasks that are intensive in skilled labor. These tasks help build transferable routines, management systems, accounting practices, technical protocols, and other forms of codified organizational knowledge. The central claim of the paper is that this organizational frontier shapes the service economy that follows.

I document this mechanism in the data and quantify it in a model. Across countries, higher peak manufacturing employment shares are associated with faster growth after the rise of services, larger gains in tertiary education, faster fertility declines, and more skill-intensive service sectors. Within the United States, counties with larger historical manufacturing booms later develop more education-intensive service occupations, with the effect concentrated in organizational and business service work. I interpret these facts through a unified growth model with structural change, endogenous fertility, education, and heterogeneous producer services. In the model, manufacturing raises demand for producer-service tasks that are relatively easy to codify and reorganize into high-skill form. Calibrated to the United States over 1850–2019, the model accounts for 30 percent of the cross-country relationship between peak manufacturing and relative income, 58 percent of the relationship with GDP per capita growth, 50 percent of the relationship with tertiary education, and 39 percent of the relationship with cumulative population growth. It also nearly matches the empirical gradient between peak manufacturing and the high-skill producer-service intensity of the modern service economy.

The results qualify a simple view of service-led development. Countries can become service economies without first becoming manufacturing economies, but the path into services matters. When the manufacturing phase is small, economies enter services with a weaker skill base and a thinner stock of organizational capabilities. This affects income and growth, but also education, fertility, and the composition of service employment over the transition. Even though manufacturing booms are associated with faster growth in the service economy, policies targeting manufacturing directly are not welfare improving in the model. Manufacturing subsidies are costly and, once countries have begun moving into services, they work against the income-driven reallocation of demand. Subsidies to high-skill producer services perform better because they target the capability that manufacturing historically helped create. The main lesson is therefore not that manufacturing is irrelevant, or that services automatically replace it as an engine of development. It is that development without industrialization requires building the organizational and human-capital base that allows services to become productive.

References

Acemoglu, Daron and David Autor, "Skills, tasks and technologies: Implications for employment and earnings," in "Handbook of labor economics," Vol. 4, Elsevier, 2011, pp. 1043–1171.

Barro, Robert J and Jong Wha Lee, "A new data set of educational attainment in the world, 1950–2010," *Journal of development economics*, 2013, *104*, 184–198.

Baumol, William J, "Macroeconomics of unbalanced growth: the anatomy of urban crisis," *The American economic review*, 1967, *57* (3), 415–426.

Becker, Gary S, Kevin M Murphy, and Robert Tamura, "Human capital, fertility, and economic growth," *Journal of political economy*, 1990, *98* (5, Part 2), S12–S37.

Bernard, Andrew B, Valerie Smeets, and Frederic Warzynski, "Rethinking deindustrialization," *Economic Policy*, 2017, *32* (89), 5–38.

Best, Mark and Duncan Neuhauser, "Walter A. Shewhart, 1924, and the Hawthorne Factory," *Quality and Safety in Health Care*, 2006, *15* (2), 142–143.

Boppart, Timo, "Structural change and the Kaldor facts in a growth model with relative price effects and non-Gorman preferences," *Econometrica*, 2014, *82* (6), 2167–2196.

Borusyak, Kirill, Xavier Jaravel, and Jann Spiess, "Revisiting event-study designs: robust and efficient estimation," *Review of Economic Studies*, 2024, *91* (6), 3253–3285.

Buera, Francisco J and Joseph P Kaboski, "The rise of the service economy," *American Economic Review*, 2012, *102* (6), 2540–2569.

_ and _ , "Scale and the origins of structural change," *Journal of Economic Theory*, 2012, *147* (2), 684–712.

_ , _ , Martí Mestieri, and Daniel G O'Connor, "The stable transformation path," Technical Report, National Bureau of Economic Research 2020.

_ , _ , Richard Rogerson, and Juan I Vizcaino, "Skill-biased structural change," *The Review of Economic Studies*, 2022, *89* (2), 592–625.

Chandler, Alfred D., *The Visible Hand: The Managerial Revolution in American Business*, Cambridge, MA: Belknap Press of Harvard University Press, 1977.

Chen, Xilu, Guangyu Pei, Zheng Song, and Fabrizio Zilibotti, "Tertiarization like china," *Annual Review of Economics*, 2023, *15*, 485–512.

Comin, Diego, Danial Lashkari, and Martí Mestieri, "Structural change with long-run income and price effects," *Econometrica*, 2021, *89* (1), 311–374.

Cunha, Flavio and James J. Heckman, "The Technology of Skill Formation," *American Economic Review*, 2007, *97* (2), 31–47.

_ , _ , and Susanne M. Schennach, "Estimating the Technology of Cognitive and Noncognitive Skill Formation," *Econometrica*, 2010, *78* (3), 883–931.

Doepke, Matthias, "Accounting for fertility decline during the transition to growth," *Journal of Economic growth*, 2004, *9*, 347–383.

Duernecker, Georg and Berthold Herrendorf, "Structural transformation of occupation employment," *Economica*, 2022, *89* (356), 789–814.

— **and Miguel Sanchez-Martinez**, "Structural change and productivity growth in Europe—Past, present and future," *European Economic Review*, 2023, *151*, 104329.

— **, Berthold Herrendorf, and Ákos Valentinyi**, "Structural Change within the Services Sector and the Future of Cost Disease," *Journal of the European Economic Association*, 2024, *22* (1), 428–473.

Engbom, Niklas, Hannes Malmberg, Tommaso Porzio, Federico Rossi, and Todd Schoellman, "Economic development according to chandler," Technical Report, National Bureau of Economic Research 2025.

Fan, Tianyu, Michael Peters, and Fabrizio Zilibotti, "Growing like India—the unequal effects of service-led growth," *Econometrica*, 2023, *91* (4), 1457–1494.

Feldman, Stanley J, David McClain, and Karen Palmer, "Sources of structural change in the United States, 1963-78: an input-output perspective," *The review of economics and statistics*, 1987, pp. 503–510.

Flesher, Tonya K., "James O. McKinsey," *The Accounting Historians Journal*, 1985, *12* (2), 1–22.

Fujiwara, Ippei and Kiminori Matsuyama, "A technology-gap model of 'premature'deindustrialization," *American Economic Review*, 2024, *114* (11), 3714–3745.

Galor, Oded, "From stagnation to growth: unified growth theory," *Handbook of economic growth*, 2005, *1*, 171–293.

— , *Unified growth theory*, Princeton University Press, 2011.

— **and David N Weil**, "Population, technology, and growth: From Malthusian stagnation to the demographic transition and beyond," *American economic review*, 2000, *90* (4), 806–828.

— **and Omer Moav**, "From physical to human capital accumulation: Inequality and the process of development," *The review of economic studies*, 2004, *71* (4), 1001–1026.

Gollin, Douglas, Remi Jedwab, and Dietrich Vollrath, "Urbanization with and without industrialization," *Journal of Economic Growth*, 2016, *21* (1), 35–70.

Graham, Laurel D., "Domesticating Efficiency: Lillian Gilbreth's Scientific Management of Homemakers, 1924–1930," *Signs*, 1999, *24* (3), 633–675.

— , "Lillian Gilbreth and the Mental Revolution at Macy's, 1925–1928," *Journal of Management History*, 2000, *6* (7), 285–305.

Graham, Laurel Diane, "Lillian Moller Gilbreth's Extensions of Scientific Management into Women's Work, 1924–1935." Phd dissertation, University of Illinois at Urbana-Champaign 1992.

Haines, Michael R, "Historical demographic, economic and social data: The united states, 1790-1970," *Department of Economics. Colgate University, Computer File*, 2001.

Hansen, Gary D and Edward C Prescott, "Malthus to solow," *American economic review*, 2002, *92* (4), 1205–1217.

Herrendorf, Berthold, Richard Rogerson, and Akos Valentinyi, "Growth and structural transformation," *Handbook of economic growth*, 2014, *2*, 855–941.

— , — **, and Ákos Valentinyi**, "New evidence on sectoral labor productivity: Implications for industrialization and development," Technical Report, National Bureau of Economic Research 2022.

Hsieh, Chang-Tai and Esteban Rossi-Hansberg, "The industrial revolution in services," *Journal of Political Economy Macroeconomics*, 2023, *1* (1), 3–42.

Huneeus, Federico and Richard Rogerson, "Heterogeneous paths of industrialization," *Review of Economic Studies*, 2024, *91* (3), 1746–1774.

Jedwab, Remi, Elena Ianchovichina, and Federico Haslop, "Consumption cities vs. production cities: New considerations and evidence," *Production Cities: New Considerations and Evidence*, 2022.

Johnson, H. Thomas, "Management Accounting in an Early Integrated Industrial: E. I. duPont de Nemours Powder Company, 1903–1912," *Business History Review*, 1975, *49* (2), 184–204.

Jones, Charles I., "R&D-Based Models of Economic Growth," *Journal of Political Economy*, 1995, *103* (4), 759–784.

Jones, Charles I, "The end of economic growth? Unintended consequences of a declining population," *American Economic Review*, 2022, *112* (11), 3489–3527.

Juhász, Réka, Nathan Lane, and Dani Rodrik, "The new economics of industrial policy," *Annual Review of Economics*, 2024, *16* (1), 213–242.

Kaplan, Robert S., "The Evolution of Management Accounting," *The Accounting Review*, 1984, *59* (3), 390–418.

Klette, Tor Jakob and Samuel Kortum, "Innovating firms and aggregate innovation," *Journal of political economy*, 2004, *112* (5), 986–1018.

Kongsamut, Piyabha, Sergio Rebelo, and Danyang Xie, "Beyond balanced growth," *The Review of Economic Studies*, 2001, *68* (4), 869–882.

Krusell, Per, Lee E Ohanian, José-Víctor Ríos-Rull, and Giovanni L Violante, "Capital-skill complementarity and inequality: A macroeconomic analysis," *Econometrica*, 2000, *68* (5), 1029–1053.

Kuznets, Simon, "Modern economic growth: findings and reflections," *The American economic review*, 1973, *63* (3), 247–258.

Kuznets, Simon Smith, *Economic growth and structure: selected essays*, Yale University Press, 1966.

Lagerlöf, Nils-Petter, "The Galor–Weil model revisited: A quantitative exercise," *Review of Economic dynamics*, 2006, *9* (1), 116–142.

Laincz, Christopher A. and Pietro F. Peretto, "Scale Effects in Endogenous Growth Theory: An Error of Aggregation Not Specification," *Journal of Economic Growth*, 2006, *11* (3), 263–288.

Levey, Stanley and E. R. Jennings, "The Use of Control Charts in the Clinical Laboratory," *American Journal of Clinical Pathology*, 1950, *20* (11), 1059–1066.

Lewis, William Arthur, "Economic development with unlimited supplies of labour," *The Manchester School*, 1954, *22* (2), 139–191.

Matsuyama, Kiminori, "A simple model of sectoral adjustment," *The Review of Economic Studies*, 1992, *59* (2), 375–387.

—, "Structural change in an interdependent world: A global view of manufacturing decline," *Journal of the European Economic Association*, 2009, *7* (2-3), 478–486.

McKinsey, James O., *Budgetary Control*, New York: The Ronald Press Company, 1922.

— , *Managerial Accounting*, Chicago: University of Chicago Press, 1924.

Miroudot, Sébastien and Charles Cadestin, "Services in Global Value Chains: From Inputs to Value-Creating Activities," OECD Trade Policy Papers 197, OECD Publishing 2017.

Ngai, L Rachel and Christopher A Pissarides, "Structural change in a multisector model of growth," *American economic review*, 2007, *97* (1), 429–443.

Peretto, Pietro F., "Technological Change and Population Growth," *Journal of Economic Growth*, 1998, *3* (4), 283–311.

Peters, Michael, Youdan Zhang, and Fabrizio Zilibotti, "Skipping the Factory: Service-Led Growth and Structural Transformation in the Developing World," Technical Report, National Bureau of Economic Research 2026.

Rodrik, Dani, "Premature Deindustrialization," *Journal of Economic Growth*, 2016, *21* (1), 1–33.

— **and Joseph E Stiglitz**, "A new growth strategy for developing nations," in "The New Global Economic Order," Routledge, 2025, pp. 89–107.

— **and Rohan Sandhu**, "Servicing Development: Productive Upgrading of Labor-Absorbing Services in Developing Economies," *Global Policy*, 2025, *16* (2), 306–318.

Romer, Paul M, "Endogenous technological change," *Journal of political Economy*, 1990, *98* (5, Part 2), S71–S102.

Rostow, Walt Whitman, "The stages of economic growth," *The economic history review*, 1959, *12* (1), 1–16.

Shewhart, Walter A., *Economic Control of Quality of Manufactured Product*, New York: D. Van Nostrand Company, 1931.

Sposi, Michael, Kei-Mu Yi, and Jing Zhang, "Trade integration, global value chains, and capital accumulation," *IMF Economic Review*, 2021, *69* (3), 505–539.

Uy, Timothy, Kei-Mu Yi, and Jing Zhang, "Structural change in an open economy," *Journal of Monetary Economics*, 2013, *60* (6), 667–682.

Vickers, Chris and Nicolas L Ziebarth, "The census of manufactures: An overview," *Handbook of Cliometrics*, 2024, pp. 2611–2634.

Westgard, James O., Patricia L. Barry, Marian R. Hunt, and Torgny Groth, "A Multi-Rule Shewhart Chart for Quality Control in Clinical Chemistry," *Clinical Chemistry*, 1981, *27* (3), 493–501.

A Data Sources and Variable Construction

This appendix describes the data sources and variable construction used in the empirical analysis. The paper combines cross-country data on income, sectoral employment, fertility, and education with U.S. county level data on historical manufacturing activity, occupational skill content, geography, patents, education, income, and employment composition. Country-level variables are harmonized at the country-year level. Historical U.S. county level variables are harmonized to 1850 county boundaries unless otherwise stated.

A.A Cross-country data

A.A.1 GDP per capita

Source note. Gapminder requests the citation: "GAPMINDER: GDP per capita v25 – constant PPP$ 2011 based on World Bank, Maddison & IMF." Documentation: https://www.gapminder.org/sources/gdp-per-capita/.

GDP per capita is from Gapminder's GDP per capita series. The variable measures output per person in purchasing-power-parity adjusted international dollars. Gapminder constructs the series to make income levels comparable across countries and over time by adjusting for inflation and cross-country price differences. For recent decades, the series relies primarily on World Bank PPP GDP per capita data. For years before 1990, Gapminder uses historical income estimates from the Maddison Project and aligns these historical series to the modern PPP benchmark. For the most recent years in which World Bank observations are not yet available, Gapminder extends the series using IMF World Economic Outlook forecasts. The historical series is available back to 1800.

In the empirical analysis, I use the log of GDP per capita. Annual log GDP per capita growth is computed as

$$g_{c,t} = \log GDPpc_{c,t} - \log GDPpc_{c,t-1}.$$

For the main cross-country growth fact, the dependent variable is the average annual log GDP per capita growth rate over 2010–2019:

$$\bar{g}_{c,2010-2019} = \frac{1}{9}\sum_{t=2011}^{2019}\left(\log GDPpc_{c,t} - \log GDPpc_{c,t-1}\right).$$

I exclude 2020 from this exercise to avoid the effect of the COVID-19 shock.

A.A.2 Fertility

Source note. Fertility is from Gapminder's "babies per woman" series, corresponding to the total fertility rate. Documentation: https://www.gapminder.org/data/documentation/gd008/.

Fertility is from Gapminder's "babies per woman" series. The variable corresponds to the standard total fertility rate, defined as the average number of children a woman would have given current age-specific fertility rates. The series combines modern demographic data with historical estimates and extends back to 1800. I use the country-year total fertility rate to study the relationship between industrialization and the demographic transition.

For the cross-country fertility regression, the dependent variable is the average fertility rate over 1800–2020:

$$\overline{TFR}_c = \frac{1}{221} \sum_{t=1800}^{2020} TFR_{c,t}.$$

A.A.3 Sectoral employment shares and manufacturing peaks

Source note. Post-1990 sectoral employment shares are from Gapminder variables based on ILO modeled employment estimates. ILOSTAT documentation: https://ilostat.ilo.org/. Historical manufacturing employment shares are supplemented using the GGDC 10-sector database: https://www.rug.nl/ggdc/structuralchange/.

Sectoral employment shares in agriculture, manufacturing, and services are taken from Gapminder/ILO for the post-1990 period and supplemented with the GGDC 10-sector database where available. The three broad sectors used in the paper are agriculture, manufacturing, and services.

The GGDC 10-sector database is used to extend manufacturing employment histories for countries covered by that database. In the GGDC data, the manufacturing employment share is constructed as

$$\text{Manu. Emp. Share}_{c,t} = \frac{EMP^{MAN}_{c,t}}{EMP^{SUM}_{c,t}},$$

where $EMP^{MAN}_{c,t}$ is manufacturing employment and $EMP^{SUM}_{c,t}$ is total employment in the GGDC 10-sector database. When both Gapminder/ILO and GGDC observations are available, I use the Gapminder/ILO series for the post-1990 harmonized employment shares and the GGDC series to fill earlier manufacturing observations.

The key cross-country treatment variable is the peak manufacturing employment share:

$$\text{Manu. Peak}_c = \max_t \text{Manu. Emp. Share}_{c,t}.$$

For countries with GGDC coverage, the peak is computed over the combined GGDC and Gapminder/ILO period. For countries without GGDC coverage, it is computed over the Gapminder/ILO period. The final regression sample contains 174 countries.

A.A.4 Education

Source note. Cross-country education is from the Barro-Lee educational attainment dataset. Data and documentation: https://barrolee.com/. The dataset is described in Barro and Lee (2013).

Cross-country education is from the Barro-Lee educational attainment dataset. I use the tertiary education measure matched to the country-year panel. Barro and Lee provide internationally comparable educational attainment measures for a large panel of countries at five-year intervals from 1950 to 2010 (Barro and Lee, 2013). In the cross-sectional figures, I average the tertiary education measure over 2010–2019 when observations are available.

For the long-difference education exercise, I compute the change in tertiary education between the first and last years in which both education and GDP per capita are observed:

$$\Delta e_c = e_{c,T_c} - e_{c,t_c},$$

and the corresponding change in log GDP per capita:

$$\Delta \log y_c = \log GDPpc_{c,T_c} - \log GDPpc_{c,t_c}.$$

Countries are then grouped by quintiles of Manu. Peak$_c$.

A.A.5 Cross-country controls

The main growth regressions include controls for average GDP per capita, the service employment share, years since the manufacturing peak, region fixed effects, and income-quintile fixed effects. Income quintiles are constructed from the cross-country distribution of GDP per capita. Years since the manufacturing peak are computed as the distance between the outcome period and the year in which country c reaches Manu. Peak$_c$.

A.B Input-output data and service sector composition

Source note. The input-output exercise uses the 2015 OECD Inter-Country Input-Output table. Data and documentation: https://www.oecd.org/sti/ind/inter-country-input-output-tables.htm.

The input-output exercise uses the 2015 OECD Inter-Country Input-Output table. I use the 2015 small ICIO table and aggregate country-industry cells to global industries. Let Z denote the matrix of intermediate flows and Y denote gross output. I construct the technical coefficient matrix

$$A_{ij} = \frac{Z_{ij}}{Y_j},$$

and compute Leontief total requirements using

$$x_j = (I - A)^{-1} f_j,$$

where f_j is a final-demand shock to sector j.

The exercise considers a \$100 final-demand shock to each of three broad using sectors: agriculture, manufacturing, and consumer services. Agriculture includes sectors $A01$, $A02$, and $A03$. Manufacturing includes all sectors whose OECD industry code begins with C. Agriculture and manufacturing shocks are allocated across detailed industries in proportion to gross output.

The market-service universe and high-skill service classification are reported in Table 9. High-skill services are computer and information services, finance and insurance, and professional, scientific, and technical activities. All remaining market services are classified as other services.

Table 9: ICIO service sector classification

Category	OECD ICIO industries
Market services	G, $H49$, $H50$, $H51$, $H52$, $H53$, I, $J58T60$, $J61$, $J62\_63$, K, L, M, N, R, S
High-skill services	$J62\_63$, K, M

Notes: $J62\_63$ denotes computer and information services, K denotes finance and insurance, and M denotes professional, scientific, and technical activities.

In the baseline exercise, consumer services are market-service industries for which final consumption exceeds 50 percent of gross output:

$$\frac{FinalConsumption_s}{GrossOutput_s} > 0.5.$$

Final consumption is measured using household final consumption expenditure, non-profit institutions serving households, and government final consumption columns in the ICIO table. The consumer-service shock is allocated across selected industries using observed final-consumption weights. In the robustness exercise, I instead allocate the consumer-services shock across all market services using observed final-consumption weights.

For each shock, I compute the induced service output and report the share of induced service output accounted for by high-skill services:

$$HSShare_j = \frac{\sum_{s \in HS} x_{j,s}}{\sum_{s \in Services} x_{j,s}}.$$

This is the object reported in the input-output figures. The purpose of the exercise is to compare the skill intensity of the service output induced by final demand in agriculture, manufacturing, and consumer services.

A.C Historical U.S. county data

A.C.1 County harmonization

Source note. Historical county boundary shapefiles are from NHGIS/IPUMS. NHGIS documentation and data access: https://www.nhgis.org/.

All historical county level variables are harmonized to 1850 county boundaries. I construct geographic crosswalks using county shapefiles for each census decade and the 1850 county shapefile. For each source-year county, I compute the area overlap with 1850 counties using an equal-area projection and assign the source county to the 1850 county with the largest overlap. The resulting crosswalk maps decade-specific county identifiers to 1850 county identifiers. For variables observed on 2000 county boundaries, I allocate counts to 1850 counties using area weighted overlays.

A.C.2 Manufacturing peaks

Source note. County manufacturing data come from the Haines historical county dataset distributed by ICPSR, study 02896: https://www.icpsr.umich.edu/web/ICPSR/studies/02896/staff.

County manufacturing data come from historical Census of Manufactures county tabulations compiled in the Haines county dataset. I use the 1880–1940 censuses to construct the manufacturing peak because these are the most reliable county level manufacturing tabulations. The 1910 Census of Manufactures did not provide a complete county tabulation; it tabulated cities instead, so 1910 is omitted from the county manufacturing panel.

The county manufacturing measure is the number of manufacturing workers divided by total population:

$$\text{Manu. Emp. Share}_{c,t} = \frac{\text{Manu. Workers}_{c,t}}{\text{Population}_{c,t}}.$$

The county manufacturing peak is

$$\text{Manu. Peak}_c = \max_{t \in \{1880,1890,1900,1920,1930,1940\}} \text{Manu. Emp. Share}_{c,t}.$$

If a county reaches the maximum in multiple years, I assign the first peak year:

$$T_c^{peak} = \min \left\{ t : \text{Manu. Emp. Share}_{c,t} = \text{Manu. Peak}_c \right\}.$$

A.C.3 Historical census sample and occupational skill content

Source note. Individual-level historical census data are from IPUMS USA full-count U.S. Census files. Data and documentation: `https://usa.ipums.org/usa/`. Occupational education scores use the 1950 IPUMS occupation-based education score variable.

The historical event-study analysis uses full-count U.S. Census data from 1850 to 1940. The baseline sample consists of fathers of children between ages 6 and 18 with non-missing county, industry, and employment information. This sample is used to construct county-year measures of school enrollment, fertility, sectoral employment, and occupational skill content.

I map industries into broad sectors using the IPUMS 1950 industry classification. Manufacturing is defined as industries with codes between 306 and 499. Services are defined as industries with codes between 506 and 946. Agriculture is the omitted primary-sector category.

Individual education is not observed in the pre-1950 full-count censuses. I therefore measure the skill content of occupations using the IPUMS occupation education score from the 1950 Census. For each occupation, this score measures the share of workers in that occupation in 1950 who had completed at least some college. I assign this fixed 1950 occupation score to workers in earlier census years and average it within county-year-sector cells. The main outcome in the county event study is the mean education score of service sector occupations:

$$EDSCOR50_{c,t}^{serv} = \frac{1}{N_{c,t}^{serv}} \sum_{i \in (c,t,serv)} EDSCOR50_{occ(i)}.$$

This variable measures whether a county's service sector is composed of occupations that later appear as high-education occupations in the 1950 Census.

Treatment groups are based on the cross-county distribution of Manu. Peak_c. Event time is measured relative to T_c^{peak}. The event-study specification is estimated using the Borusyak, Jaravel, and Spiess imputation estimator (Borusyak et al., 2024). The implementation uses county and year fixed effects and the default standard-error procedure in `did_imputation`.

A.D Geographic characteristics

Source note. Coal-field locations are from the NHGIS/USGS coal-field shapefile. Physiographic provinces are from the USGS physiographic provinces shapefile. River and lake geographic files are from the GIS files used in the county level construction.

The geographic characteristics used in the IV-style exercises are measured from 1850 county centroids. Distances are measured in kilometers and transformed as $\log(1 + d)$. Distance to coal is the distance from the 1850 county centroid to the nearest coal-field point. Coal-field locations come from the USGS/NHGIS coal-field shapefile, restricted to points within the 1850 county coverage. Distance to the fall line is computed from the boundary between the Piedmont and Atlantic Plain physiographic regions in the USGS physiographic provinces shapefile. I also use distance to the Ohio River and an indicator for whether the county is adjacent to the Great Lakes. These variables are merged into the county panel and interacted with the manufacturing peak decade in the instrument construction described in the main text.

A.E Long-run county outcomes from NHGIS

Source note. Long-run county outcomes are constructed from 2000 Census county files downloaded from NHGIS. The relevant local extract files are `population_nhgis0030_ts_nominal_county.csv`, `nhgis0031_ds151_2000_county.csv`, `nhgis0032_ds153_2000_county.csv`, `nhgis0036_ts_nominal_county.csv`, and `nhgis0041_ds142_2000_county.csv`. NHGIS documentation: https://www.nhgis.org/.

I construct long-run county outcomes using 2000 Census data from NHGIS and harmonize them to 1850 county boundaries using area weighted overlays. Count variables are allocated by the area share of each 2000 county that overlaps an 1850 county. Mean income is first converted to an income sum by multiplying mean income by population, then allocated by area share and divided by allocated population.

College attainment is the fraction of the population age 25 and older with a bachelor's degree or graduate/professional degree. School enrollment is constructed from NHGIS counts of enrolled and not-enrolled school-age individuals. Per-capita income is constructed by allocating total income, defined as mean income times population, to 1850 county boundaries and dividing by allocated population.

Employment shares in agriculture, manufacturing, and services are constructed from 2000 NHGIS industry counts. I use the broad industry-by-sex counts from the 2000 Census. Agriculture combines employment in agriculture-related industries. Manufacturing combines manufacturing employment. Services combine transportation, information, finance, professional/business services, education/health, arts/accommodation, and other service categories.

For the long-run composition outcomes in Table 3, I also use the detailed 2000 NHGIS industry file `nhgis0041_ds142_2000_county.csv`. Table 10 reports the definitions of high-skill services, total services, innovation activities, advanced manufacturing, and legacy routine manufacturing.

Table 10: Long-run industry classifications

Variable	Included activities
High-skill services	Information; finance and insurance; real estate; professional, scientific, and technical services; management of companies and enterprises.
Total services	Information; finance and insurance; real estate; professional, scientific, and technical services; management; administrative support; education; health; arts; accommodation and food; other services.
Innovation activities	Data processing and information services; architectural and engineering services; computer systems design; management, scientific, and technical consulting; scientific research and development; computer and electronic products manufacturing; pharmaceutical manufacturing.
Advanced manufacturing	Computer and electronic products manufacturing; pharmaceutical manufacturing.
Legacy routine manufacturing	Textile mills; textile product mills; apparel; leather; wood; paper; printing; primary metals; fabricated metals; furniture manufacturing.

The high-skill services share is high-skill service employment divided by total service employment. The innovation employment share is innovation employment divided by total employment. The advanced manufacturing share is advanced manufacturing employment divided by total manufacturing employment.

A.F Patents

Source note. Patent data are from PatentsView bulk data files. Data and documentation: https://patentsview.org/download/data-download-tables. The files used are `g_patent.tsv`, `g_ipc_at_issue.tsv`, `g_assignee_disambiguated.tsv`, and `g_location_disambiguated.tsv`.

Patent counts are constructed from PatentsView bulk data. I classify patents into broad technology sectors using IPC codes at issue. I construct the IPC subclass by combining IPC section, class, and subclass. Patents with an IPC subclass beginning with *A*01 are classified as agriculture. Patents with IPC subclasses beginning with *G*06, *G*07, *G*08, or *H*04 are classified as services, corresponding primarily to computing, data processing, control, and communications technologies. All remaining patents are classified as manufacturing. If a patent appears in multiple technology groups, I assign sectors using the priority order agriculture, services, and then manufacturing.

I then merge patents to assignee locations. PatentsView assignee records are linked to disambiguated location records using `location_id`. I keep assignee locations with non-missing geographic information and merge them to the patent table using `patent_id`. Each patent-assignee location contributes one patent count. I then aggregate patent counts by county FIPS, state FIPS, and broad technology sector.

To merge patents to the historical county panel, I use the longitude and latitude attached to the PatentsView assignee location records. Patent-location observations are converted to point geometries and spatially joined to 1850 county polygons. Patent counts are then summed within 1850 counties:

$$\text{Patents}_c^k = \sum_{p \in c} \mathbb{1}\{sector(p) = k\}, \qquad k \in \{A, M, S\}.$$

The main long-run outcome in Table 3 is log patents per capita, constructed using total patent counts and the 2000 population harmonized to 1850 county boundaries.

A.G University openings

Source note. University and college openings are collected from Wikidata and supplemented with information from English Wikipedia pages. Wikidata query service: `https://query.wikidata.org/`.

I collect university and college openings from Wikidata. The query retrieves institutions with a founding year, coordinates, and an English Wikipedia page. I keep institutions with non-missing founding years and locations. Institutions are assigned to counties using their coordinates. For descriptive exercises, I group counties by their historical manufacturing peak and compute the distribution of university openings across peak-manufacturing groups.

I also classify institutions by topic using the first 250 words of their Wikipedia page. Topic categories include clergy, agriculture, liberal arts, commerce, business, law, mechanical arts, medicine, military, teacher training, and other.

B Additional Cross-Country Evidence

This appendix reports additional cross-country evidence supporting the facts documented in Section II. The exercises below are not used in the calibration. They show that the relationship between peak manufacturing employment and development outcomes is robust to alternative specifications and that the input-output mechanism is not driven by the baseline definition of consumer services. Variable definitions and data sources are described in Appendix A.

B.A GDP per capita growth without service employment controls

The baseline growth regressions in Section II control directly for the average service employment share over 2010–2019. This is a demanding specification because the service share is itself an outcome of structural change and is mechanically related to the decline of agriculture and manufacturing. Table 11 reports the same set of regressions after omitting the service employment-share control.

The coefficient on Manu. Peak_c remains positive and is only marginally insignificant at the 10% level when controlling only for income quintile and regional fixed effects. Thus, the positive association between peak manufacturing employment and subsequent GDP per capita growth is not driven by directly conditioning on the size of the service sector. Instead, countries with larger manufacturing peaks grow faster even when the specification compares them without holding fixed the contemporaneous service employment share; that is, countries with a higher peak tend to grow faster even when compared to countries which are at an earlier stage of the development process which, ceteris paribus, would enjoy the advantages of catch up growth and grow faster than countries at the frontier.

Table 11: Dep. Var. is Avg GDPPC growth rate 2010-2019

	(1)	(2)	(3)	(4)
Peak Emp. % Man.	3.334*	5.164**	4.201	5.002*
	(1.83)	(2.10)	(1.54)	(1.71)
N	174	174	174	174
R2	0.0143	0.0495	0.105	0.130
Income Quintile FE	no	yes	yes	yes
Region FE	no	no	yes	yes
Controls	no	no	no	yes

t statistics in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

B.B human capital quality

Section II shows that countries with larger manufacturing peaks accumulate more tertiary education over development. This subsection reports a complementary exercise using human capital quality, measured by test scores. The dependent variable is human capital quality in 2015. The regressions control for average log GDP per capita over 2010–2019 and progressively add income-quintile fixed effects, region fixed effects, and additional controls.

Table 12: Dep. Var. is Human Capital Quality, 2015

	(1)	(2)	(3)	(4)
Avg. log GDP per capita, 2010–2019	0.618***	0.523***	0.571*	0.462
	(5.36)	(2.74)	(1.82)	(1.42)
Peak Emp. % Man.	4.943***	4.866***	3.554***	3.237***
	(3.97)	(4.05)	(2.91)	(2.66)
N	79	79	79	79
R2	0.552	0.559	0.697	0.702
Income Quintile FE	no	no	yes	yes
Region FE	no	no	yes	yes
Controls	no	no	no	yes

t statistics in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Across specifications, Manu. Peak$_c$ is positively associated with human capital quality. The relationship remains statistically significant after controlling for income levels, income-quintile fixed effects, regional fixed effects, and additional controls. This suggests that the link between manufacturing and human capital is not only about years of schooling or tertiary enrollment. Countries with larger manufacturing peaks also tend to have higher measured education quality.

B.C Input-output robustness: alternative consumer-service demand

The baseline input-output exercise in Section II defines consumer services as market-service industries for which final consumption exceeds 50 percent of gross output. This subsection reports a robustness exercise using a broader definition. Instead of selecting only consumer-oriented service industries, I allocate the consumer-services final-demand shock across all market services using observed final-consumption weights. This allows final consumers to purchase services that are also heavily used as intermediate inputs, such as finance, computer and information services, and professional services.

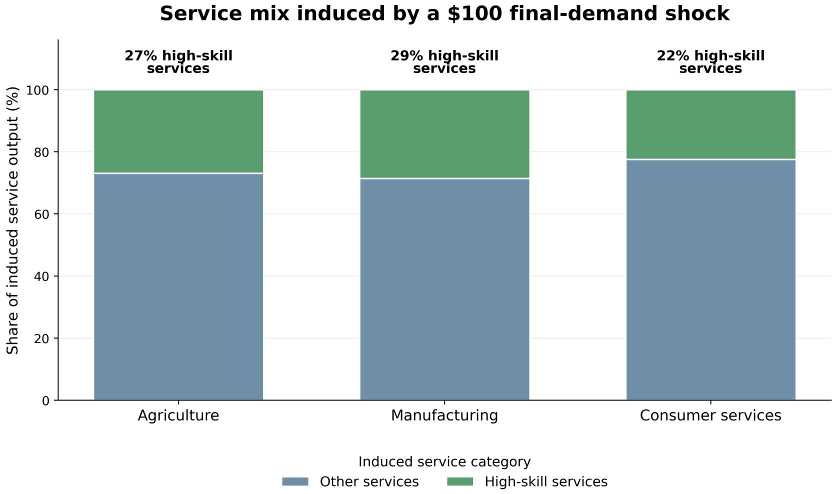


Figure 15: Robustness of the input-output service mix exercise. Bars show the service sector mix induced by a $100 final-demand increase in agriculture, manufacturing, or consumer services, computed using Leontief total requirements from the 2015 OECD ICIO table. In this robustness exercise, the consumer-services shock is allocated across all market services using observed final-consumption weights.

The qualitative pattern is unchanged. Manufacturing induces a more high-skill intensive service mix than consumer-service demand, even when consumer-service demand is allowed to include high-skill market services. This supports the interpretation that manufacturing matters not only through direct manufacturing employment, but also through its intermediate-input linkages to high-skill producer services.

B.D Input-output robustness by country income group

Figure 16 repeats the input-output service mix exercise by country income group. This checks whether the baseline pattern is driven by a small set of high-income economies or whether manufacturing intensive demand is associated with a more high-skill service mix more broadly.

The manufacturing shock remains relatively intensive in high-skill services across income groups. The magnitude varies across groups, but the broad ordering is consistent with the mechanism emphasized in the paper: manufacturing demand pulls on producer-service activities that are more skill-intensive than the services generated by consumer-facing final demand.

C Additional U.S. County Evidence

This appendix provides additional evidence for the U.S. county analysis in Section III. I first document the timing and distribution of historical manufacturing peaks across counties. I then report a robustness exercise that splits counties into finer bins of the manufacturing peak distribution. I next decompose the service sector education-score result into the occupations that

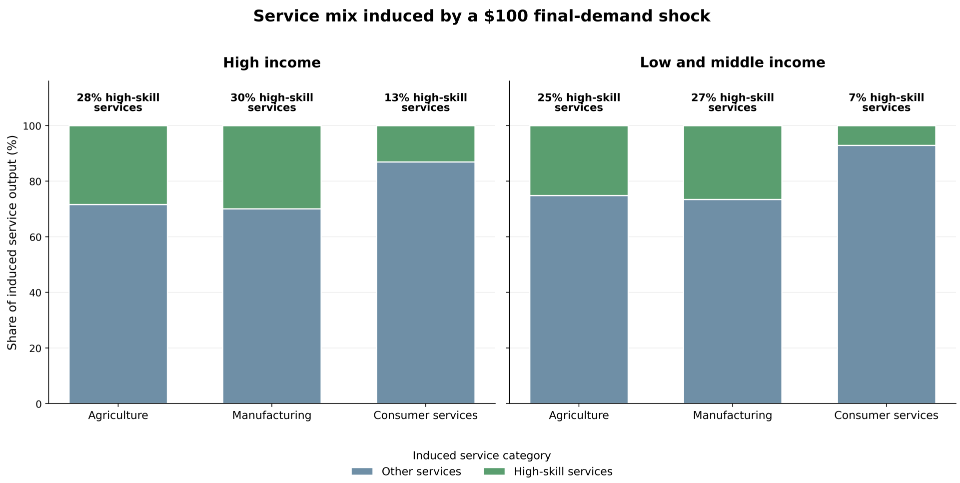


Figure 16: Input-output service mix exercise by income group. Bars show the service sector mix induced by a $100 final-demand increase in agriculture, manufacturing, or consumer services, computed from Leontief total requirements using the 2015 OECD ICIO table. Consumer services are defined as service industries for which final consumption exceeds 50 percent of gross output.

drive it. Finally, I provide complementary evidence from university openings and historical help-wanted advertisements.

C.A Timing and size of county manufacturing peaks

Figure 17 summarizes the timing and size of county manufacturing peaks. For each county, I define Manu. Peak$_c$ as the maximum manufacturing employment share observed in the Census of Manufactures county tabulations over 1880, 1890, 1900, 1920, 1930, and 1940. The 1910 Census of Manufactures is omitted because it does not provide complete county level manufacturing labor tabulations. The figure reports, by peak decade, the average peak manufacturing employment share among counties peaking in that decade and the number of counties reaching their peak in that decade.

The distribution shows substantial variation in the timing of industrialization across counties. Some counties reach their manufacturing peak during the late nineteenth century, while others peak only in the interwar period. This variation is useful for the event-study design because treatment timing differs across counties while outcomes are observed in common census decades.

C.B Panel evidence during the U.S. transition

Before turning to the event-study design, I examine whether manufacturing intensive counties experience different changes in fertility, schooling, and the skill content of services during the U.S. transition from a high fertility/agrarian/low schooling economy to the highly educated and industrialized country it was by the mid 20$^{\text{th}}$ century. This exercise uses the full-count Census sample from 1850 to 1940 and estimates panel regressions of the form

$$\Delta y_{ct} = \alpha + \beta \, \text{Manu. Emp. Share}_{c,t-1} + \Gamma' X_{c,t-1} + \varepsilon_{ct},$$

where y_{ct} is fertility, school enrollment, or the mean education score of service occupations in county c and decade t. The dependent variables are measured in first differences, and the

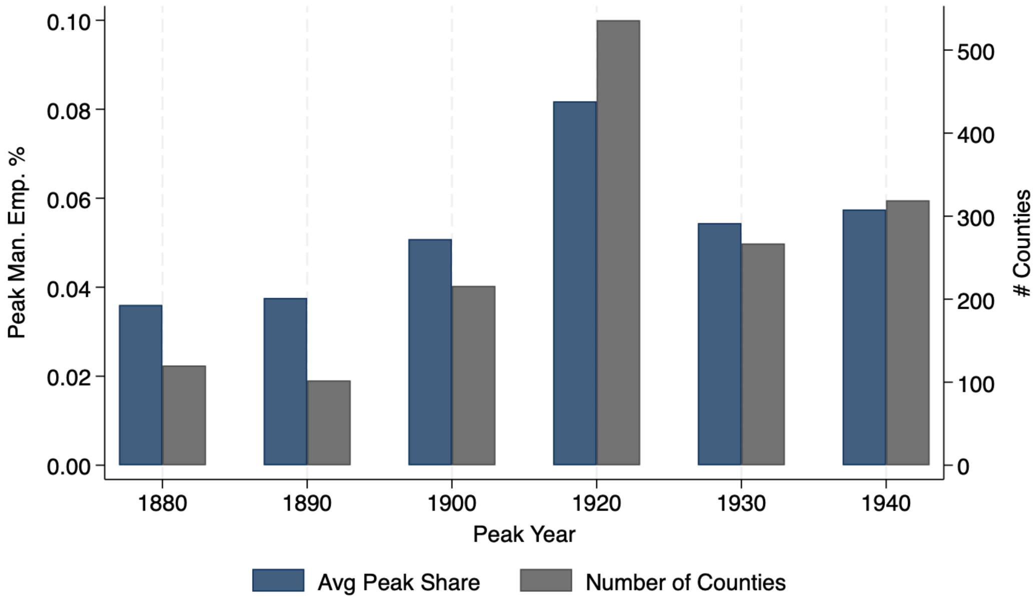


Figure 17: Timing and size of county manufacturing peaks. The figure reports the average value of Manu. Peak$_c$ among counties peaking in each decade and the number of counties reaching their manufacturing peak in that decade. County manufacturing peaks are constructed from Census of Manufactures county tabulations over 1880, 1890, 1900, 1920, 1930, and 1940.

main regressor is the lagged county manufacturing employment share. The controls include the urbanization rate, log population, and the agricultural employment share. The coefficient β therefore asks whether, conditional on broad measures of county development, more manufacturing intensive counties subsequently experience larger or smaller changes in the outcome. In the more demanding specifications, I also include state-by-year fixed effects to absorb state-level shocks and policy changes, including changes in compulsory schooling laws.

Table 13: Panel Data (1850–1940)

	(1) Δ Fertility	(2) Δ Enrollment	(3) Δ Ed. Score	(4) Δ Fertility	(5) Δ Enrollment	(6) Δ Ed. Score
L.% Man Emp.	-0.282***	0.109***	6.267***	-0.312***	0.086***	2.181***
	(0.067)	(0.035)	(0.504)	(0.064)	(0.025)	(0.567)
State × Year FE	No	No	No	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	12758	12758	12748	12758	12758	12748
Within R2	0.022	0.060	0.075	0.522	0.546	0.568

Standard errors are clustered at the county level. Dependent variables are measured in first differences. Data source: US Census of Population 1850-1940.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 13 reports the results. Counties with higher lagged manufacturing employment shares subsequently experience larger fertility declines, larger increases in school enrollment, and larger increases in the education score of service occupations. The signs are stable across specifications with and without state-by-year fixed effects. The magnitudes decline once state-by-year fixed effects are included, but the relationships remain statistically significant. This evidence is descriptive, but it shows that the three outcomes emphasized in the paper—fertility decline, schooling, and the skill content of services—move together with manufacturing intensity within U.S. counties during the transition.

C.C Alternative peak-manufacturing bins

The baseline event study in Figure 5 compares mutually exclusive groups of counties in the tails of the manufacturing peak distribution. Figure 18 reports a finer grouping. Counties are divided into seven mutually exclusive bins: the bottom decile, the 10th–20th percentiles, the 20th–30th percentiles, the middle 30th–70th percentiles, the 70th–80th percentiles, the 80th–90th percentiles, and the top decile of the Manu. Peak_c distribution.

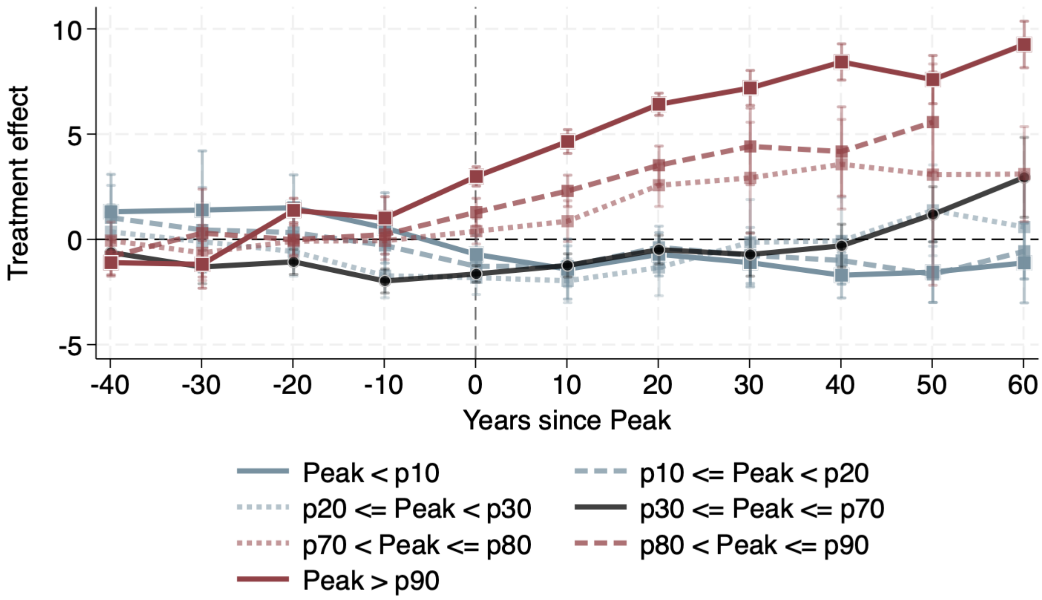


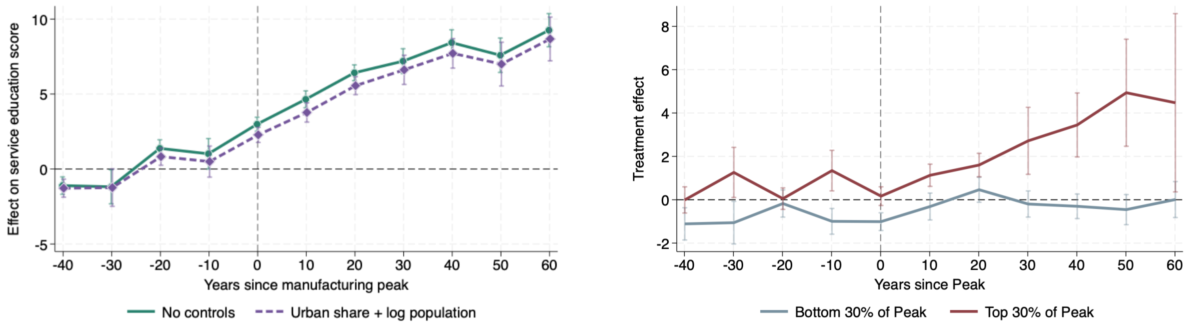
Figure 18: Event-study estimates by mutually exclusive peak-manufacturing bins. The outcome is the mean 1950 occupation education score among service sector workers in county c and decade t. Counties are grouped by their position in the cross-county distribution of Manu. Peak_c. Event time is measured in years relative to the county manufacturing peak. Estimates are obtained using the Borusyak, Jaravel, and Spiess imputation estimator with county and census-decade fixed effects.

The results confirm that the baseline pattern is concentrated in the upper tail of the manufacturing peak distribution. Counties with the largest manufacturing peaks experience the strongest post-peak increase in the educational content of service occupations. Counties in the middle of the distribution display much smaller changes, while counties in the lower tail do not exhibit a comparable increase. This pattern supports the interpretation that the intensity of the manufacturing phase matters for the subsequent composition of the service economy.

C.D Robustness to Urbanization, Market Size, and IV Specification

A natural concern is that the event-study estimates reflect the growth of larger and more urban counties rather than the legacy of manufacturing itself. manufacturing intensive counties may have become urban centers, and urban counties may have developed more educated service sectors for reasons unrelated to the organizational capabilities emphasized in the paper.

A second concern is that the peak-manufacturing measure may be endogenous to local service sector trajectories. Figure 19 addresses these concerns by reporting two complementary robustness exercises.



(a) Urbanization and market-size controls

(b) IV event-study specification

Figure 19: Robustness of service sector event-study estimates. Panel (a) compares the top-decile manufacturing peak event study with and without controls for contemporaneous urban share and log county population. Panel (b) reports the IV event-study specification. The outcome is the mean 1950 occupation education score among service sector workers in county c and decade t. Estimates are obtained using the Borusyak, Jaravel, and Spiess imputation estimator with county and census-decade fixed effects.

Panel (a) compares the baseline event-study path for counties in the top decile of the manufacturing peak distribution with an otherwise identical specification that controls for contemporaneous urban share and log county population. Urban share is measured as the population living in places above 25,000 residents divided by total county population. Panel (b) reports the corresponding IV event-study specification. Across both exercises, the dynamic pattern remains similar to the baseline estimates, suggesting that the rise in service sector education scores is not explained by differential urbanization, county market size, or the direct endogeneity of the peak-manufacturing measure alone. Because urbanization and population growth may themselves be consequences of industrial development, the controls in Panel (a) are interpreted as a robustness check rather than as the preferred estimate of the total effect.

C.E Which occupations drive the service sector education-score result?

The event-study outcome $EDSCOR50^{serv}_{c,t}$ increases when service employment shifts toward occupations that had higher college intensity in the 1950 Census. Figure 20 decomposes this change by occupation. I compare counties in the top and bottom deciles of the Manu. Peak$_c$ distribution and compute, for each occupation, the difference-in-differences in employment-share changes between 1850 and 1940. The dark blue bars report this occupation-level difference-in-differences in percentage points. The light blue bars report the occupation's 1950 education score.

The occupations with the largest positive differential growth in high-peak manufacturing counties include physicians and surgeons, clerical workers, stenographers and secretaries, lawyers and judges, teachers, accountants, office-machine operators, and other professional or HSPS occupations. These occupations are not manufacturing production jobs, but many are organizationally complementary to manufacturing expansion: they involve accounting, administration, legal services, communication, technical support, record keeping, and coordination. By

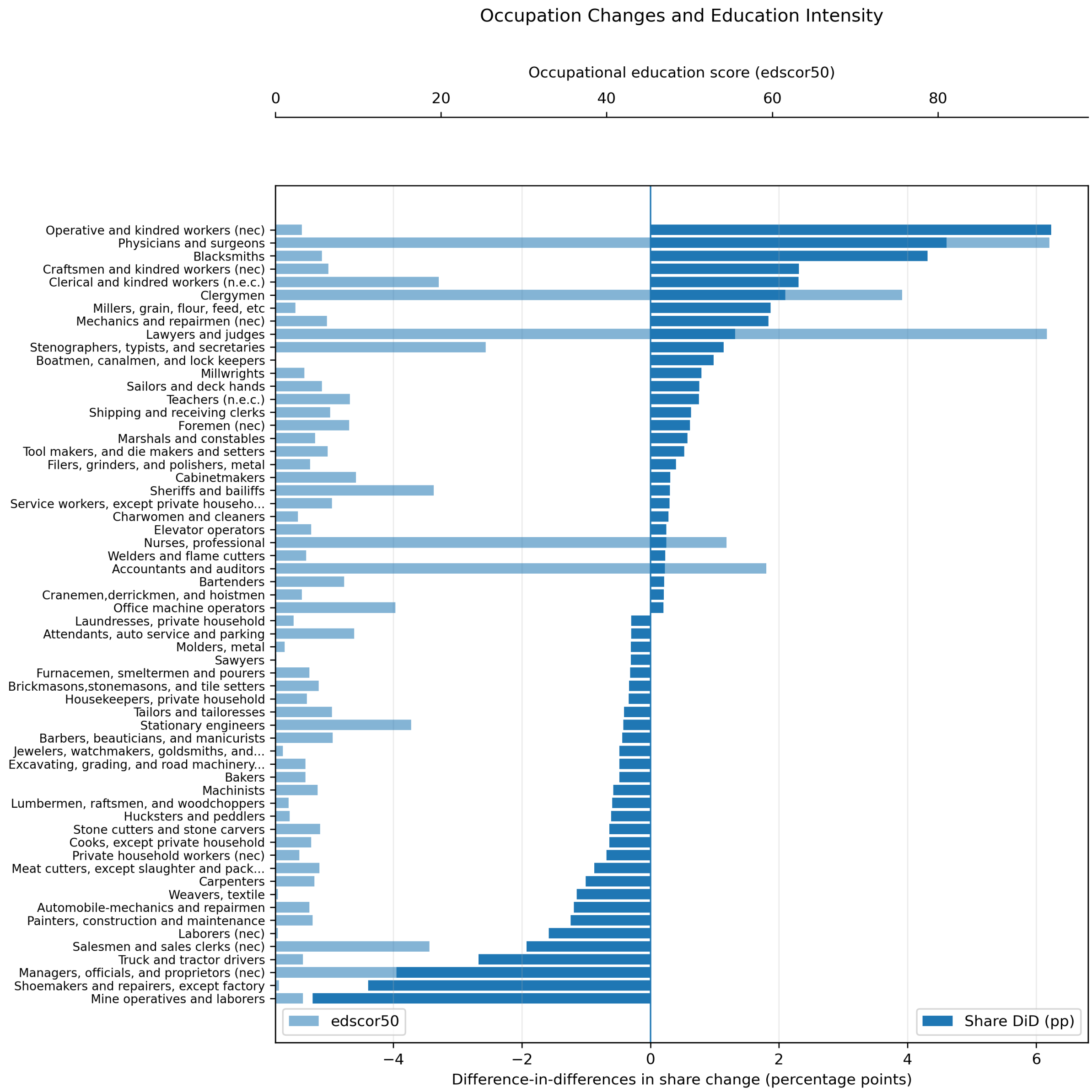


Figure 20: Occupation-level decomposition of changes in service sector education intensity. Blue bars report the difference-in-differences in occupation employment-share changes between high- and low-peak manufacturing counties from 1850 to 1940. Orange bars report the 1950 occupation education score, defined as the share of workers in the occupation with at least one year of college. Farm occupations are excluded.

contrast, occupations with negative differential growth include laborers, salesmen, shoemakers, operatives, and other lower-education manual or routine occupations. The decomposition therefore suggests that historical manufacturing peaks are associated with a later shift in services toward the kinds of professional and organizational occupations emphasized by the mechanism.

C.F Service industries and occupations by education intensity

Tables 14 and 15 provide additional detail on the service industries and occupations with low and high educational content in the 1950 Census. These tables help interpret the education-score outcome by showing the types of activities that receive high and low *EDSCOR*50 values.

Table 14: Service sub-industries by college intensity in 1950

Low college intensity		**High college intensity**	
Sub-industry	%	Sub-industry	%
Private households	3.5	Educational services	75.8
Shoe repair	4.3	Legal services	73.5
Trucking	5.3	Medical, excl. hospitals	71.2
Taxicabs	5.7	Welfare and religious services	59.1
Dressmaking	6.1	Engineering and architecture	58.4
Rail and bus transit	6.4	Misc. professional services	53.9
Misc. repair services	7.1	Drug stores	45.7
Sanitary services	7.2	Radio and television	43.0
Auto repair	8.0	Accounting and auditing	42.7
Water transport	8.1	Hospitals	38.5

Notes: Entries report the fraction of workers in each service sub-industry with at least one year of college in the 1950 Census.

Table 15: Service occupations by college intensity in 1950

Low college intensity		**High college intensity**	
Occupation	%	Occupation	%
Teamsters	0.0	Mathematicians	100.0
Boatmen, canalmen, lock keepers	0.0	Biological scientists	96.0
Bootblacks	0.0	Chiropractors	93.9
Telegraph messengers	0.0	Professors and instructors, n.e.c.	93.8
Midwives	0.0	Professors: engineering	93.8
Hucksters and peddlers	1.7	Professors: chemistry	93.8
Longshoremen and stevedores	2.1	Professors: biological sciences	93.8
Laundresses, private household	2.2	Professors: mathematics	93.8
Charwomen and cleaners	2.7	Professors: agricultural sciences	93.8
Private household workers, n.e.c.	2.9	Professors: physics	93.8

Notes: Entries report the fraction of workers in each occupation with at least one year of college in the 1950 Census. manufacturing heavy occupations are excluded from the low-college list.

The high-education service activities are concentrated in legal, medical, educational, accounting, engineering, architectural, professional, and technical occupations. This is consistent with the mechanism in the paper: manufacturing exposure predicts the later emergence of service activities that are complementary to organization, coordination, technical knowledge, and professional expertise.

C.G University openings

I next examine whether counties with larger historical manufacturing peaks were also more likely to develop local higher-education institutions. University and college openings are collected from Wikidata, assigned to counties using institution coordinates, and grouped by founding year. Figure 21 plots the fraction of universities founded between 1860 and 2000 across counties grouped into quartiles of historical peak manufacturing intensity.

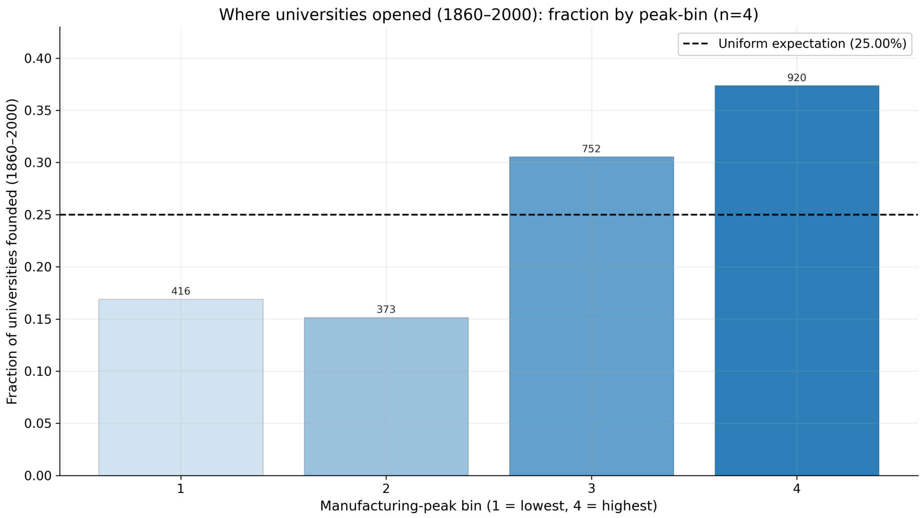


Figure 21: University openings and historical manufacturing exposure. The figure reports the fraction of universities founded between 1860 and 2000 across counties grouped into quartiles of Manu. Peak$_c$. University openings are collected from Wikidata and assigned to counties using institution coordinates.

Counties with larger manufacturing peaks account for a disproportionate share of university openings. This evidence is descriptive, but it is consistent with the idea that industrialization is associated with the local development of institutions that support skill formation and human capital accumulation.

C.H Help-wanted advertisements and demand for skilled labor

I also use historical help-wanted advertisements to study whether manufacturing expansion is associated with demand for skilled labor. The data are constructed from the Library of Congress *Chronicling America* newspaper repository. The underlying corpus begins from pages containing a proximity match for "help wanted." Newspapers are geolocated and grouped into geographic regions. Region-year outcomes are constructed from occupation keyword counts.

I distinguish between skilled manufacturing occupations and skilled service or HSPS occupations. Skilled manufacturing keywords include occupations such as machinist, millwright, and boilermaker. Skilled service and HSPS keywords include occupations such as clerk, bookkeeper, and stenographer. To normalize for changes in newspaper coverage and general labor-market advertising intensity, I compare these keyword counts to a baseline basket of common lower-skill occupations: cook, waiter, janitor, and laborer.

I estimate an event study around the appearance of new or expanding factory mentions in local newspapers. A region is treated in the first year in which factory-expansion mentions exceed the threshold used in the text-classification procedure. Figure 22 reports the long-run event-study estimates. The dependent variable is the log of occupation keyword hits normalized by the baseline basket of lower-skill occupations.

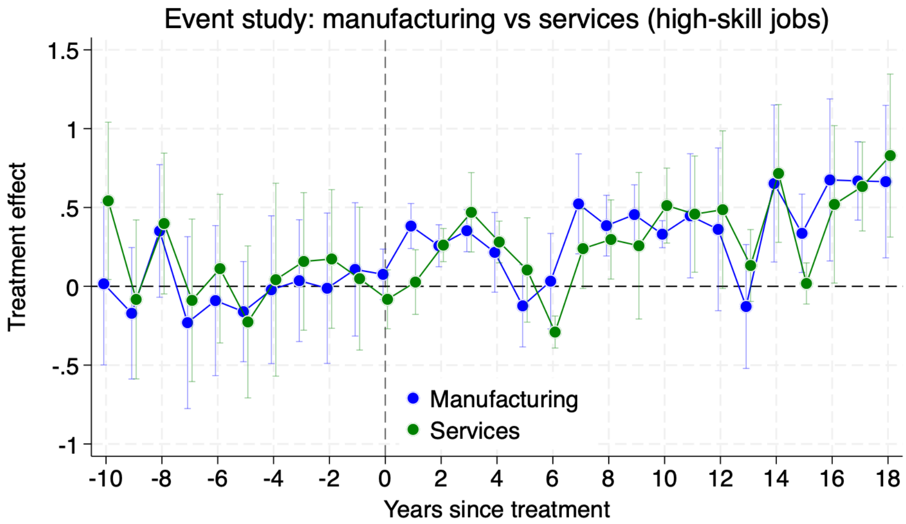


Figure 22: Long-run event-study effect of new factory mentions on demand for skilled labor. The dependent variable is the log of occupation keyword hits normalized by a baseline basket of lower-skill occupations. Estimates are obtained using the Borusyak, Jaravel, and Spiess imputation estimator on region-year newspaper data. Vertical bars report 95 percent confidence intervals clustered by region. Data source: author's calculations using Library of Congress *Chronicling America*.

The estimates show that new factory mentions are followed by an increase in advertisements for skilled labor. Importantly, the increase is not limited to skilled manufacturing occupations. Skilled service and HSPS occupations also rise, consistent with the idea that manufacturing generates demand for organizational and administrative tasks. The help-wanted evidence therefore supports the occupational mechanism emphasized in the paper: manufacturing expansion is associated not only with greater demand for production-related skills, but also with greater demand for the skilled service and HSPS occupations that help organize production.

D Model Details, Equilibrium, and Asymptotic Results

This appendix supplies the formal material used in Section IV. Section D.A defines a competitive equilibrium. Section D.B derives the fertility and education rules. Section D.C characterizes the relation between final expenditure, input–output production, and employment, and proves Propositions 2 and 3.

D.A Competitive equilibrium

The household-origin index is $i \in \{a, m, s\}$, where s includes consumer- and producer-service workers. The final-good sector index is $j \in \{a, m, c\}$, where c denotes consumer services. These two indices must be kept distinct: a producer-service worker serving manufacturing is attributed to manufacturing in $\ell_{m,t}$, but belongs to a service-origin household in $\lambda_{s,t}$.

The initial state

$$\Omega_0 = \left(L_0, S_0, \bar{e}_0, H_0, g_0, \{A_{j,0}\}_{j\in\{a,m,c\}}\right)$$

is given. The sequences of gross-output productivity $\{A_{go,j,t}\}_{j,t}$, producer-service productivities $\{A_{hs,t}, A_{ls,t}\}_t$, and gross-output cost shares $\{\theta_{j,t}, \gamma_{a,j,t}, \gamma_{m,j,t}, \gamma_{ps,j,t}\}_{j,t}$ are exogenous primitives known to households and firms. They are strictly positive where used and satisfy

$$\theta_{j,t} + \gamma_{a,j,t} + \gamma_{m,j,t} + \gamma_{ps,j,t} = 1. \tag{55}$$

In the quantitative implementation, the common gross-output productivity growth parameter g_{GO} generates

$$A_{go,j,t+1} = (1 + g_{GO})A_{go,j,t}, \qquad j \in \{a, m, c\}, \tag{56}$$

Producer-service productivity grows with consumer-service value-added productivity:

$$\frac{A_{hs,t+1}}{A_{hs,t}} = \frac{A_{ls,t+1}}{A_{ls,t}} = \frac{A_{c,t+1}}{A_{c,t}}. \tag{57}$$

The quantitative specification of the search-productivity map is the bounded Hill function

$$\phi(H) = \frac{H^{\eta_H}}{H^{\eta_H} + H_{1/2}^{\eta_H}}, \qquad \eta_H > 0, \quad H_{1/2} > 0. \tag{58}$$

Thus $\phi(0) = 0$, $\phi'(H) > 0$ for $H > 0$, and $\lim_{H\to\infty} \phi(H) = 1$. The normalization of $H_{1/2}$ is described in the calibration appendix.

The input shares follow the interpolation described in the calibration appendix. If different laws are used in the numerical code, equation (56) must be replaced by that law.

For each date t, a competitive equilibrium consists of household choices

$$\{C_{ij,t}, C_{i,t}, n_{i,t}, e_{i,t}\}_{i,j},$$

final-good producer choices

$$\{Y_{j,t}, VA_{j,t}, L_{s,j,t}, L_{u,j,t}, M_{a,j,t}, M_{m,j,t}, PS_{j,t}\}_j,$$

producer-service task choices and labor allocations

$$\{y_{j,t}(z), HS^d_{j,t}, LS^d_{j,t}, L_{s,hs,j,t}, L_{u,ls,j,t}\}_{j,z},$$

prices

$$\{p_{a,t}, p_{m,t}, p_{c,t}, w_{s,t}, w_{u,t}, p_{hs,t}, p_{ls,t}, P_{ps,j,t}\}_j,$$

and aggregate variables

$$\{L_t, S_t, \bar{e}_t, H_t, g_t, HS_t, LS_t, \lambda_{i,t}, \ell_{j,t}\}_{i,j},$$

such that the following conditions hold.

Households and conditional final demand. The closed-form demographic block uses the separability convention stated in Assumption D.B: conditional final-good demand is determined by the non-homothetic system (25), while fertility and education are chosen using the expenditure-equivalent consumption composite $C_{i,t}$. Given this convention, households maximize (24) subject to (28), and their choices satisfy (30)–(31). The derivation is provided in Section D.B.

Aggregate final consumption is

$$C_{j,t} = L_t \sum_{i \in \{a,m,s\}} \lambda_{i,t} C_{ij,t}, \qquad j \in \{a, m, c\}. \tag{59}$$

The pooled wage entering the demographic block is $w_t(S_t) = (1 - S_t)w_{u,t} + S_t w_{s,t}$. Under perfect foresight, the next cohort's skilled share satisfies

$$S_{t+1} = p_{s,t} = \frac{1}{1 + \exp\left[-\{\beta_\pi \log(w_{s,t+1}/w_{u,t+1}) - c_0\}/\sigma_{\text{skill}}\right]}. \tag{60}$$

Final-good producers. For every $j \in \{a, m, c\}$, final-good producers minimize the cost of producing $Y_{j,t}$ subject to (34) and (35). Constant returns and free entry imply zero profits. Manufacturing is the numeraire:

$$p_{m,t} = 1. \tag{61}$$

The value-added and gross-output unit-cost functions satisfy

$$p_{j,t}^{VA} = \mathcal{C}_j^{VA}\left(w_{s,t}, w_{u,t}; A_{j,t}, \alpha_j, h_s, h_u, \sigma_L\right), \tag{62}$$

$$p_{j,t} = \mathcal{C}_j^{GO}\left(p_{j,t}^{VA}, p_{a,t}, p_{m,t}, P_{ps,j,t}; A_{go,j,t}, \theta_{j,t}, \gamma_{a,j,t}, \gamma_{m,j,t}, \gamma_{ps,j,t}\right). \tag{63}$$

These conditions, together with (61), determine relative final-good prices whenever the associated input–output matrix is productive.

Producer-service supply and task demand. Formal and informal producer-service efficiency units satisfy

$$HS_t = A_{hs,t} h_s L_{s,hs,t}, \qquad LS_t = A_{ls,t} h_u L_{u,ls,t}, \tag{64}$$

where

$$L_{s,hs,t} = \sum_j L_{s,hs,j,t}, \qquad L_{u,ls,t} = \sum_j L_{u,ls,j,t}.$$

Competitive efficiency-unit prices are

$$p_{hs,t} = \frac{w_{s,t}}{A_{hs,t} h_s}, \qquad p_{ls,t} = \frac{w_{u,t}}{A_{ls,t} h_u}. \tag{65}$$

The unit price of task z is

$$p_t^{PS}(z) = \min\left\{p_{ls,t}, p_{hs,t} q_\alpha(z, H_t)\right\}. \tag{66}$$

The sector-specific producer-service price index is

$$P_{ps,j,t} = \left[\int_0^\infty p_t^{PS}(z)^{1-\sigma_{ps}}\, dF_j(z)\right]^{1/(1-\sigma_{ps})}. \tag{67}$$

Cost minimization gives

$$y_{j,t}(z) = PS_{j,t}\left(\frac{p_t^{PS}(z)}{P_{ps,j,t}}\right)^{-\sigma_{ps}}. \tag{68}$$

Tasks are assigned to the least-cost mode. The common cutoff is z_t^* from (44); it is not indexed by j. Define

$$I_{j,t}^{HS} \equiv \int_0^{z_t^*} q_\alpha(z, H_t)^{1-\sigma_{ps}}\, dF_j(z). \tag{69}$$

Sector j's demands for formal and informal producer-service efficiency units are

$$\begin{aligned}
HS_{j,t}^d &= \int_0^{z_t^*} q_\alpha(z, H_t) y_{j,t}(z)\, dF_j(z) \\
&= PS_{j,t} P_{ps,j,t}^{\sigma_{ps}} p_{hs,t}^{-\sigma_{ps}} I_{j,t}^{HS}, \qquad (70) \\
LS_{j,t}^d &= \int_{z_t^*}^\infty y_{j,t}(z)\, dF_j(z) \\
&= PS_{j,t} P_{ps,j,t}^{\sigma_{ps}} p_{ls,t}^{-\sigma_{ps}} \left[1 - F_j(z_t^*)\right]. \qquad (71)
\end{aligned}$$

Under the homogeneous method process, $1 - F_j(z_t^*) = \exp(-z_t^*/\kappa_j)$. Producer-service markets clear when

$$\sum_{j\in\{a,m,c\}} HS^d_{j,t} = HS_t, \qquad \sum_{j\in\{a,m,c\}} LS^d_{j,t} = LS_t. \tag{72}$$

The corresponding worker allocations are

$$L_{s,hs,j,t} = \frac{HS^d_{j,t}}{A_{hs,t}h_s}, \qquad L_{u,ls,j,t} = \frac{LS^d_{j,t}}{A_{ls,t}h_u}. \tag{73}$$

Labor-market clearing and employment shares. The skilled and unskilled labor markets clear when

$$S_tL_t = \sum_{j\in\{a,m,c\}} L_{s,j,t} + \sum_{j\in\{a,m,c\}} L_{s,hs,j,t}, \tag{74}$$

$$(1-S_t)L_t = \sum_{j\in\{a,m,c\}} L_{u,j,t} + \sum_{j\in\{a,m,c\}} L_{u,ls,j,t}. \tag{75}$$

Employment attributed to final-sector activity j is

$$L^{\text{attr}}_{j,t} \equiv L_{s,j,t} + L_{u,j,t} + L_{s,hs,j,t} + L_{u,ls,j,t}, \tag{76}$$

and

$$\ell_{j,t} = \frac{L^{\text{attr}}_{j,t}}{L_t}, \qquad \sum_{j\in\{a,m,c\}} \ell_{j,t} = 1. \tag{77}$$

Household-origin shares are

$$\lambda_{a,t} = \frac{L_{s,a,t} + L_{u,a,t}}{L_t}, \qquad \lambda_{m,t} = \frac{L_{s,m,t} + L_{u,m,t}}{L_t},$$
$$\lambda_{s,t} = 1 - \lambda_{a,t} - \lambda_{m,t}. \tag{78}$$

Thus producer-service workers belong to service-origin households even though their employment is attributed in $\ell_{j,t}$ to the final sector they serve.

Goods-market clearing. Agricultural and manufacturing output is used for final consumption and as an intermediate input:

$$Y_{a,t} = C_{a,t} + \sum_{j\in\{a,m,c\}} M_{a,j,t}, \tag{79}$$

$$Y_{m,t} = C_{m,t} + \sum_{j\in\{a,m,c\}} M_{m,j,t}. \tag{80}$$

Consumer services have no intermediate use, so

$$Y_{c,t} = C_{c,t}. \tag{81}$$

Aggregate consistency and dynamics. Population and cohort-average education satisfy

$$L_{t+1} = n_t L_t, \qquad n_t = \sum_{i\in\{a,m,s\}} \lambda_{i,t} n_{i,t}, \tag{82}$$

$$\bar{e}_{t+1} = \frac{\sum_{i\in\{a,m,s\}} \lambda_{i,t} n_{i,t} e_{i,t}}{n_t}. \tag{83}$$

The skilled share satisfies (60). Sectoral value-added productivity, aggregate productivity growth, and organizational knowledge satisfy (51), (52), and (20). Initial conditions are given, and the perfect-foresight price sequence is consistent with current market clearing and future skill choice.

D.B Household fertility and education

The non-homothetic final-demand system and the closed-form demographic rules must be connected by an explicit separability convention.

[Demographic separability] Conditional final-good demands $\{C_{ij,t}\}_j$ are allocated according to (25). For the fertility–education choice, $C_{i,t}$ is measured in expenditure-equivalent units with price normalized to one. Consequently, the resources available for this composite are

$$C_{i,t} = [1 - n_{i,t}\{\tau_i + \tau_e(S_t)e_{i,t}\}]\, w_t(S_t). \tag{84}$$

The conditional non-homothetic allocation affects the composition of final demand but not the marginal demographic tradeoff.

Without Assumption D.B, the expenditure function associated with (25) is nonlinear in $C_{i,t}$, and the closed-form rules below do not follow from (24) and (25) alone.

Let

$$\chi_t \equiv \tau_e(S_t), \qquad a_{i,t}(e) \equiv \tau_i + \chi_t e.$$

Using (84), household utility can be written, up to terms independent of (n, e), as

$$(1-\gamma)\log[1 - n a_{i,t}(e)] + \gamma \log n + \gamma \log\left(\frac{e+\tau_h}{e+\tau_h+g_t}\right). \tag{85}$$

[Fertility and education rules] Suppose Assumption D.B holds. For any interior education choice, optimal fertility is

$$n_{i,t} = \frac{\gamma}{\tau_i + \chi_t e_{i,t}}. \tag{86}$$

Optimal education is

$$e_{i,t} = \max\left\{\sqrt{\frac{g_t(\tau_i - \chi_t\tau_h)}{\chi_t}} - \tau_h, 0\right\}. \tag{87}$$

An interior positive choice requires $\tau_i > \chi_t \tau_h$ and a value of the square-root expression greater than τ_h.

Proof. For a fixed education choice, the first-order condition for fertility from (85) is

$$-\frac{(1-\gamma)a_{i,t}(e)}{1-na_{i,t}(e)} + \frac{\gamma}{n} = 0.$$

Solving gives $na_{i,t}(e) = \gamma$, which proves (86). At this solution, disposable time is $1-\gamma$, so the education choice maximizes

$$\log\left(\frac{e+\tau_h}{e+\tau_h+g_t}\right) - \log(\tau_i + \chi_t e).$$

For an interior choice, the first-order condition is

$$\frac{g_t}{(e+\tau_h)(e+\tau_h+g_t)} = \frac{\chi_t}{\tau_i + \chi_t e}.$$

After canceling the common term $g_t \chi_t e$, this becomes

$$(e+\tau_h)^2 = \frac{g_t(\tau_i - \chi_t \tau_h)}{\chi_t}.$$

The nonnegativity constraint gives (87). □

D.C Limiting structure and proofs

This subsection first separates three distinct limiting objects: final expenditure, gross output, and employment. It then proves the terminal aggregate result and the manufacturing-versus-services comparison.

D.C.1 Final expenditure and the production network

Let

$$E_{j,t} \equiv p_{j,t} C_{j,t}$$

denote final expenditure on sector $j \in \{a, m, c\}$. The non-homothetic CES demand system implies

$$\frac{E_{j,t}}{E_{k,t}} = \frac{\omega_j}{\omega_k}\left(\frac{p_{j,t}}{p_{k,t}}\right)^{1-\sigma} C_t^{(1-\sigma)(\varepsilon_j - \varepsilon_k)}. \tag{88}$$

[Long-run service demand] Aggregate real consumption satisfies $C_t \to \infty$, and, for $j \in \{a, m\}$,

$$\left(\frac{p_{j,t}}{p_{c,t}}\right)^{1-\sigma} C_t^{(1-\sigma)(\varepsilon_j - \varepsilon_c)} \longrightarrow 0. \tag{89}$$

[Concentration of final expenditure] Under Assumption D.C.1,

$$\frac{E_{a,t}}{E_{c,t}} \longrightarrow 0, \qquad \frac{E_{m,t}}{E_{c,t}} \longrightarrow 0.$$

Consequently, if

$$b_{j,t} \equiv \frac{E_{j,t}}{\sum_{k\in\{a,m,c\}} E_{k,t}},$$

then $(b_{a,t}, b_{m,t}, b_{c,t}) \to (0, 0, 1)$.

Proof. Set $k = c$ in (88). The result follows from (89) and the fact that final-expenditure shares sum to one. □

Let (V$_{j,t} \equiv p_{j,t}Y_{j,t}$)*denotenominalgrossoutputanddefine*

$$\mathbf{V}_t = (V_{a,t}, V_{m,t}, V_{c,t})', \qquad \mathbf{E}_t = (E_{a,t}, E_{m,t}, E_{c,t})'.$$

The nominal intermediate-use matrix is

$$(B_t)_{kj} \equiv \begin{cases} \gamma_{k,j,t}, & k \in \{a, m\}, \\ 0, & k = c, \end{cases} \tag{90}$$

where column j records sector j's expenditure shares on agricultural and manufacturing intermediates. Goods-market clearing and cost minimization give

$$\mathbf{V}_t = B_t\mathbf{V}_t + \mathbf{E}_t, \qquad \mathbf{V}_t = (I - B_t)^{-1}\mathbf{E}_t. \tag{91}$$

Define the direct attributed-employment requirement per unit of nominal gross output by

$$d_{j,t} \equiv \frac{L^{\text{attr}}_{j,t}}{V_{j,t}}. \tag{92}$$

[Terminal production network] The matrices B_t are productive, their Leontief inverses are uniformly bounded, and

$$B_t \longrightarrow B^*.$$

Moreover, for every $j \in \{a, m, c\}$,

$$d_{j,t} \longrightarrow d_j^* \in (0, \infty).$$

[Limiting gross output and employment] Under Assumptions D.C.1 and D.C.1, define

$$\mathbf{q}^* \equiv (I - B^*)^{-1}\mathbf{e}_c.$$

Then

$$\frac{\mathbf{V}_t}{V_{c,t}} \longrightarrow \frac{\mathbf{q}^*}{q_c^*}, \tag{93}$$

$$\ell_{j,t} \longrightarrow \ell_j^* \equiv \frac{d_j^* q_j^*}{\sum_{k\in\{a,m,c\}} d_k^* q_k^*}. \tag{94}$$

Proof. By Lemma D.C.1, $\mathbf{E}_t/E_{c,t} \to \mathbf{e}_c$. Continuity of matrix inversion on the set of productive matrices and uniform boundedness of the inverses imply

$$\frac{\mathbf{V}_t}{E_{c,t}} = (I - B_t)^{-1}\frac{\mathbf{E}_t}{E_{c,t}} \longrightarrow (I - B^*)^{-1}\mathbf{e}_c = \mathbf{q}^*.$$

Dividing by the c-component proves (93). Since $L_{j,t}^{\text{attr}} = d_{j,t}V_{j,t}$, dividing each sector's attributed employment by total employment and taking limits gives (94). $\square$

[Exact concentration of employment in consumer services] Under the assumptions of Lemma D.C.1,

$$\ell_{c,t} \longrightarrow 1$$

if and only if

$$d_a^* q_a^* = d_m^* q_m^* = 0.$$

Because Assumption D.C.1 imposes $d_a^*, d_m^* > 0$, this is equivalent to

$$\mathbf{e}_a'(I - B^*)^{-1}\mathbf{e}_c = \mathbf{e}_m'(I - B^*)^{-1}\mathbf{e}_c = 0. \tag{95}$$

Proof. The result follows immediately from (94). For a nonnegative productive input matrix, $q_j^* = \mathbf{e}_j'(I - B^*)^{-1}\mathbf{e}_c$. Thus $q_j^* = 0$ exactly when there is no direct or indirect input path from sector j into consumer services. $\square$

Corollary D.C.1 clarifies the scope of the service-employment branch used in Proposition 2. Final expenditure can converge entirely to consumer services while agriculture and manufacturing retain positive employment because they supply intermediate inputs. Accordingly, the exact limit $\ell_{c,t} \to 1$ is an additional branch restriction; it is not implied by final-demand concentration alone.

[Exact service-employment branch] For the analytical terminal result in Proposition 2, the economy is on a branch satisfying

$$\ell_{a,t} \longrightarrow 0, \qquad \ell_{m,t} \longrightarrow 0, \qquad \ell_{c,t} \longrightarrow 1. \tag{96}$$

Equivalently, under Assumption D.C.1, the terminal network satisfies the condition in Corollary D.C.1. If the calibrated terminal network instead contains positive nonservice input paths into consumer services, its employment limit is (94), not (96).

D.C.2 Terminal skill equilibrium

Let $r = w_s/w_u$ be the terminal skill premium. Conditional on $H > 0$, let $D_s(r; H)$ denote the share of total employment demanded in skilled labor and let

$$S^{\text{sup}}(r) = \frac{1}{1 + \exp[-(\beta_\pi \log r - c_0)/\sigma_{\text{skill}}]}. \tag{97}$$

[Terminal skill-market regularity] For every $H > 0$, $D_s(r; H)$ is continuous and strictly decreasing in r, $S^{\text{sup}}(r)$ is continuous and strictly increasing in r, and their endpoint values cross. Moreover, $D_s(r; H)$ is weakly increasing in H, and the increase is strict at the equilibrium whenever producer-service demand is positive.

[Terminal skilled share] Under Assumption D.C.2, for every $H > 0$ there is a unique skill premium $r^*(H)$ satisfying

$$D_s(r^*(H); H) = S^{\text{sup}}(r^*(H)).$$

The associated skilled share $S(H) = S^{\text{sup}}(r^*(H))$ is strictly increasing in H.

Proof. Existence follows from continuity and the endpoint conditions. Uniqueness follows because $D_s(r; H) - S^{\text{sup}}(r)$ is strictly decreasing in r. A higher H shifts this excess-demand function upward at the equilibrium. Its unique root therefore increases, and so does $S^{\text{sup}}(r^*(H))$. □

D.C.3 Proof of Proposition 2

The proposition is conditional on the exact service-employment branch in Assumption D.C.1. This branch implies $\lambda_{a,t}, \lambda_{m,t} \to 0$ and $\lambda_{s,t} \to 1$, because direct agricultural and manufacturing employment is bounded above by attributed employment in the corresponding final sector. Lemma D.C.2 gives the unique terminal skilled share

$$S^* = S(H^*), \qquad S'(H) > 0. \tag{98}$$

Replacement fertility and Lemma D.B imply

$$1 = \frac{\gamma}{\tau_s + \tau_e(S^*)e^*}, \tag{99}$$

and hence

$$e^* = \frac{\gamma - \tau_s}{\tau_e(S^*)}. \tag{100}$$

The restrictions $\gamma > \tau_s$ and $\tau_s > \tau_e(S^*)\tau_h$, together with the stated interiority condition, ensure that this education choice is positive and interior. The education rule then gives

$$e^* = \sqrt{\frac{g^*[\tau_s - \tau_e(S^*)\tau_h]}{\tau_e(S^*)}} - \tau_h.$$

Substituting (100) and solving for growth yields

$$g^* = \frac{[\gamma - \tau_s + \tau_e(S^*)\tau_h]^2}{\tau_e(S^*)[\tau_s - \tau_e(S^*)\tau_h]}. \tag{101}$$

If

$$H^* \geq \frac{e^*}{\kappa_c}, \tag{102}$$

service-era learning does not expand the inherited organizational frontier. On the exact service-employment branch, the terminal productivity law is

$$g^* = H^*(S^* L^*)^{\varphi_c}.$$

Therefore

$$L^* = \frac{1}{S^*}\left(\frac{g^*}{H^*}\right)^{1/\varphi_c}. \tag{103}$$

Equations (98), (100), (101), and (103) uniquely determine the aggregate terminal tuple (S^*, e^*, g^*, L^*), conditional on H^* and on the existence of a terminal competitive equilibrium satisfying the branch and regularity conditions. This proves Proposition 2. □

When $\tau_h \to 0$, equation (101) becomes

$$g^*(H) = \frac{(\gamma - \tau_s)^2}{\tau_s \tau_e(S(H))}. \tag{104}$$

Since $S'(H) > 0$ and $\tau_e'(S) < 0$, $g^*(H)$ is strictly increasing in H.

D.C.4 Proof of Proposition 3

The canonical branching experiment compresses a sectoral phase into one node in Figure 7. To make its timing consistent with (83) and (20), the intermediate node represents two consecutive model dates. At the first date, production in sector $q \in \{m, c\}$ generates productivity growth and parents choose education. At the second date, the educated cohort operates in the same sector q, so its education is applied to sector q's codification problem. The economy enters the terminal service regime only after this two-date phase.

Both paths inherit the same state (S_1, L_1, H_1). During the first date of the intermediate phase, full attributed specialization in sector q generates

$$g_q = H_1(S_1 L_1)^{\varphi_q}. \tag{105}$$

Under the household-origin mapping in (78), the education carried by the next cohort is, in general,

$$\bar{e}_q = \frac{\sum_{i\in\{a,m,s\}} \lambda_{i,q} n_{i,q} e_{i,q}}{\sum_{i\in\{a,m,s\}} \lambda_{i,q} n_{i,q}}, \tag{106}$$

where $e_{i,q}$ and $n_{i,q}$ are given by Lemma D.B with growth g_q. The candidate frontier generated during the second date of the phase is

$$\widetilde{H}_q = \frac{\bar{e}_q}{\kappa_q}. \tag{107}$$

Consequently, without any additional demographic restriction, the exact comparison is

$$H_A^* > H_B^* \quad \Longleftrightarrow \quad \frac{\bar{e}_m}{\kappa_m} > \frac{\bar{e}_c}{\kappa_c}, \tag{108}$$

provided neither candidate is truncated by the common inherited frontier.

Proposition 3 reports the closed-form special case of (108). For that result, the following restriction is required.

[Canonical-origin restriction] During the stylized intermediate phase, all households whose employment is attributed to sector q face the common demographic cost τ_q, where $\tau_c \equiv \tau_s$. This restriction is satisfied if producer-service employment is negligible during the phase. It is also immaterial for the baseline calibration when $\tau_m = \tau_s$.

Under Assumption D.C.4 and in the limit $\tau_h \to 0$, Lemma D.B gives

$$e_q = \sqrt{\frac{H_1 (S_1 L_1)^{\varphi_q} \tau_q}{\tau_e(S_1)}}. \tag{109}$$

Full specialization during the second date of the phase implies

$$\widetilde{H}_q = \frac{e_q}{\kappa_q}. \tag{110}$$

Assume

$$\min\{\widetilde{H}_m, \widetilde{H}_c\} \geq H_1. \tag{111}$$

The max law then implies

$$H_A^* > H_B^* \quad \Longleftrightarrow \quad \widetilde{H}_m > \widetilde{H}_c.$$

Using (109) and (110), this comparison becomes

$$\frac{1}{\kappa_m}\sqrt{H_1 (S_1 L_1)^{\varphi_m} \tau_m / \tau_e(S_1)} > \frac{1}{\kappa_c}\sqrt{H_1 (S_1 L_1)^{\varphi_c} \tau_s / \tau_e(S_1)}.$$

Canceling common terms and squaring gives

$$(S_1 L_1)^{\varphi_m - \varphi_c} \frac{\tau_m}{\tau_s} > \left(\frac{\kappa_m}{\kappa_c}\right)^2. \tag{112}$$

Finally, equation (104) implies

$$g_A^* > g_B^* \quad \Longleftrightarrow \quad H_A^* > H_B^*.$$

Combining this equivalence with (112) proves Proposition 3. □

D.C.5 Arbitrary finite transition paths

For a finite pre-terminal path P, define the candidate frontier generated at date t by

$$\widetilde{H}_t(P) = \bar{e}_t(P) \sum_{j \in \{a,m,c\}} \frac{\ell_{j,t}(P)}{\kappa_j}. \tag{113}$$

Repeated application of the max law gives

$$H^*(P) = \max\left\{H_0, \max_{t \in P} \widetilde{H}_t(P)\right\}. \tag{114}$$

[Path comparison] Consider two finite pre-terminal paths P and P' with the same initial frontier and the same terminal primitives. Suppose both paths subsequently enter the same exact service-employment branch, satisfy the terminal non-expansion condition (102), and satisfy the regularity conditions of Proposition 2. Then

$$g^*(P) > g^*(P') \quad \iff \quad \max_{t\in P} \widetilde{H}_t(P) > \max_{t\in P'} \widetilde{H}_t(P'),$$

provided both maxima exceed the common initial frontier.

Proof. Equation (114) maps each path into its inherited terminal frontier. Under the stated qualification on the initial frontier, the ordering of the inherited frontiers is the ordering of the two path maxima. Equation (104), together with Lemma D.C.2, makes terminal growth strictly increasing in the inherited frontier. Combining these two observations proves the result. □

Lemma D.C.5 is the arbitrary-path counterpart of the canonical comparison in Proposition 3. It does not require full specialization during the transition: interior sectoral employment shares enter directly through (113).

D.D Historical examples of producer-service codification

This appendix gives historical examples that clarify the interpretation of the codification friction κ_j and the organizational state H_t. The examples are not used to identify or calibrate the model. They illustrate the economic content of the mechanism: producer-service tasks can be performed informally in low-skill mode, or they can be reorganized into codified routines that require skilled workers and can be reused across firms and sectors.

The key distinction is between a function and the mode in which that function is performed. Work organization, quality control, inventory management, accounting, procurement, and personnel management exist in many economies. In low-skill mode, these functions are carried out through local experience, tacit knowledge, and informal supervision. In high-skill mode, the same functions are transformed into records, procedures, metrics, job categories, and decision rules. Manufacturing and industrial procurement are natural settings for this transformation because many of the relevant tasks are repeated, measurable, and costly to perform informally. Once codified, however, the resulting routines can travel into service organizations such as department stores, hospitals, banks, retailers, offices, and universities.

Table 16 summarizes the mapping from the model to several historical examples.

Table 16: Examples of producer-service tasks moving from low-skill to high-skill mode

Producer-service task	Low-skill mode	Industrial codification	Later service sector use
Work organization	Foreman judgment, local rules of thumb, informal supervision	Motion study, workflow design, timing of tasks, written procedures, training systems	Department stores, offices, hospitals, and other service organizations use workflow design, scheduling, clerical organization, and personnel practices
Quality control	Inspection by eye, informal rejection of defective output	Testing protocols, defect records, statistical process control, control charts	Clinical laboratories and hospitals use control charts and statistical quality-control routines to monitor measurement processes
Accounting and managerial control	Bookkeeping and transaction recording	Cost accounting, budgeting, return-on-investment calculations, divisional reporting, performance metrics	Retail chains, department stores, banks, hospitals, universities, and other multi-unit service organizations use budgeting and management-control systems
Inventory, procurement, and logistics	Visual stock checks, ad hoc ordering, relationship-based purchasing	Stock-control systems, procurement planning, supplier coordination, delivery scheduling	Retail chains, wholesalers, hospitals, restaurants, and distribution firms use inventory and logistics routines
Personnel and training	Informal apprenticeship, personal hiring, on-the-job learning	Job analysis, hiring rules, training manuals, personnel departments, performance evaluation	Department stores, hotels, hospitals, offices, and other service organizations use formal hiring and training systems

Work organization: from factory motion study to department stores. A first example is work organization. In low-skill mode, the task of organizing work is performed by foremen and supervisors using experience and local knowledge. Manufacturing made this task easier to codify because factory operations were repeated and measurable: worker motions, delays, output, and defects could be observed, timed, compared, and redesigned. The scientific-management tradition transformed work organization into a high-skill producer-service function involving motion study, workflow design, written procedures, and training systems.

Lillian Gilbreth provides a useful historical example of diffusion across sectors. Frank and Lillian Gilbreth developed motion-study methods within the scientific-management tradition, initially oriented toward industrial work. Historical accounts emphasize that Lillian Gilbreth later adapted motion study and industrial psychology to service and household settings, including department-store employees and consumers (Graham, 1992, 1999). In particular, her work at Macy's between 1925 and 1928 applied scientific-management and personnel-management ideas to hiring, training, employee handling, clerical organization, and the organization of work within the department store (Graham, 2000). This is the mechanism in the model: manufacturing and industrial work make a producer-service task easier to codify, and the resulting routine can later be used by service organizations.

Quality control: from industrial process control to clinical laboratories. A second example is quality control. In low-skill mode, quality is monitored through direct inspection or the judgment of experienced workers. Manufacturing made quality problems measurable because defects, tolerances, downtime, and output variation could be recorded. This gave rise to statistical process control. Walter Shewhart's work at Bell Laboratories in the 1920s introduced the control chart, and his later book formalized statistical control as a method for monitoring variation in production processes (Shewhart, 1931; Best and Neuhauser, 2006).

Once codified, the same routine was portable outside manufacturing. Clinical laboratories later adopted statistical quality-control methods to monitor the reliability of laboratory measurements. Levey and Jennings introduced statistical quality-control charts for clinical laboratories in 1950, adapting the logic of process control to repeated measurement in medical testing (Levey and Jennings, 1950; Westgard et al., 1981). The task changed from informal inspection to a high-skill routine based on records, thresholds, and statistical decision rules. This example maps directly into the model: codified organizational knowledge H_t makes it easier for a service organization to adopt a high-skill version of a task that was first formalized in an industrial setting.

Accounting and managerial control: from industrial cost control to retail management. A third example is accounting and managerial control. In low-skill mode, accounting records transactions and keeps the books. Large industrial firms and industrial procurement systems created a different problem: managers needed to compare plants, products, inventories, costs, divisions, and capital allocations. This transformed accounting into a high-skill producer-service activity: cost accounting, budgeting, return-on-investment analysis, divisional reporting, and management control. The historical literature on management accounting emphasizes the development of cost accounting and managerial control in large industrial and transportation enterprises, the influence of scientific management, and the later use of formal budgeting and performance measures in large corporations such as DuPont and General Motors (Johnson, 1975; Kaplan, 1984; Chandler, 1977).

The same managerial routines later became portable tools for service firms. James O. McKinsey helped professionalize budgeting and managerial accounting as a consulting service through books such as McKinsey (1922) and McKinsey (1924). In 1935, McKinsey completed a management consulting engagement for Marshall Field's, a large Chicago department-store chain, and then became its chairman and chief executive (Flesher, 1985). The relevant diffusion is not a machine or a product, but a producer-service routine: a way of measuring, budgeting, and managing a complex organization. In the language of the model, management accounting is a task that moves from bookkeeping in low-skill mode to codified high-skill managerial control.

E Calibration Details and Counterfactual Implementation

This appendix provides additional details on the numerical calibration and clarifies the interpretation of the counterfactual variation in manufacturing peaks.

E.A Numerical procedure

The model is simulated at an annual frequency and calibrated to the U.S. transition over 1850–2019. The simulation horizon is longer than the data window. The model date corresponding to 1850 is fixed at $t_0 = 60$, and all simulated moments are evaluated after applying this fixed calendar alignment. Gross-output input shares are interpolated so that the modern input-output structure is reached in calendar year 2025.

Let m^{data} denote the vector of empirical moments and $m^{model}(\Theta)$ the corresponding model moments. The calibrated parameter vector minimizes the loss

$$\mathcal{L}(\Theta) = \frac{1}{N}\sum_{\ell=1}^{N}\left(\frac{m_\ell^{model}(\Theta) - m_\ell^{data}}{\max\{|m_\ell^{data}|, \underline{s}_\ell\}}\right)^2,$$

for share and rate moments. Strictly positive level moments, such as normalized GDP per capita and the agriculture-to-service fertility ratio, are matched in logs. The constants $\underline{s}_\ell$ are moment-specific scale floors used to avoid overweighting moments close to zero.

The numerical minimization uses a global dual-annealing search followed by a Powell local search. I add small regularization penalties for high-frequency oscillations in the skill premium and in the aggregate skilled share. These penalties are not targeted economic moments; they rule out parameter vectors that fit isolated moments through implausibly volatile transition paths.

The half-saturation parameter in the codification frontier, $H_{1/2}$, is normalized relative to the simulated scale of codified organizational knowledge accumulation. For each candidate parameter vector, I first run a probe simulation with a fixed input value of $H_{1/2}$. I then set $H_{1/2}$ equal to the average of the final observations of the probe H_t path, subject to lower and upper bounds. This prevents the arbitrary units of H_t from mechanically determining the location of the frontier.

E.B Calibrated parameters and targeted moments

The internally calibrated parameter vector is

$$\Theta = \left(\gamma, \tau_a, \varphi_a, \varphi_m, \varphi_c, g^{GO}, \kappa_a, \kappa_m, \kappa_c, \eta_H, c_{col}, \alpha_a, \alpha_m, \alpha_c, \varepsilon_a, \varepsilon_c, w_a, w_c\right).$$

The manufacturing demand weight is fixed at $w_m = 2.5$. Manufacturing is the numeraire and $\varepsilon_m = 1$.

The targeted moments are: sectoral employment shares in agriculture, manufacturing, and services; the peak manufacturing employment share; the fertility path; the agriculture-to-service fertility ratio; normalized GDP per capita; long-run value added productivity growth in agriculture, manufacturing, and consumer services; the aggregate skilled employment share; within-value added skilled employment shares by sector; high-skill employment shares in producer-service inputs by using sector; and the aggregate change in the high-skill producer-service share between 1960 and 2019. The calibration is overidentified: the number of targeted moments exceeds the number of internally calibrated parameters.

E.C High-skill producer-service inputs by final sector

This subsection describes the construction of the high-skill producer-service moments used in the calibration. These moments discipline the sector-specific codification parameters κ_j, which determine how easily producer-service tasks used by final sector $j \in \{a, m, c\}$ can be reorganized into high-skill form.

The object I need is not the average education of all workers in agriculture, manufacturing, or consumer services. The model distinguishes final-good production from the producer-service tasks used as intermediate inputs. The relevant empirical counterpart is therefore the education intensity of producer-service occupations employed within each broad final sector. In the model, these occupations correspond to managerial, professional, technical, administrative, accounting, engineering, planning, and coordination tasks.

I construct the U.S. moments using the 2019 IPUMS USA Census sample. I classify workers into broad industries using the 1950 Census industry classification. Agriculture includes agriculture, forestry, and fisheries. Manufacturing includes durable and nondurable manufacturing. Consumer services include retail trade, repair services, private households, hotels and lodging, laundry and cleaning services, personal services, and entertainment and recreation services. I then identify producer-service occupations using the 1950 Census occupation classification. The baseline definition includes occupations that clearly perform professional, technical, managerial, or organizational tasks, such as accountants, architects, engineers, lawyers, statisticians, personnel workers, technicians, purchasing agents, inspectors, insurance agents, real-estate agents, and related business-facing occupations. The replication files report the exact industry and occupation codes.

For each broad final sector j, I compute the weighted share of producer-service workers with at least one year of college:

$$HSPS_j = \frac{\sum_{n \in j} \omega_n \mathbb{1}\{\text{PS occupation}_n = 1\}\mathbb{1}\{\text{college}_n = 1\}}{\sum_{n \in j} \omega_n \mathbb{1}\{\text{PS occupation}_n = 1\}},$$

where ω_n is the person weight. These are the empirical counterparts of the high-skill shares within producer-service inputs used by agriculture, manufacturing, and consumer services in the model. The resulting U.S. moments are

$$HSPS_a = 0.393, \qquad HSPS_m = 0.653, \qquad HSPS_c = 0.529.$$

Thus, producer-service occupations employed in manufacturing are substantially more college-intensive than producer-service occupations employed in agriculture or consumer services. This ranking is the key moment used to discipline the ordering of codification resistance in the model. The calibrated model chooses $\kappa_m < \kappa_c < \kappa_a$, so manufacturing related producer-service tasks are easiest to reorganize into high-skill form.

I also examine whether this pattern is specific to the United States. Using IPUMS International, I construct the same object for 298 census samples spanning countries at very different stages of development, including low, middle and high income countries. In then repeat the exercise by classifying industries into agriculture, manufacturing, and consumer services using the harmonized IPUMS industry variable. I classify producer-service occupations using ISCO major groups corresponding to managerial, professional, and technical occupations.[11] I then compute college shares separately by broad sector and occupation type. Figure 23 reports the resulting averages for agriculture, manufacturing, and consumer services. Producer-service occupations are much more college-intensive than other occupations in every broad sector. Among producer-service occupations, the college share is also higher in manufacturing than in consumer services or agriculture. This supports the interpretation of the U.S. calibration moments: the relevant margin is not that all manufacturing workers are uniformly more educated, but that manufacturing uses a set of producer-service occupations that is especially skill-intensive.

E.D Untargeted producer-service response

As an external validation of the producer-service block, I compare the model to a final-demand-shock exercise in the 2015 OECD ICIO table. In the data, I compute the service output induced by a $100 final-demand shock to agriculture, manufacturing, and consumer services. In the model, I apply the analogous shock to the calibrated economy and compute the high-skill share of induced producer-service output. This moment is not directly targeted in the calibration.

E.E Interpreting the manufacturing demand counterfactual

The counterfactuals in the main text vary the manufacturing demand shifter w_m to generate economies with different peak manufacturing employment shares. This is a reduced-form way to change the relative pull of manufacturing during the transition while holding technology, fertility and education preferences, and production primitives fixed.

This counterfactual has an equivalent price-wedge representation. Final demand over agriculture, manufacturing, and consumer services implies relative demand terms of the form

$$K_a = \left(\frac{p_a}{p_m}\right)^{1-\sigma} Y^{(1-\sigma)(k_a-1)} \frac{w_a}{w_m^{k_a}}, \qquad K_c = \left(\frac{p_c}{p_m}\right)^{1-\sigma} Y^{(1-\sigma)(k_c-1)} \frac{w_c}{w_m^{k_c}},$$

11 This corresponds to ISCO code 3 or less.

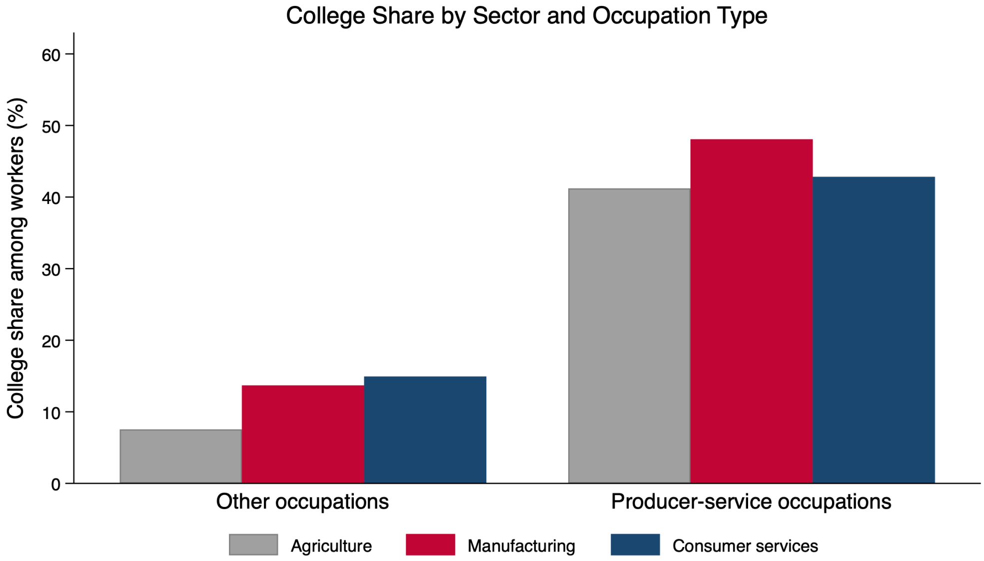


Figure 23: College share by sector and occupation type in IPUMS International. The figure uses 298 census samples covering economies across the development spectrum. Producer-service occupations are defined using managerial, professional, and technical occupation groups. Bars report average college shares separately for other occupations and producer-service occupations within agriculture, manufacturing, and consumer services.

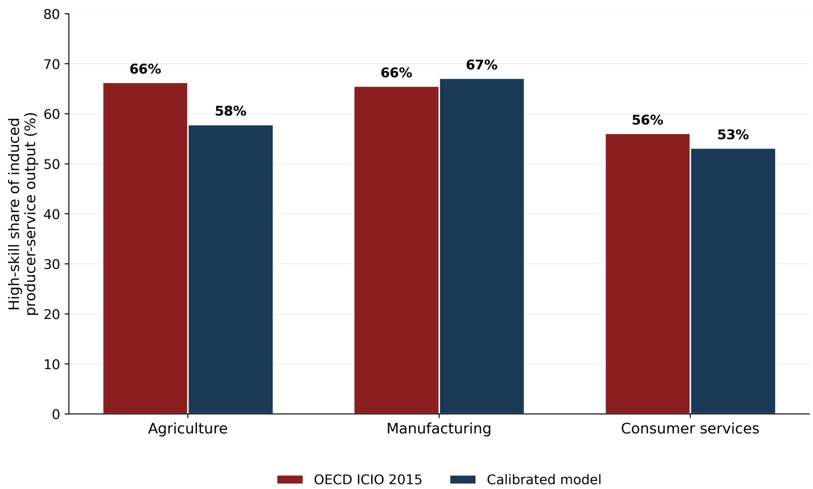


Figure 24: High-skill share of producer-service output induced by a $100 final-demand shock. The data exercise uses Leontief total requirements from the 2015 OECD ICIO table; the model exercise applies the same shock to the calibrated economy.

where $k_j \equiv \varepsilon_j/\varepsilon_m$. Changing the manufacturing demand weight from w_m to $f_m w_m$ therefore changes the relative demand terms to

$$K_a(f_m) = K_a(1) f_m^{-k_a}, \qquad K_c(f_m) = K_c(1) f_m^{-k_c}.$$

Now consider an economy with the original w_m, but in which households face effective final-good prices

$$\widetilde{p}_{a,t} = (1+\tau_a) p_{a,t}, \qquad \widetilde{p}_{m,t} = p_{m,t}, \qquad \widetilde{p}_{c,t} = (1+\tau_c) p_{c,t}.$$

The wedge economy generates the same relative demand schedules as the $f_m w_m$ economy if

$$(1+\tau_a)^{1-\sigma} = f_m^{-k_a}, \qquad (1+\tau_c)^{1-\sigma} = f_m^{-k_c}.$$

Equivalently,

$$\boxed{1+\tau_a = f_m^{-k_a/(1-\sigma)}, \qquad 1+\tau_c = f_m^{-k_c/(1-\sigma)}.}$$

Thus the w_m counterfactual can be interpreted either as a manufacturing demand shifter or as an equivalent system of final-good wedges. The two representations are algebraically identical for household final-demand allocations. The purpose of varying w_m is therefore not to assign structural content to tastes, but to provide a parsimonious parameterization of a relative-price distortion that changes the size of the manufacturing phase.

E.F Construction of input-output coefficients

This subsection describes how I construct the modern input-output coefficients used in the calibration. The model production function for each final-good sector $j \in \{a, m, c\}$ uses value added, agricultural intermediates, manufacturing intermediates, and producer-service intermediates:

$$Y_{j,t} = A_{go,j,t} VA_{j,t}^{\theta_{j,t}} M_{a,j,t}^{\gamma_{a,j,t}} M_{m,j,t}^{\gamma_{m,j,t}} PS_{j,t}^{\gamma_{ps,j,t}}, \qquad \theta_{j,t} + \gamma_{a,j,t} + \gamma_{m,j,t} + \gamma_{ps,j,t} = 1.$$

The modern - 2025 - coefficients are constructed from the BEA Use Table, after redefinitions, at purchasers' prices. I use the sector-level table and map BEA industries into the model sectors as follows. Agriculture is BEA sector 11, manufacturing is BEA sector 31G, and consumer services are the sum of educational services, health care, and social assistance; arts, entertainment, recreation, accommodation, and food services; and other services except government, corresponding to BEA sectors 6, 7, and 81. Producer-service inputs are the sum of transportation and warehousing, information, finance, insurance, real estate, rental, and leasing, and professional and business services, corresponding to BEA sectors 48TW, 51, $FIRE$, and $PROF$.

For each destination sector j, I compute gross output Y_j and four raw shares:

$$\theta_j^{raw} = \frac{VA_j}{Y_j}, \qquad \gamma_{a,j}^{raw} = \frac{U_{A,j}}{Y_j}, \qquad \gamma_{m,j}^{raw} = \frac{U_{M,j}}{Y_j}, \qquad \gamma_{ps,j}^{raw} = \frac{U_{PS,j}}{Y_j},$$

where $U_{A,j}$ is the use of agricultural inputs by destination sector j, $U_{M,j}$ is the use of manufacturing inputs, and $U_{PS,j}$ is the use of producer-service inputs. Since the model abstracts from residual input categories such as mining, utilities, construction, government, scrap, and noncomparable imports, I normalize the four model-relevant shares to sum to one:

$$(\theta_j, \gamma_{a,j}, \gamma_{m,j}, \gamma_{ps,j}) = \frac{\left(\theta_j^{raw}, \gamma_{a,j}^{raw}, \gamma_{m,j}^{raw}, \gamma_{ps,j}^{raw}\right)}{\theta_j^{raw} + \gamma_{a,j}^{raw} + \gamma_{m,j}^{raw} + \gamma_{ps,j}^{raw}}.$$

For agriculture, the BEA table gives

$$\theta_a^{raw} = 0.394, \qquad \gamma_{a,a}^{raw} = 0.295, \qquad \gamma_{m,a}^{raw} = 0.222, \qquad \gamma_{ps,a}^{raw} = 0.051.$$

After normalizing the four model-relevant components, the coefficients are

$$\theta_a = 0.410, \qquad \gamma_{a,a} = 0.307, \qquad \gamma_{m,a} = 0.230, \qquad \gamma_{ps,a} = 0.053.$$

For manufacturing, the BEA table gives

$$\theta_m^{raw} = 0.354, \qquad \gamma_{a,m}^{raw} = 0.060, \qquad \gamma_{m,m}^{raw} = 0.407, \qquad \gamma_{ps,m}^{raw} = 0.070.$$

After normalizing the four model-relevant components, the unadjusted coefficients are

$$\theta_m^{BEA} = 0.397, \qquad \gamma_{a,m}^{BEA} = 0.068, \qquad \gamma_{m,m}^{BEA} = 0.457, \qquad \gamma_{ps,m}^{BEA} = 0.078.$$

This raw producer-service share captures producer services purchased from outside the manufacturing sector, but it does not fully capture service tasks performed inside manufacturing establishments.

I therefore make a conservative adjustment for in-house producer-service activity in manufacturing. A growing literature emphasizes that the boundary between manufacturing and services is blurred because manufacturing firms both purchase services from outside suppliers and perform many service tasks internally (Miroudot and Cadestin, 2017; Bernard et al., 2017). Miroudot and Cadestin (2017) report that, across countries, between 25 and 60 percent of employment in manufacturing firms is in service support functions such as R&D, engineering, transport, logistics, distribution, marketing, sales, after-sale services, IT, management, and back-office support. To capture this margin without mechanically assigning a large share of manufacturing to producer services, I reallocate only 20 percent of the normalized manufacturing value added share and 20 percent of the normalized manufacturing intermediate share to producer services:

$$\theta_m = 0.8\,\theta_m^{BEA}, \qquad \gamma_{m,m} = 0.8\,\gamma_{m,m}^{BEA},$$

$$\gamma_{ps,m} = \gamma_{ps,m}^{BEA} + 0.2\,\theta_m^{BEA} + 0.2\,\gamma_{m,m}^{BEA}, \qquad \gamma_{a,m} = \gamma_{a,m}^{BEA}.$$

This gives the adjusted manufacturing coefficients

$$\theta_m = 0.318, \qquad \gamma_{a,m} = 0.068, \qquad \gamma_{m,m} = 0.365, \qquad \gamma_{ps,m} = 0.249.$$

The adjustment is conservative relative to the OECD evidence: the resulting producer-service share in manufacturing is about 25 percent, which is at the lower end of the 25–60 percent range documented for service support employment inside manufacturing firms.

For consumer services, the BEA table gives

$$\theta_c^{raw} = 0.613, \qquad \gamma_{a,c}^{raw} = 0.002, \qquad \gamma_{m,c}^{raw} = 0.108, \qquad \gamma_{ps,c}^{raw} = 0.224.$$

After normalizing the four model-relevant components, the coefficients are

$$\theta_c = 0.647, \qquad \gamma_{a,c} = 0.002, \qquad \gamma_{m,c} = 0.114, \qquad \gamma_{ps,c} = 0.237.$$

The final modern input-output coefficients used in the calibration are therefore

Destination sector	θ_j	$\gamma_{a,j}$	$\gamma_{m,j}$	$\gamma_{ps,j}$
Agriculture	0.410	0.307	0.230	0.053
Manufacturing	0.318	0.068	0.365	0.249
Consumer services	0.647	0.002	0.114	0.237

In the calibrated transition, the economy starts in 1850 with value added production, $\theta_{j,1850} = 1$, and the input-output coefficients gradually converge to these modern values by 2025. This allows the model to capture the increasing role of intermediate inputs and producer services over development while preserving a parsimonious three-sector structure.

E.G Robustness: Recalibrating Producer-Service Intensities from Spending Shocks

The baseline calibration disciplines the codification parameters $\kappa_a, \kappa_m, \kappa_c$ using worker-level Census moments: within each broad final sector, I restrict attention to producer-service occupations and measure the share of workers with at least one year of college. This appendix reports a robustness exercise that instead disciplines the same parameters using input-output spending-shock moments from the 2015 OECD ICIO table.

I aggregate the 2015 OECD ICIO table to global sectoral industries. For each final sector $j \in \{a, m, c\}$, I apply a \$100 final-demand shock and compute the induced gross-output vector x^j using the Leontief inverse. I then focus only on selected producer-service output. The denominator is

$$PS = \{J58T60, J61, L, N, J62\_63, K, M\},$$

where $J58T60$ is publishing, motion picture, and broadcasting; $J61$ is telecommunications; L is real estate; N is administrative and support services; $J62\_63$ is IT and other information services; K is finance and insurance; and M is professional, scientific, and technical activities. The high-skill producer-service subset is

$$HS = \{J62\_63, K, M\}.$$

The ICIO target is

$$\chi_j^{data} = \frac{\sum_{s \in HS} x_s^j}{\sum_{s \in PS} x_s^j}.$$

Thus, χ_j^{data} is the high-skill share of induced selected producer-service output. It is not a worker-level education moment, but an input-output proxy for the high-skill producer-service intensity of final demand for sector j.

The agriculture and manufacturing shocks are allocated across ICIO industries in proportion to gross output. I compute two consumer-service targets. The first uses a strict consumer-service definition, selecting market-service sectors whose final-consumption-to-gross-output ratio exceeds 50%. The second uses final-expenditure weights over all market-service sectors, allowing mixed-use service industries such as finance, information, and professional services to enter the consumer-service shock in proportion to their final-expenditure importance.

For each definition, I recalibrate only

$$(\kappa_a, \kappa_m, \kappa_c),$$

holding all other parameters fixed at their baseline values. The model analogue is computed in 2015, the same year as the ICIO table. A \$100 model final-demand shock to sector j induces gross-output responses x_k^j across sectors k. The corresponding model moment is

$$\chi_j^{model} = \frac{\sum_k h_k^{PS} \gamma_k^{PS} x_k^j}{\sum_k \gamma_k^{PS} x_k^j},$$

where γ_k^{PS} is sector k's producer-service input share and h_k^{PS} is the high-skill share of producer-service tasks used by sector k. The recalibration solves

$$\min_{\kappa_a, \kappa_m, \kappa_c} \sum_{j \in \{a,m,c\}} \left(\chi_j^{model}(\kappa_a, \kappa_m, \kappa_c) - \chi_j^{data} \right)^2.$$

After each recalibration, I repeat the standardized counterfactual experiment. I choose the manufacturing demand weight w_m so that the economy reaches a 10 percent manufacturing employment peak, and then choose a second w_m so that the economy reaches a 30 percent peak. I compare 2019 outcomes across these two economies: GDP per capita, the skill entry rate, and the FIRE/high-skill producer-service share of service employment.

Figure 25 reports the results. The baseline Census calibration and both ICIO spending-shock recalibrations imply that raising the manufacturing peak from 10 percent to 30 percent increases GDP per capita, skill entry, and the high-skill producer-service share of services. The final-expenditure-weighted specification produces the largest effect because it implies that broad service final demand is sufficiently high-skill intensive in its producer services inputs to take advantage of the knowledge base created by the manufacturing boom. In contrast, the strict consumer services definition based on final demand being larger than 50% of gross output yields high-skill share of producer services targets which are much lower hence leading to a recalibration of κ_c around 3 times larger than the baseline. This implies that the consumer service sector has relatively little use for the high-skill inputs used by the manufacturing sector and hence lowers the overall effects.

This exercise is a robustness check, not a replacement for the baseline calibration. The baseline directly targets the education intensity of workers in producer-service occupations within each broad final sector. The spending-shock exercise instead infers high-skill producer-service intensity from industry-level input-output relationships and from the classification of high-skill producer-service sectors. It is less direct, but it shows that the main counterfactual conclusion is not driven by the baseline's strict consumer-service classification: manufacturing continues to matter because it expands demand for high-skill producer-service tasks and leaves the economy with a more skill-intensive service sector.

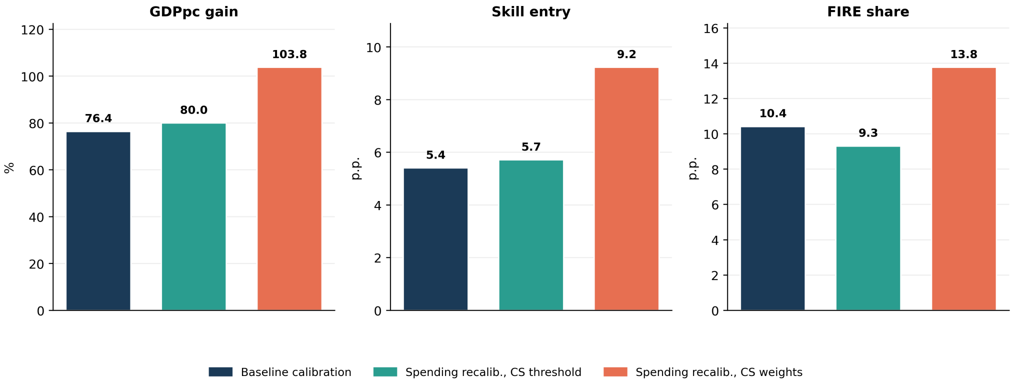


Figure 25: Robustness to alternative producer-service intensity targets. The figure compares the effect of increasing the manufacturing employment peak from 10 percent to 30 percent under three calibrations: the baseline Census-based calibration, a recalibration that matches 2015 ICIO spending-shock targets using a strict consumer-service definition, and a recalibration that matches 2015 ICIO spending-shock targets using final-expenditure weights over market services. GDP per capita is reported as a percent change. Skill entry and FIRE/high-skill producer-service shares are reported in percentage points.

F Policy Extensions

This appendix studies how trade policy affects the development path in the calibrated economy. The main counterfactuals in Section 6 vary the size of the manufacturing phase directly. Here I instead introduce an explicit trade environment and ask whether openness amplifies or weakens the manufacturing based development mechanism.

F.A Trade environment

I extend the calibrated economy to a small-open-economy setting. Home can trade agriculture and manufacturing with a rest-of-world economy. Consumer services and producer services remain non-traded. The rest of the world follows the same calibrated transition as Home, but may be shifted forward or backward in time. Let Δ denote Home's development head start relative to the rest of the world. If $\Delta > 0$, Home is more advanced than its trading partner; if $\Delta < 0$, Home trades with a more advanced partner.

Trade is introduced through an Armington aggregator for the two traded final goods. For $j \in \{a, m\}$, the consumer price of the traded composite is

$$p^A_{j,t} = \left[(1-\lambda_j)p_{j,t}^{1-\theta} + \lambda_j(\tau p^*_{j,t})^{1-\theta}\right]^{\frac{1}{1-\theta}},$$

where $p_{j,t}$ is the Home producer price, $p^*_{j,t}$ is the foreign producer price, τ is the iceberg trade cost, θ is the Armington elasticity, and λ_j governs the weight on foreign varieties. The domestic expenditure share is

$$d_{j,t} = (1-\lambda_j)\left(\frac{p_{j,t}}{p^A_{j,t}}\right)^{1-\theta}.$$

Before the opening date, trade costs are prohibitive and $d_{j,t} = 1$. After the opening date, trade costs fall to the counterfactual level.

The production side is kept close to the closed-economy model. Domestic production responds to domestic absorption, not to foreign demand. Thus, trade affects employment and dynamics by changing the domestic expenditure share on Home agriculture and manufacturing. Exports are computed as an accounting residual from domestic gross output net of domestic final absorption. Imports are then scaled so that the value of imports equals the value of exports. Hence trade is balanced period by period:

$$\sum_{j\in\{a,m\}} M_{j,t} = \sum_{j\in\{a,m\}} X_{j,t}.$$

This balanced-trade restriction prevents the welfare effects from being driven by external borrowing or lending. It also makes the exercise conservative: trade affects the development path only through prices, domestic absorption, and sectoral reallocation.

F.B Sectoral effects of opening

Figure 26 shows the first-order effect of the benchmark trade opening. Relative to the protected counterfactual, openness reduces agriculture and manufacturing employment and increases service employment. This is the key margin in the model. Trade does not matter only because it changes prices; it changes the sectoral path through which the economy develops.

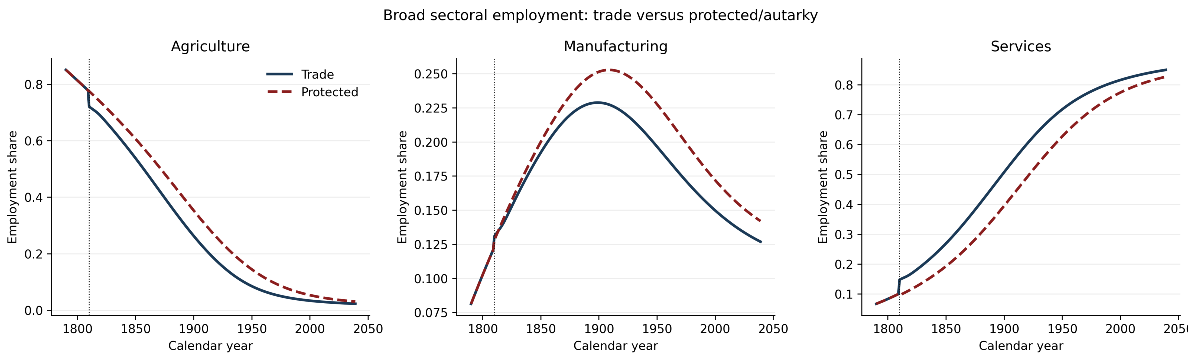


Figure 26: Broad sectoral employment under trade and protection.

Figure 27 shows that the service expansion induced by trade is also compositionally different. In the open economy, services expand more toward consumer services and less toward producer services. Within producer services, the economy also develops a lower high-skill producer-service share for much of the transition. This is the relevant dynamic channel: trade moves labor into services, but into a less skill-intensive service economy.

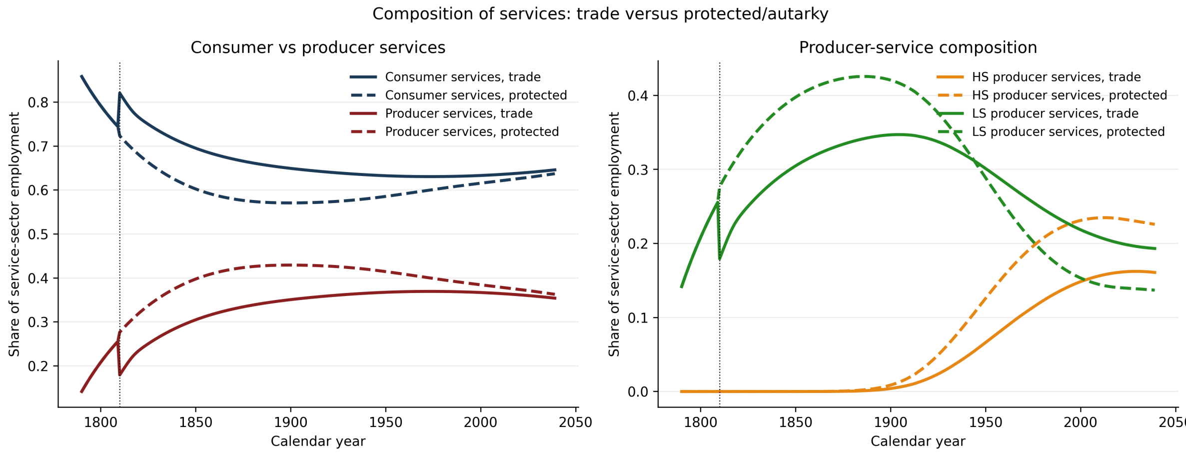


Figure 27: Composition of services under trade and protection.

This prediction is directly connected to the empirical validation exercise discussed later in the paper. Using subnational trade shocks constructed from IPUMS International, I test whether trade-induced reductions in manufacturing exposure shift local labor markets toward less producer-service intensive and less skill-intensive service employment.

F.C Welfare gains from trade

I compute welfare gains as the difference between the open economy and the protected counterfactual:

$$\Delta W_t = W_t^{open} - W_t^{protected}.$$

The total discounted gain is summarized as a consumption-equivalent variation:

$$\text{CEV} = \exp\left(\frac{\sum_t \beta^t \Delta W_t}{(1-\gamma)\sum_t \beta^t}\right) - 1.$$

This expresses the average discounted welfare gain as a permanent proportional increase in consumption.

Figure 28 plots the dynamic welfare path in the benchmark opening. Trade raises welfare on impact because households gain access to cheaper traded goods. Over time, however, the welfare gain declines and eventually turns negative. The reason is that the static consumption gain is partly offset by a weaker manufacturing phase and a less skill-intensive service transition.

Figure 28: Dynamic gains from trade in the benchmark opening.

Thus, the model separates two forces. The static gains from trade are positive at the opening date. The dynamic effect can be negative if trade reduces the manufacturing phase that builds human capital, codified organizational knowledge, and high-skill producer-service capacity.

F.D Manufacturing comparative advantage

I next vary Home's static comparative advantage in manufacturing. The exercise changes the gross-output productivity wedge

$$\frac{A_m^{GO}}{A_a^{GO}},$$

holding fixed the dynamic spillover parameters. Hence the exercise changes Home's static specialization pattern without mechanically changing the strength of manufacturing spillovers.

Figure 29 shows that the gains from trade rise sharply with Home's static manufacturing comparative advantage. When Home has weak manufacturing comparative advantage, the gains from trade are small or negative. When Home has strong manufacturing comparative advantage, the CEV becomes large. Figure 30 confirms that the wedge works through the intended channel: stronger manufacturing comparative advantage raises export intensity in tradables.

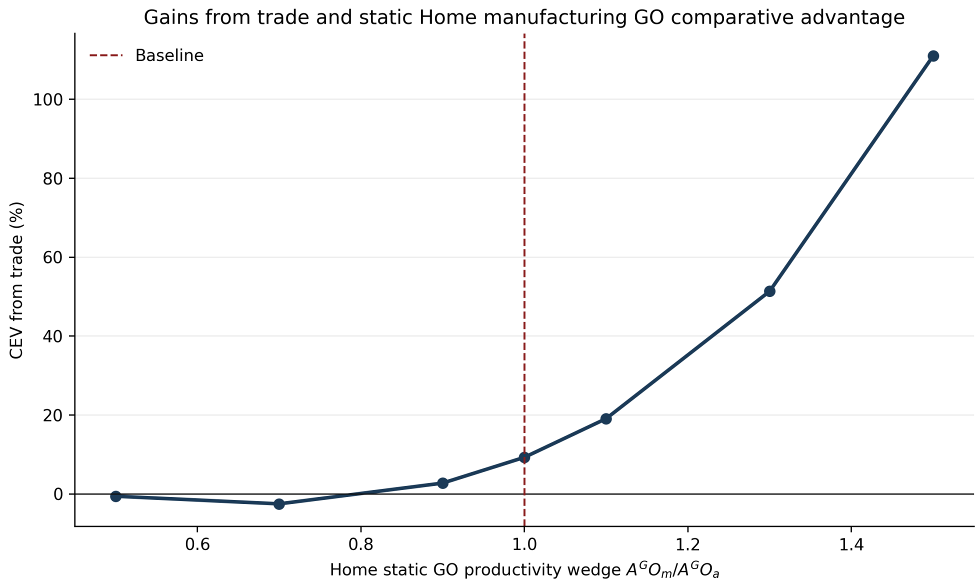


Figure 29: Gains from trade and static Home manufacturing comparative advantage.

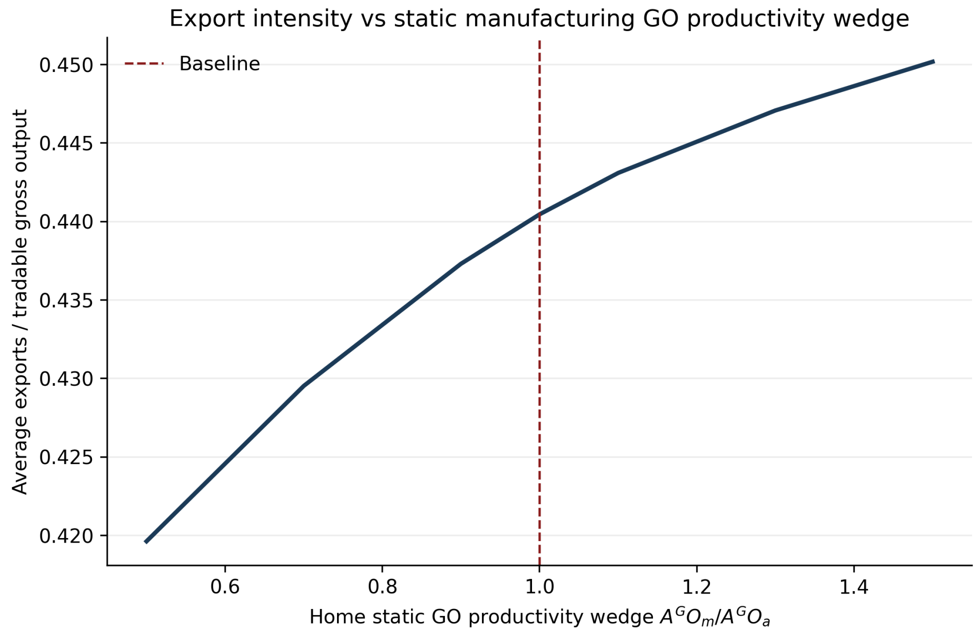


Figure 30: Export intensity and static manufacturing comparative advantage.

The lesson is that openness is most beneficial when it supports, rather than displaces, the manufacturing phase. Comparative advantage in manufacturing makes trade less likely to generate premature movement into low-skill services.

F.E Development gaps and timing

Finally, I vary both the timing of trade opening and the development gap between Home and the rest of the world. Figure 31 reports two heatmaps. The left panel reports total discounted gains from trade. The right panel reports the long-run dynamic gain, measured by the final value of $W_t^{open} - W_t^{protected}$.

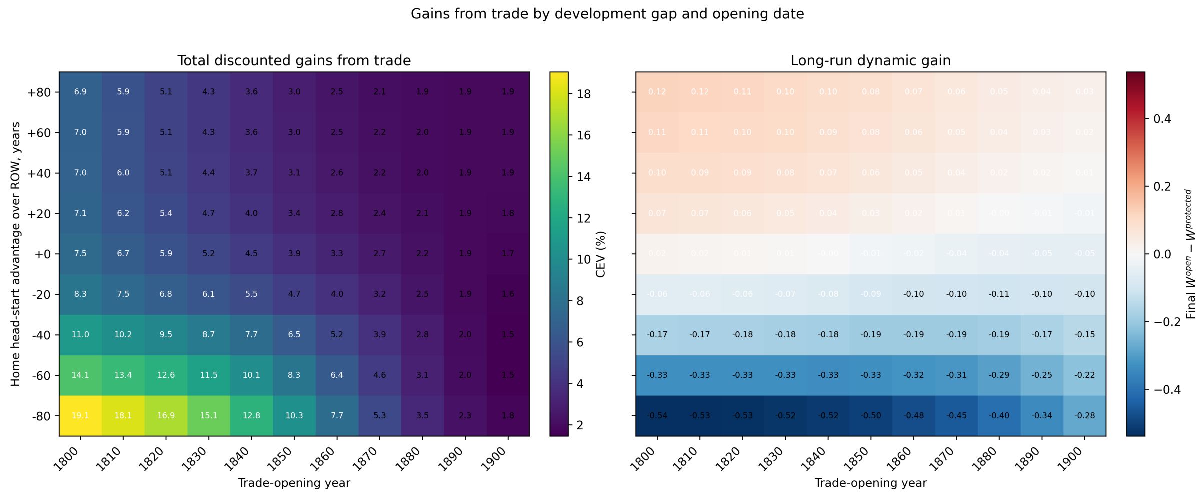


Figure 31: Gains from trade by development gap and opening date. Positive values on the vertical axis mean that Home is more advanced than the rest of the world. Negative values mean that Home opens to a more advanced partner.

The two panels reveal a sharp distinction between total and dynamic gains. Total discounted gains are largest when Home opens early to a more advanced partner. This reflects the large static benefits from cheaper traded goods. Long-run dynamic gains display the opposite pattern. They are positive when Home is more advanced than its partner, and negative when Home opens to a more advanced partner.

This asymmetry follows directly from the model mechanism. A less advanced economy opening to a more advanced partner imports cheaper manufactures, contracts domestic manufacturing, and enters services earlier. This raises consumption in the short run but weakens the domestic accumulation of high-skill producer-service capabilities. By contrast, an advanced economy opening to a less advanced partner preserves its manufacturing and producer-service base while still benefiting from trade. In that case, the long-run dynamic welfare effect can remain positive.

These results should however not be read as an argument for autarky. This exercise abstracts from technology diffusion, foreign direct investment, imported capital goods, and learning through exporting. Instead, we isolate one force: when manufacturing builds the human capital and organizational foundations of high-growth services, trade policy affects development not only through prices, but also through the type of service economy that emerges.